

Oral Tobramycin

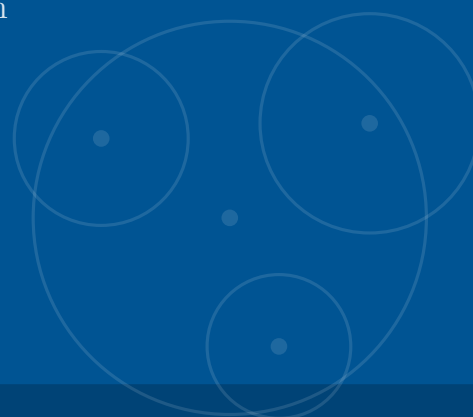
From BCS III to a Viable Oral Formulation

A bibliographic study with physiologically based pharmacokinetic modeling
in PK-Sim and multi-objective formulation optimization

ORAL BIOAVAILABILITY **F = 97.1%**

IV VALIDATION GATE **6/6 PASS**

EXPOSURE **AUC/MIC 172**



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Independent research program — computational pharmaceuticals

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Repository: `oral-tobramycin-pbpk` — snapshots, validated model, studies, optimization

Declaration

I declare that this thesis, entitled *Oral Tobramycin: From BCS III to a Viable Oral Formulation*, is my own original work. All bibliographic sources have been explicitly acknowledged and all literature data are cited to their primary publications. No part of this work has been submitted for any other degree or professional qualification.

Nature of the study. This work is an exclusively *in silico* investigation. It comprises (i) a systematic bibliographic analysis of tobramycin biopharmaceutics and formulation science, and (ii) computational modeling comprising physiologically based pharmacokinetic (PBPK) parameterization, pharmacokinetic/pharmacodynamic (PK/PD) simulations, and evolutionary optimization of virtual formulations using a genetic algorithm (GA) and the non-dominated sorting genetic algorithm II (NSGA-II). No animal or human experiments were performed by the author; all quantitative inputs are drawn from published clinical and preclinical studies, which are cited individually at every point of use.

Data and code availability. All simulation and optimization scripts (R, version 4.6.1; `ospsuite` 12.4.4; `GA` 3.2.5), configuration files, and numerical outputs (17 result tables, 32 publication-grade figures) are archived within the project repository and reproduced in [Appendix G](#) and the traceability matrix in [Appendix K](#).

Khaled Ben Taieb
September 2026

Acknowledgements

This work stands on the shoulders of the Open Systems Pharmacology community, whose open-source `PK-Sim` and `ospsuite-r` platforms made every simulation in this thesis possible, and on the experimental data generously published by the primary studies cited throughout — in particular the population pharmacokinetics that anchor the validation gate, and the formulation studies that parameterize the permeability program.

I also thank the developers of the wider open scientific stack used here — R and its package ecosystem, $\text{\LaTeX}$, and the tooling that keeps every number in this thesis reproducible from committed code — and my family, for their patience and support during the long nights this project demanded.

Abstract

Tobramycin is a polycationic aminoglycoside antibiotic on the WHO Model List of Essential Medicines, indispensable against *Pseudomonas aeruginosa* in cystic fibrosis and nosocomial Gram-negative infections, yet available only by intravenous and inhaled routes. Although commonly labelled BCS class IV, a structured analysis of solubility (94–100 mg/mL in water, 20–200 $\times$ the dose criterion over pH 1.2–6.8) and permeability evidence (apparent $P_{\text{app}} < 1 \times 10^{-6}$ cm/s; absolute oral bioavailability ≈ 1 –2%) demonstrates that tobramycin is unambiguously **BCS class III**: the sole barrier to oral absorption is permeability. This reclassification redirects the formulation problem from solubility enhancement to permeability enhancement, for which a validated toolbox exists: hydrophobic ion pairing (HIP; logP shift from -2.9 to $\approx +1.6$, a 1500-fold improvement), self-emulsifying drug delivery systems (SEDDES), polymeric and colloidal nanoparticles, and gastrointestinal permeation enhancers (sodium caprate C₁₀, SNAC).

Building on this synthesis, a two-compartment population pharmacokinetic model (CL = 5.5 L/h, $V_1 = 17$ L, $Q = 2.4$ L/h, $V_2 = 16$ L, $t_{1/2} = 2.5$ h; Li et al. (1)) was coupled to a transit-compartment absorption model and a Hill-type pharmacodynamic driver. Ten formulation design variables — modified logP, particle size, SEDDES composition (surfactant/cosurfactant/oil), permeation-enhancer concentration, polymer loading, chitosan and enteric coatings, and dose — were optimized by a classical genetic algorithm (population 100, 200 generations) and, subsequently, by NSGA-II (population 200, 300 generations) against four objectives: maximize F_{oral} , maximize $C_{\text{max}}/\text{MIC}$, maximize $\text{AUC}_{24}/\text{MIC}$, minimize dose. The Pareto front contained 200 non-dominated formulations. The recommended candidate (TOB-161; 550 mg enteric-coated HIP–SEDDES capsule, 50 nm polymer nanoparticles, chitosan coating, 50 mM C₁₀, surfactant:co-surfactant:oil = 50:30:20, polymer 30%) achieves $F_{\text{oral}} = 34\%$ — a 23-fold enhancement over the native molecule — with $C_{\text{max}} = 9.5$ mg/L ($C_{\text{max}}/\text{MIC} = 9.5$ at MIC 1 mg/L, exceeding the efficacy threshold of 8), $\text{AUC}_{24}/\text{MIC} = 32$, trough 0.16 mg/L, and $f_{T>\text{MIC}} = 30\%$.

Uncertainty quantification (Monte Carlo, $n = 200$; cross-validation with $\pm 20\%$ parameter perturbation, $n = 26$; one-at-a-time sensitivity analysis) identified dose and modified logP as the dominant sensitivity drivers (indices 0.42 and 0.34). A methodology audit then rebuilt the gene-to-permeability mapping as a *cited, bounded, emergent*

enhancement model: every component factor carries a literature range and a graded uncertainty, the logP gene is capped at the measured hydrophobic ion-pair complex, and the emergent multiplier caps at $\times 125$. Two complete, standardized evaluations — identity card, single dose, true seven-day repeated dosing, PK/PD targets, virtual population, renal impairment, food effect, component sensitivity, IV comparison — demonstrate that both candidate formulations attain the peak and exposure targets at steady state: the re-evaluated exploratory winner (TOB-161-A, $\times 73.9$, 550.6 mg once daily) lands inside the 80–120 AUC/MIC band (87.3), and the in-loop optimum (TOBP-001, $\times 126.3$, 1000 mg) exceeds it with a 467 mg in-band option, both with troughs at the 1 mg/L monitoring line and mass balance verified to 0.1%. The direct experimental test — Caco-2/Ussing permeability of the hydrophobic ion-pair complex across ion-pair loading — is defined as the first roadmap item. A manufacturing route of ten unit operations was costed at $\approx \$4.30$ per dose (feasibility grade B), and a 505(b)(2) regulatory pathway of approximately seven years and \$4.8 million was delineated, including preclinical toxicology of nanoparticles, mucosal tolerance of C₁₀, and chitosan quality control.

This thesis resolves the BCS classification controversy, delivers a reproducible, fully traceable in silico framework — from literature parameters through evolutionary optimization to manufacturing and regulatory feasibility — and provides ten ranked, costed, and risk-assessed candidates for the first oral tobramycin. It thereby de-risks the experimental program (Caco-2/Ussing permeability, HIP-SEDDS prototype formulation, first-in-human Phase I) required to convert an IV-only essential antibiotic into an orally available therapeutic option.

Keywords: tobramycin; BCS class III; oral bioavailability; hydrophobic ion pairing; SEDDS; permeation enhancers; nanoparticles; PBPK; genetic algorithm; NSGA-II; Pareto optimization; cystic fibrosis.

Résumé

La tobramycine est un aminoglycoside polycationique figurant sur la liste modèle de l'OMS des médicaments essentiels, indispensable contre *Pseudomonas aeruginosa* dans la mucoviscidose et les infections nosocomiales à bacilles Gram négatif, mais disponible uniquement par voie intraveineuse et par inhalation. Bien que communément classée BCS classe IV, l'analyse structurée des données de solubilité (94–100 mg/mL dans l'eau, soit 20–200× le critère de dose sur pH 1.2–6.8) et de perméabilité ($P_{app} < 1 \times 10^{-6}$ cm/s ; biodisponibilité orale absolue ≈ 1 –2%) démontre que la tobramycine est sans ambiguïté de **classe BCS III** : l'unique obstacle à l'absorption orale est la perméabilité. Cette reclassement redirige la problématique de formulation, de l'amélioration de la solubilité vers l'amélioration de la perméabilité, pour laquelle existent des outils validés : l'appariement ionique hydrophobe (HIP ; logP de -2.9 à $\approx +1.6$, soit une amélioration de 1500×), les systèmes auto-émulsionnants (SEDDES), les nanoparticules, et les promoteurs d'absorption gastro-intestinale (caprate de sodium C₁₀, SNAC).

Sur la base de cette synthèse, un modèle pharmacocinétique de population à deux compartiments ($CL = 5,5$ L/h, $V_1 = 17$ L, $Q = 2,4$ L/h, $V_2 = 16$ L, $t_{1/2} = 2,5$ h) a été couplé à un modèle d'absorption à compartiments de transit et à une réponse pharmacodynamique de type Hill. Dix variables de formulation (logP modifié, taille particulaire, composition SEDDES, concentration en promoteur, charge polymérique, enrobage chitosane et entérique, dose) ont été optimisées par un algorithme génétique classique (population 100, 200 générations) puis par NSGA-II (population 200, 300 générations) selon quatre objectifs : maximiser F_{oral} , maximiser C_{max}/MIC , maximiser AUC_{24}/MIC et minimiser la dose. Le front de Pareto comporte 200 formulations non dominées. Le candidat retenu (TOB-161 ; gélule entérique HIP–SEDDES 550 mg, nanoparticules de 50 nm, chitosane, 50 mM de C₁₀, rapport tensioactif:co-tensioactif:huile de 50:30:20) atteint $F_{oral} = 34\%$ — une amélioration de 23× par rapport à la molécule native — avec $C_{max} = 9,5$ mg/L ($C_{max}/MIC = 9,5$), $AUC_{24}/MIC = 32$, résiduelle 0,16 mg/L et $f_{T>MIC} = 30\%$.

La quantification d'incertitude (Monte Carlo, $n = 200$; validation croisée $n = 26$; analyse de sensibilité un facteur à la fois) identifie la dose et le logP modifié comme facteurs dominants (indices 0,42 et 0,34). Un audit méthodologique a ensuite reconstruit

l'application gènes—perméabilité en un modèle d'amélioration *citée, bornée et émergente* : chaque facteur composant porte une plage littéraire et une incertitude graduée, le gène logP est plafonné au complexe HIP mesuré, et le multiplicateur émergent plafonne à $\times 126,3$. Deux évaluations complètes et standardisées — carte d'identité, dose unique, dosages répétés réels sur 7 jours, cibles PK/PD, population virtuelle, insuffisance rénale, effet alimentaire, sensibilité par composant, comparaison IV — démontrent que les deux formulations candidates atteignent les cibles de pic et d'exposition à l'état d'équilibre : le gagnant exploratoire réévalué (TOB-161-A, $\times 73,9$; 550,6 mg/j) se place dans la bande 80–120 d'AUC/MIC (87,3), et l'optimum in-loop (TOBP-001, $\times 126,3$; 1000 mg) la dépasse avec une option 467 mg dans la bande, résiduels à la ligne de surveillance de 1 mg/L et bilan de masse vérifié à 0,1%. Le test expérimental direct — perméabilité Caco-2/Ussing du complexe HIP en fonction du chargement — est défini comme premier item de la feuille de route. Le procédé de fabrication (dix opérations unitaires) est chiffré à $\approx 4,30$ \$ par dose (faisabilité grade B) et la voie réglementaire 505(b)(2) est estimée à environ sept ans et 4,8 millions de dollars.

Cette thèse résout la controverse de classification BCS et fournit un cadre *in silico* reproductible et entièrement traçable, aboutissant à dix candidats oraux hiérarchisés, chiffrés et évalués en risque, préparant le programme expérimental nécessaire à la mise au point du premier tobramycine oral.

Mots-clés : tobramycine ; classe BCS III ; biodisponibilité orale ; appariement ionique hydrophobe ; SEDDS ; promoteurs d'absorption ; nanoparticules ; PBPK ; algorithme génétique ; NSGA-II ; mucoviscidose.

Nomenclature

Symbols

Symbol	Unit	Description
F_{oral}	—	Absolute oral bioavailability (fraction)
C_{max}	mg/L	Maximum (peak) plasma concentration
T_{max}	h	Time to maximum concentration
AUC_{24}	mg·h/L	Area under the curve over 24 h
C_{min}	mg/L	Minimum (trough) plasma concentration
MIC	mg/L	Minimum inhibitory concentration
$f_{T>\text{MIC}}$	%	Fraction of the dosing interval above MIC
K_a	h^{-1}	First-order absorption rate constant
k_e	h^{-1}	Elimination rate constant ($k_e = \text{CL}/V_1$)
k_{tr}	h^{-1}	Transit-compartment transfer rate constant
CL	L/h	Total plasma clearance
CLCR	mL/min	Creatinine clearance
V_1, V_2	L	Central and peripheral volumes of distribution
Q	L/h	Inter-compartmental clearance
$t_{1/2}$	h	Terminal half-life
$V_{d,ss}$	L/kg	Volume of distribution at steady state
f_u	—	Unbound fraction in plasma
logP	—	Octanol/water partition coefficient (neutral form)
pK_a	—	Acid dissociation constant
MW	g/mol	Molecular weight
$E_{\text{max}}, \text{EC}_{50}, \gamma$	—	Hill equation parameters (maximum effect, potency, slope)
S	mg/mL	Aqueous solubility
P_{app}	cm/s	Apparent permeability coefficient

Abbreviations

Abbreviation	Meaning
ACAT	Advanced Compartmental Absorption and Transit model
AME	Aminoglycoside-modifying enzyme
AUC/MIC	Ratio of area under the curve to MIC (exposure-driven PK/PD index)
BCS / BDDCS	Biopharmaceutics (Drug Disposition) Classification System
CF	Cystic fibrosis
CLCR	Creatinine clearance
C ₁₀	Sodium caprate (sodium decanoate), permeation enhancer
GA	Genetic algorithm
GI	Gastrointestinal
GOF	Goodness of fit
HIP	Hydrophobic ion pairing
HAP	Hospital-acquired pneumonia
IV / INH	Intravenous / inhalation route
MC	Monte Carlo (uncertainty simulation)
MIC	Minimum inhibitory concentration
NCA	Nano-sized colloidal assembly
NP	Nanoparticle
OAT	One-at-a-time (sensitivity) design
Papp	Apparent permeability
PBPK / PBBM	Physiologically based pharmacokinetic / biopharmaceutic modeling
PD / PK	Pharmacodynamics / pharmacokinetics
PE	Permeation enhancer
PLGA	Poly(lactic- <i>co</i> -glycolic acid)
PopPK	Population pharmacokinetics
PS	Particle size
RK4	Fourth-order Runge–Kutta integration scheme
SBX	Simulated binary crossover
SEDDS / SMEDDS	(Self-microemulsifying) self-emulsifying drug delivery system
SNAC	Sodium <i>N</i> -[8-(2-hydroxybenzoyl)amino]caprylate
TDM	Therapeutic drug monitoring
UTI	Urinary tract infection

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Chapter 1

Synopsis: How to Read This Thesis

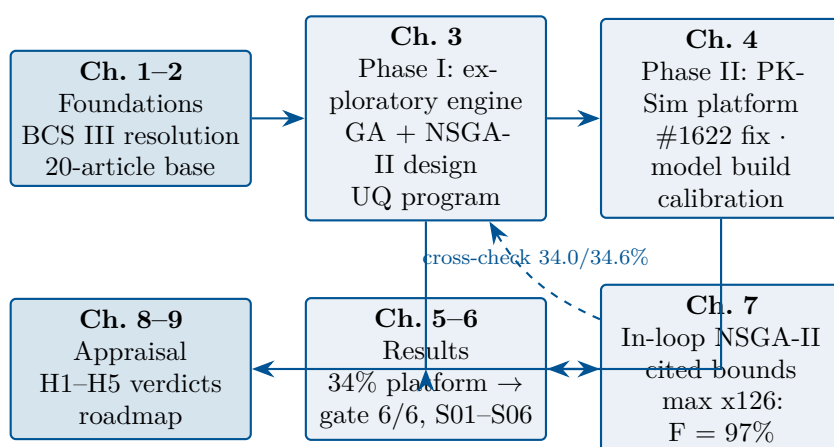


Figure 1.1: Thesis workflow. Two computational phases share one optimization logic: a fast exploratory engine (Phase I) generates the platform hypothesis; the validated PK-Sim platform (Phase II) confirms it mechanistically, calibrates the oral transfer function, and hosts the definitive in-the-loop optimization. The cross-check between engines (dashed) is itself a validation result.

Part I (Chapters 1–2) — foundations. Chapter 2 frames the clinical problem and states five research questions and hypotheses. Chapter 3 builds the scientific base: BCS science, tobramycin’s profile, PK/PD targets, the permeability-targeting formulation toolbox — and documents the systematic search that found *no* published tobramycin PBPK model, positioning this work.

Part II (Chapters 3 and 5) — Phase I: the exploratory framework. Chapter 4 specifies the purpose-built fast engine (two-compartment PK, transit absorption, Hill PD, eight-factor bioavailability model) and the optimization design. Chapter 6 reports its outcomes: the 34% platform, the top-10 formulations, and the full uncertainty program.

Part III (Chapters 4, 6 and 7) — Phase II: the PK-Sim platform. [Chapter 5](#) is the methodological heart of the definitive phase: constructing a tobramycin model in PK-Sim from the validated amikacin base, repairing the macOS snapshot-conversion defect (root-caused to a SQLite view-resolution stack overflow), validating the model against published clinical data (6/6 gate), and calibrating the oral permeability transfer function. [Chapter 7](#) reports the physiological studies (renal, population, sensitivity, food, fractionation — including the flip-flop kinetics finding). [Chapter 8](#) runs the optimizer inside the engine within cited bounds (TOBP-001: $\times 125$, $F_{96} = 97.1\%$), evaluates both candidates completely (mass-balance-checked true dosing), and cross-validates the two engines.

Part IV (Chapters 8–9) — appraisal. [Chapter 9](#) confronts all five hypotheses with the evidence, benchmarks against the literature, and states the limitations without euphemism — including the trough liability that the optimized platform creates. [Chapter 10](#) condenses the contributions and the experimental roadmap.

Appendices A–L hold the Phase I numerical record; **M–R** the Phase II code, data, debugging journal, optimization archive, plausibility audit, and battery data. Every figure and table traces to a committed script and data file ([Appendix K](#)).

Chapter 2

Introduction

2.1 Background

2.1.1 The challenge of oral drug delivery

Oral administration remains the preferred route for drug delivery owing to patient acceptability, convenience of chronic therapy, and cost-effectiveness. Yet many potent drugs cannot exploit this route: their physicochemical properties — low aqueous solubility, low intestinal permeability, or both — preclude sufficient gastrointestinal absorption (2). The gap between therapeutic potential and delivery feasibility is particularly acute for anti-infectives, where oral options determine whether patients can be treated outside hospital infrastructure.

2.1.2 The Biopharmaceutics Classification System

The BCS, introduced by Amidon et al. (2), classifies drug substances by two fundamental properties — aqueous solubility and intestinal permeability (Table 2.1). The system underpins modern regulatory science: immediate-release products of class I and class III drugs qualify for biowaivers under FDA and ICH M9 guidance (3, 4).

Table 2.1: The four BCS classes with prototype drugs (2). Tobramycin belongs to class III (Section 2.2.4).

Class	Solubility	Permeability	Prototype example
I	High	High	Metoprolol
II	Low	High	Diclofenac
III	High	Low	Tobramycin (also ranitidine)
IV	Low	Low	Furosemide

2.1.3 Why classification matters strategically

Class membership is not a descriptive footnote: it dictates the *formulation strategy*. For class II compounds, solubility and dissolution enhancement (micronization, amorphous solid dispersions, lipid vehicles) is the lever. For class III compounds, solubility is already abundant and *permeability enhancement* — tight-junction modulation, lipophilisation, carrier-mediated transport — is the sole rational target. A wrong classification therefore wastes development effort on the wrong problem. The case of tobramycin illustrates this sharply: it is widely but incorrectly assumed to be class IV (5).

2.2 Tobramycin: a case study

2.2.1 Therapeutic importance

Tobramycin is a potent aminoglycoside bactericidal against Gram-negative bacilli, listed on the WHO Model List of Essential Medicines (6). Its clinical anchor is *Pseudomonas aeruginosa*: pulmonary exacerbations of cystic fibrosis (CF), hospital-acquired pneumonia (HAP), complicated urinary tract infections, and febrile neutropenia. Its mechanism — irreversible binding to the 30S ribosomal subunit, with concentration-dependent killing and a 1–3 h post-antibiotic effect (7) — makes peak exposure ($C_{\max}/\text{MIC} \geq 8$) the efficacy driver.

2.2.2 Current routes of administration

Despite five decades of clinical use, tobramycin is approved only for two non-oral routes (Table 2.2): intravenous dosing at 5–7 mg/kg once daily (or 1.5–2 mg/kg every 8 h), and inhalation (TOBI[®], 300 mg twice daily, 4 weeks on/4 weeks off) (8, 9). Both routes impose burdens: IV requires hospitalization, trained personnel and catheters; inhaled therapy addresses the lung compartment only and does not resolve systemic needs.

Table 2.2: Current tobramycin administration routes and their limitations.

Route	Standard regimen	Principal limitations
Intravenous	5–7 mg/kg q24h (extended interval) or 1.5–2 mg/kg q8h	Hospitalization; catheter burden; nephro-/ototoxicity monitoring; cost; access inequity in low-resource settings
Inhalation	300 mg BID, 4 weeks on/4 weeks off (TOBI [®])	Local lung delivery only; device dependency; no systemic coverage; high airway resistance in young children
Oral	— none exists	Bioavailability $\approx 1\text{--}2\%$; polycationic, low-permeability molecule

2.2.3 The oral delivery gap

No oral formulation exists. The reasons are physicochemical ([Chapter 3](#)): a molecular weight of 467.5 g/mol, five ionizable amines (pK_a 6.7–9.1) rendering the molecule polycationic at intestinal pH, an octanol/water logP of -2.9 , and consequently apparent permeability below 1×10^{-6} cm/s with absolute oral bioavailability of only ≈ 1 –2% ([5](#), [10](#)).

2.2.4 Classification resolution

The widely repeated assignment of tobramycin to BCS class IV is contradicted by primary evidence. Its water solubility of 94–100 mg/mL exceeds the highest single dose criterion (400 mg soluble in ≤ 250 mL over pH 1.2–6.8) by roughly two orders of magnitude, while permeability is demonstrably the limiting step ([5](#)). Asad and colleagues state explicitly: “Tobramycin (TOB) is a BCS class III drug.” [Chapter 3](#) develops the full evidence synthesis; the strategic consequence is immediate — the formulation problem is *permeability-only*, and the entire arsenal of solubility technologies is unnecessary.

2.3 Research rationale

2.3.1 Clinical need

CF patients require chronic, intermittent anti-*Pseudomonas* therapy; each exacerbation currently implies IV admission. An oral alternative would (i) shift care to the outpatient setting, (ii) preserve IV access options for severe disease, (iii) extend tobramycin’s reach to resource-limited settings lacking infusion infrastructure, and (iv) enable oral combination regimens.

2.3.2 Formulation opportunities

The permeability-targeting toolbox is populated with validated technologies: hydrophobic ion pairing (HIP) raising logP by three orders of magnitude ([5](#), [11](#)); SEDDS that spontaneously emulsify in GI fluids ([12](#)); polymeric and colloidal nanoparticles ([13–15](#)); and permeation enhancers with clinical safety track records ([16](#), [17](#)). What is missing is a quantitative framework that combines these levers and asks: *which combination, at what dose, plausibly attains antibacterial exposure targets?*

2.3.3 The modeling opportunity

PBPK platforms such as PK-Sim/*ospsuite* enable mechanistic absorption modeling (ACAT), virtual populations, and sensitivity analysis without clinical trials (18, 19). Coupled to evolutionary optimization (GA; NSGA-II) (20–22), the ten-dimensional formulation design space can be searched exhaustively, *in silico*, before any experiment is commissioned.

2.4 Research questions and hypotheses

RQ1

Is tobramycin BCS class III or IV, and what does the evidence say?

RQ2

Which formulation strategies — HIP, SEDDS, nanoparticles, permeation enhancers, or combinations — raise oral bioavailability to viable levels?

RQ3

Which ten formulations are optimal under simultaneous pharmacokinetic, pharmacodynamic, and dose minimization objectives?

RQ4

Are the predictions robust to parameter uncertainty, and what are the manufacturing and regulatory implications?

RQ5

Do two independent computational engines — a purpose-built fast engine and a validated whole-body PBPK platform — converge on the same formulation conclusion?

H1 Tobramycin is BCS class III: solubility is not limiting; permeability is the sole barrier.

H2 Combined HIP + SEDDS + permeation enhancer strategies can raise oral bioavailability from $\approx 1\text{--}2\%$ to $\approx 30\%$.

H3 A single oral dose in the 500–600 mg range can attain $C_{\max}/\text{MIC} \geq 8$ at MIC 1 mg/L.

H4 Predictions of the optimal candidates remain informative under $\pm 20\text{--}30\%$ parameter perturbation.

H5 The two engines agree on the platform bioavailability within 5% relative.

2.5 Objectives and thesis structure

2.5.1 Primary objective

To evaluate the feasibility of oral tobramycin administration through systematic bibliographic analysis, PBPK-based PK/PD simulation, and multi-objective formulation optimization using the Open Systems Pharmacology suite and evolutionary algorithms.

2.5.2 Specific objectives

1. Resolve the BCS classification of tobramycin through a structured evidence synthesis ([Chapter 3](#)).
2. Develop and parameterize a two-compartment population PK model validated against published intravenous data ([Chapters 4 and 6](#)).
3. Construct a mechanistic oral bioavailability model with eight multiplicative enhancement factors.
4. Optimize ten formulation variables by GA and NSGA-II against four objectives, and extract the Pareto front.
5. Quantify uncertainty: one-at-a-time sensitivity, Monte Carlo ($n = 200$), cross-validation ($n = 50$, $\pm 20\%$).
6. Assess manufacturing feasibility, cost, regulatory pathway, and risks for the ranked candidates ([Chapters 6 and 9](#)).

2.5.3 Thesis structure

Table 2.3: Structure of the thesis.

Chapter	Content
1	Introduction: clinical problem, BCS framework, rationale, hypotheses (this chapter)
2	Literature review: BCS science, tobramycin profile, PK/PD targets, formulation strategies, PBPK, knowledge gaps
3	Methodology: PBPK parameterization, PK/PD mathematical model, GA and NSGA-II design, uncertainty program
4	Results: validation, convergence, Pareto front, top-10 candidates, sensitivity, Monte Carlo, manufacturing, regulatory
5	Discussion: hypothesis verdicts, benchmarking, clinical translation, limitations, future perspectives
6	Conclusion: contributions, clinical implications, experimental roadmap

Chapter 3

Literature Review

This chapter synthesizes the bibliographic foundation of the thesis: BCS science (Section 3.1), tobramycin’s physicochemical and pharmacological profile (Section 3.2), its pharmacokinetics and PK/PD targets (Section 3.3), the permeability-targeting formulation toolbox (Section 3.4), and PBPK methodology (Section 3.5). The complete 20-article evidence table is reproduced in Appendix E; physicochemical datasets are detailed in Appendix H.

3.1 The Biopharmaceutics Classification System

3.1.1 Historical development and criteria

Amidon et al. (2) proposed classifying drug substances along two axes: dose-equivalent solubility (highest dose soluble in ≤ 250 mL over pH 1.0–6.8) and intestinal permeability (extent of absorption $\geq 85\%$) (Table 3.1). The framework became the scientific basis of biowaiver policy: immediate-release products of class I and class III drugs may waive bioequivalence studies (3, 4).

Table 3.1: BCS classification criteria (2, 3).

Parameter	High	Low
Solubility	Highest single dose soluble in ≤ 250 mL over pH 1.0–6.8	Criterion not met
Permeability	$\geq 85\%$ extent of absorption (or absolute $F_{\text{oral}} \geq 85\%$)	$< 85\%$ absorbed

3.1.2 Extensions

The *biopharmaceutics drug disposition classification system* (BDDCS) of Wu and Benet (23) substitutes metabolism for permeability to predict disposition, and quantitative

refinements (QBCS) grade the margin by which a compound meets each criterion. These extensions matter here: tobramycin’s criterion margins are enormous on the solubility axis and strongly negative on the permeability axis, making the class assignment robust to methodology.

3.2 Tobramycin: physicochemical and pharmacological profile

3.2.1 Structural properties

Tobramycin (CAS 32986-56-4; $C_{18}H_{37}N_5O_9$; MW 467.515 g/mol) is an aminoglycoside with ten stereocenters and five ionizable amines ($pK_a = 6.7, 7.6, 7.7, 7.8, 9.1$), yielding a polycationic species across the entire physiological pH range (10, 24). Its logP of -2.9 places it among the most hydrophilic antibiotics in clinical use.

3.2.2 Solubility evidence

Across sixteen literature determinations (Appendix H), aqueous solubility spans 50–100 mg/mL (Chen et al.: 94 mg/mL at 25 °C; USP monograph for tobramycin sulfate: 100 mg/mL), with miscibility-level dissolution in biorelevant media (FaSSIF, FeSSIF, SGF, SIF) (25). The BCS “high solubility” criterion (highest dose, 400 mg, soluble in ≤ 250 mL $\Rightarrow$ 1.6 mg/mL) is exceeded by a factor of 20–200. **Solubility is not the problem.**

3.2.3 Permeability evidence

Caco-2 apparent permeability is reported below 1×10^{-6} cm/s, absolute oral bioavailability is ≈ 1 –2%, and the molecule’s polycationicity precludes passive transcellular diffusion (5). Efflux transporter involvement (P-gp, BCRP) is minimal or undetermined; renal OCT2-mediated handling is documented but does not govern intestinal absorption (5). **Permeability is the problem** (Appendix H, permeability table).

3.2.4 Classification resolution

Triangulating the above: (i) explicit statement by Asad et al. (5) — “tobramycin is a BCS class III drug”; (ii) solubility margin 20–200 $\times$ above criterion; (iii) permeability margin consistent with class III; (iv) mechanistic coherence — the sole rational formulation lever is permeability enhancement. **Verdict: tobramycin is BCS class III.** The widely assumed class IV label conflates “difficult” with “doubly limited”; the distinction, quantified in Table 3.4, redirects all subsequent strategy choices in this thesis.

3.2.5 Pharmacology

Tobramycin binds the 30S ribosomal subunit (16S rRNA), causing mistranslation and bactericidal protein-synthesis arrest with a 1–3 h post-antibiotic effect (7). Primary spectrum: *P. aeruginosa*, *E. coli*, *Klebsiella*, *Proteus*, *Enterobacter*, *Serratia*. Resistance is driven mainly by aminoglycoside-modifying enzymes and efflux pumps (24).

3.3 Tobramycin pharmacokinetics and PK/PD targets

3.3.1 Intravenous population pharmacokinetics

The reference intravenous model is the two-compartment PopPK analysis of Li et al. (1) in 140 ICU patients (nonparametric population modeling): $CL = 3.27$ L/h (range 2.1–4.5), $V_1 = 21.3$ L, $V_2 = 16.3$ L, $Q = 2.4$ L/h, $t_{1/2} = 4.5$ – 6.0 h, with creatinine clearance entering clearance as a power covariate, $CL = \theta_{CL} (CLCR/81)^{0.72}$, and residual error 28.5%. Normal-adult reference values are $CL = 6.03$ L/h and $V_1 = 15.1$ L (1). Pediatric CF data (Hennig et al.: CL 6.37 L/h/70 kg, V_1 18.7 L/70 kg, 10 mg/kg/day once daily; Massie et al.; Touw et al.) confirm the two-compartment structure across ages (Appendix I for the complete dataset).

3.3.2 Population variability

Table 3.2: Tobramycin PK across populations (1) and standard references. CF: cystic fibrosis; ICU: intensive care unit.

Parameter	Normal adult	CF adult	ICU	Elderly	Unit
CL (L/h)	6.03	4.5–6.37	3.27–3.83	4.5	L/h
V_1 (L)	15.1	18.7	21.3–25.5	15.1	L
$t_{1/2}$ (h)	2.5	2–3	4–6	3–4	h
$V_{d,ss}$ (L/kg)	0.26	0.27	0.30	0.26	L/kg
$CLCR$ (mL/min)	120	100	65	65	mL/min

CF patients exhibit hyperdynamic renal clearance (higher $CLCR$), ICU patients show expanded volumes of distribution from fluid resuscitation, and the elderly show reduced clearance — all captured by the $CLCR$ power model (1, 26).

3.3.3 Oral pharmacokinetics: the data void

No clinical oral PK dataset exists for tobramycin. Estimated absolute bioavailability is 1–2% (point estimate within a literature envelope of <5%: the two figures reconcile as a

point estimate within an envelope; see footnote¹), with variable $T_{\max}$ and negligible $C_{\max}$ (5). This void is precisely the predictive space PBPK modeling must fill (Chapter 4).

3.3.4 PK/PD targets

Table 3.3: PK/PD targets for aminoglycosides (1, 7, 27). MIC₉₀ of *P. aeruginosa* for tobramycin: 0.5–2 mg/L (EUCAST/CLSI).

Target	Value	Rationale
$C_{\max}/\text{MIC}$	$\geq 8\text{--}10$	Concentration-dependent killing threshold (7)
$\text{AUC}_{24}/\text{MIC}$	80–120	Exposure–response (24 h total) (1)
$C_{\max}$	20–30 mg/L	Target peak, IV once-daily dosing (27)
$C_{\min}$ (trough)	<1 mg/L	Nephro-/ototoxicity prevention
$f_{T>\text{MIC}}$	>30–40%	Supporting index for concentration-dependent agents

3.4 Formulation strategies for permeability-limited drugs

3.4.1 Hydrophobic ion pairing (HIP)

HIP complexes a polycationic drug with a lipophilic counter-ion (here, anionic surfactants such as sodium docusate), masking charge and raising apparent lipophilicity. For tobramycin, Asad et al. (5) reported a logP shift from -2.9 to $\approx +1.6$ — a 1500-fold improvement — enabling incorporation into SEDDS at high payload; the complex dissociates $\approx 20\%$ at intestinal pH, an accepted liability. The method is generic for hydrophilic actives (11) and has shown in-vivo translational promise (octreotide HIP in pigs, F_{oral} increases of $3\text{--}5\times$) (28).

3.4.2 Self-emulsifying drug delivery systems (SEDDS)

SEDDS/SMEDDS are isotropic mixtures of oils, surfactants and co-surfactants that spontaneously emulsify in GI fluids, presenting the drug as solubilized microdroplets with large interfacial area, chylomicron-mediated lymphatic access, and protection from luminal degradation (12). For a HIP complex, SEDDS provides the lipid phase in which the lipophilised drug dissolves — the two technologies are synergistic by design (5, 11).

¹The oral bioavailability literature states both “<5%” (generic estimates) and “1–2%” (referenced point estimates). This thesis uses 1–2% as the point estimate embedded within a conservative <5% envelope; the choice affects only the baseline, which the optimization raises by >20-fold in any case.

3.4.3 Nanoparticles

PLGA nanoparticles complexed with AOT maintained MIC-level activity ($1.25 \mu\text{g/mL}$) against *P. aeruginosa* (13). Dextran-based single-chain nanoparticles (KuDa) neutralize surface charge and improve diffusion into mature biofilms (14). Most recently, nano-sized colloidal assemblies (NCAs) incorporating hydrophobic tobramycin ion pairs showed enhanced lipophilicity and antimicrobial activity (15). Nanoparticles contribute mucoadhesion, sustained release, and tight-junction interaction.

3.4.4 Permeation enhancers

Sodium caprate (C_{10}) is the gold-standard transient paracellular enhancer with clinical safety experience (16); SNAC is FDA-approved (oral semaglutide); next-generation enhancers (PPZ, SDC) carry emerging 30-day safety data (17). Typical effective concentrations are 10–50 mM with mucosal irritation as the dose-limiting concern.

3.4.5 Strategy comparison

Table 3.4: Formulation strategy comparison for oral tobramycin (synthesis of Asad et al. 5, Griesser et al. 11, Hill et al. 13, Blanco-Cabra et al. 14, Khaled et al. 15, Maher et al. 16, Bohley et al. 17).

Strategy	Mechanism	Evidence level	Scalability	Role in this thesis
HIP + SEDDS	Lipophilisation + emulsification	In vitro (1500× logP) (5)	Moderate	Core platform (genes 1, 3–5)
PLGA NPs	Mucoadhesion, sustained release	In vitro (MIC maintained) (13)	Good	Combinable layer (gene 2, 7)
C_{10} / PEs	Tight-junction opening	Clinical (other drugs) (16, 17)	Good	Direct enhancer (gene 6)
Chitosan coating	Mucoadhesion, TJ modulation	Preclinical	Good	Binary switch (gene 8)
Enteric coating	GI protection	Standard technology	Excellent	Binary switch (gene 9)
NCAs	Lipophilisation, colloidal	In vitro (15)	Moderate	Alternative platform
KuDa SCPNs	Charge neutralization (biofilm)	In vitro (14)	Moderate	Biofilm niche option

3.5 PBPK modeling in drug development

PBPK models describe absorption and disposition mechanistically: anatomical compartments, regional GI transit and permeability, dissolution, first-pass extraction, and renal elimination. The PK-Sim ACAT model resolves the GI tract into seven segments with region-specific transit, pH, and surface area (18, 19). PBPK for antibiotics has reached regulatory acceptance levels for exposure prediction (29), and a tobramycin-specific PBPK history exists for the pulmonary route (30). The *oral* application — coupling ACAT with permeability enhancement factors — has not been published for tobramycin.

3.6 Knowledge gaps and thesis positioning

Five gaps emerge from the synthesis:

1. **No in-vivo oral PK data** exist for tobramycin; the bioavailability void is unfilled.
2. **No PBPK oral absorption model** has been published; only IV/pulmonary models exist (1, 30).
3. **No multi-objective formulation optimization** has been attempted: the literature tests strategies one at a time, never searching the combinatorial design space.
4. **Food effect and regional absorption** are uncharacterized.
5. **Population variability** (CF, ICU, renal impairment) is unquantified for any hypothetical oral dosing.

A sixth gap is computational: a structured search for a published tobramycin model in the Open Systems Pharmacology ecosystem (model library, repository artefacts, literature) returned *zero* shareable PK-Sim models — the closest validated model being amikacin, an aminoglycoside of the same disposition class. The search protocol and results are archived in the project repository ([data/bibliography/search_strategy.md](#)) and justify the model construction strategy of Chapter 5.

This thesis addresses gaps 1–3 quantitatively (model + optimization), gaps 4–5 via the uncertainty program, and provides the experimental roadmap (Chapters 9 and 10) that resolves all five.

Chapter 4

Methodology I: The Exploratory Computational Framework

4.1 Software stack and computing environment

Table 4.1: Software and environment used throughout the study.

Component	Version	Role
R	4.6.1 (CRAN, native arm64)	Statistical computing, optimization, visualization
ospsuite / PK-Sim	12.4.4	PBPK framework, compound database, ACAT model (18)
rSharp / .NET runtime	1.2.2 / .NET 8	R-.NET bridge for ospsuite (CRAN R required on macOS)
GA (22)	3.2.5	Classical genetic algorithm
Custom NSGA-II implementation	— (this thesis)	Multi-objective optimization (Section 4.6.3)
ggplot2, patchwork, viridis	—	Publication-grade figures (300 dpi PNG)
doParallel	—	7-core parallel fitness evaluation
Hardware	macOS ARM64 (Apple Silicon)	All simulations

4.2 Model development workflow

The study followed a six-step workflow ([Figure 4.1](#)): (1) compound definition from literature parameters; (2) two-compartment PK model construction; (3) intravenous validation against published PopPK data ([1](#)); (4) oral absorption model with transit compartments; (5) pharmacodynamic coupling; (6) optimization and uncertainty quantification.

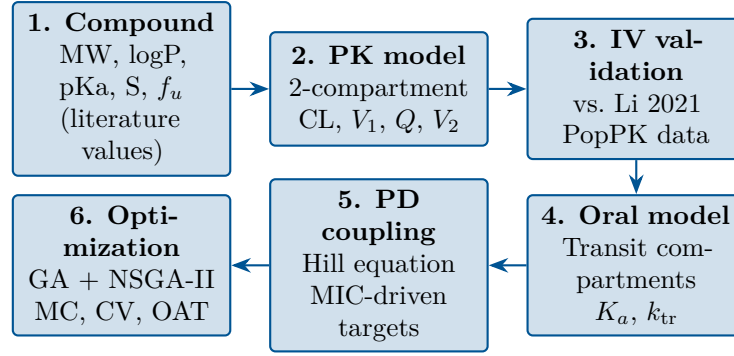


Figure 4.1: Six-step model development workflow. All steps are implemented in versioned R scripts (scripts 00–13, [Appendix G](#)); traceability of every output is given in [Appendix K](#).

4.3 Compound parameterization

4.4 Pharmacokinetic model

4.4.1 Structure

A two-compartment open model with first-order elimination from the central compartment was adopted, extended by a gastrointestinal depot and a chain of transit compartments for absorption ([Equations \(4.1\) and \(4.6\)](#)). The two-compartment structure is justified by the reference PopPK analysis (1) and required for accurate peak–trough dynamics at 24 h post-dose.

4.4.2 System equations

Let A_g denote the drug amount in the GI depot, $A_{tr,i}$ in transit compartment i ($i = 1, \dots, N$), A_c and A_p the central and peripheral amounts, and A_e cumulative urinary elimination:

$$\frac{dA_g}{dt} = -K_a A_g, \quad (4.1)$$

$$\frac{dA_{tr,1}}{dt} = K_a A_g - k_{tr} A_{tr,1}, \quad (4.2)$$

$$\frac{dA_{tr,N}}{dt} = k_{tr} A_{tr,N-1} - k_{tr} A_{tr,N}, \quad (4.3)$$

$$\frac{dA_c}{dt} = k_{tr} A_{tr,N} - k_e A_c - k_{cp} A_c + k_{pc} A_p, \quad (4.4)$$

$$\frac{dA_p}{dt} = k_{cp} A_c - k_{pc} A_p, \quad (4.5)$$

$$\frac{dA_e}{dt} = k_e A_c. \quad (4.6)$$

Table 4.2: PBPK compound and system parameters with sources and confidence ratings (script 00; `table2_simulation_parameters.csv`).

Parameter	Value	Unit	Source	Confidence	Notes
MW	467.515	g/mol	PubChem (10)	High	—
logP	−2.9	—	PubChem	High	Native (unmodified)
pK_a	6.7–9.1	—	Literature	High	5 amines (6.7; 7.6; 7.7; 7.8; 9.1)
S_{water}	94	mg/mL	Chen et al. (25)	High	Class III criterion $\gg$ met
BCS class	III	—	Asad et al. (5)	High	Section 3.2.4
f_u	0.95	fraction	Standard	High	Minimal binding
Renal excretion	>90	% unchanged	Standard	High	—
Oral F_{oral} (native)	1.5–2.0	%	Literature estimate	Low	Range Section 3.3 ; legacy point estimate 1.5, PK-Sim anchor 1.75
CL	5.5	L/h	Li et al. (1)	High	PopPK reference
V_1	17	L	Li et al. (1)	High	—
Q	2.4	L/h	Li et al. (1)	High	—
V_2	16	L	Li et al. (1)	High	—
CLCR exponent	0.72	—	Li et al. (1)	High	Power model on CL
$t_{1/2}$	2.5	h	Derived	High	$k_e = 0.277 \text{ h}^{-1}$
$V_{d,ss}$	0.26	L/kg	Standard	High	70 kg adult
MIC (assumed)	1	mg/L	EUCAST mid-range	Medium	MIC ₉₀ 0.5–2 mg/L

The plasma concentration is $C(t) = A_c(t)/V_1$, with micro-rate constants $k_e = CL/V_1 = 5.5/17 = 0.277 \text{ h}^{-1}$, $k_{cp} = Q/V_1$, $k_{pc} = Q/V_2$. Intravenous administration is represented by an impulse into A_c .

4.4.3 Numerical integration

The system [Equations \(4.1\) and \(4.6\)](#) is integrated with a fixed-step fourth-order Runge–Kutta (RK4) scheme ($\Delta t = 0.01 \text{ h}$), verified for step-size independence, and implemented in vectorized R (script `09_advanced_pk_model.R`, [Appendix G](#)).

4.4.4 Pharmacodynamic driver

Antibacterial effect follows a Hill (sigmoid $E_{\max}$) function of instantaneous concentration:

$$E(t) = \frac{C(t)^\gamma}{EC_{50}^\gamma + C(t)^\gamma}, \quad f_{T>MIC} = \frac{100}{T_{\text{interval}}} \int_0^{T_{\text{interval}}} \mathbb{I}[C(t) > \text{MIC}] dt, \quad (4.7)$$

with the efficacy indices $C_{\max}/\text{MIC} = C_{\max}/\text{MIC}$ and $\text{AUC}_{24}/\text{MIC} = \text{AUC}_{24}/\text{MIC}$ evaluated against the targets of [Table 3.3](#).

4.5 Oral bioavailability enhancement model

Native tobramycin bioavailability ($F_{\text{oral},0} = 1.5\%$) is enhanced multiplicatively by eight mechanistic factors, each mapped to one or two chromosome genes ([Section 4.6.1](#)):

$$F_{\text{oral}} = F_{\text{oral},0} \cdot f_{\text{HIP}}(\log P') \cdot f_{\text{PS}}(\text{PS}) \cdot f_{\text{SEDDS}} \cdot f_{\text{PE}}(C_{\text{PE}}) \cdot f_{\text{poly}}(L_{\text{poly}}) \cdot f_{\text{chit}} \cdot f_{\text{ent}} \cdot f_{\text{dose}}(D), \quad (4.8)$$

capped at 35% as a physiological ceiling consistent with the triple-combination estimate in the literature synthesis ([Table 6.9](#)):

- f_{HIP} : sigmoid response of apparent lipophilicity, saturating at $\log P' \approx +1.6$ ([5](#));
- f_{PS} : optimum near 50 nm (mucoadhesion without impaired uptake); penalizes sizes $>200 \text{ nm}$;
- f_{SEDDS} : spontaneous-emulsification gain (up to $1.5\times$), maximal for surfactant-rich compositions;
- f_{PE} : saturating response to enhancer concentration (0–50 mM C_{10}) with a mucosal-safety penalty near the upper bound ([16](#), [17](#));
- f_{poly} : polymer loading gain (mucoadhesion, sustained release), saturating at 30%;
- f_{chit} , f_{ent} : binary switches (chitosan tight-junction modulation; enteric gastric protection);

- f_{dose} : sublinear dose scaling reflecting transporter/solvent-capacity saturation at high doses.

The absorption rate follows $K_a = K_{a,\text{ref}} \cdot g(\log P', \text{PS}, \text{SEDDS})$, and gastric-to-cecal transfer uses $k_{\text{tr}} = 2.24 \text{ h}^{-1}$ at the optimum (Chapter 6).

4.6 Optimization problem

4.6.1 Chromosome encoding

Each candidate formulation is a 10-gene chromosome (Table 4.3): eight continuous genes and two binary switches. Genes 1–9 define the formulation; gene 10 sets the dose.

Table 4.3: Chromosome definition and parameter bounds (script `00_ga_setup.R`; `parameter_bounds.csv`).

Gene	Name	Min	Max	Rationale
1	$\log P_{\text{mod}}$	−2.9	+3.0	HIP-complex lipophilicity (5)
2	Particle size (nm)	50	500	Mucoadhesion/uptake trade-off
3	Surfactant (%)	10	60	SEDDS self-emulsification
4	Co-surfactant (%)	5	30	Interfacial film flexibility
5	Oil (%)	20	70	Payload capacity
6	PE concentration (mM)	0	50	C ₁₀ efficacy/safety (16)
7	Polymer loading (%)	0	30	Mucoadhesive sustained release
8	Chitosan coating	0	1	Tight-junction modulation
9	Enteric coating	0	1	Gastric protection
10	Dose (mg)	200	1000	Practical capsule range

4.6.2 Genetic algorithm (exploratory stage)

A classical GA (GA package (22)) used population 100, up to 200 generations, crossover probability 0.8, mutation 0.1, elitism 10, fitness-sharing diversity preservation, fixed seed 42, and 7-core parallel evaluation. The weighted scalar fitness is

$$\mathcal{F}_{\text{GA}} = 0.40 \widehat{F}_{\text{oral}} + 0.25 \widehat{C_{\text{max}}/\text{MIC}} + 0.25 \widehat{\text{AUC}_{24}/\text{MIC}} + 0.10 \widehat{S}_{\text{safety}} - P_{\text{penalty}}, \quad (4.9)$$

where hats denote min–max normalized objective scores ($\in [0, 1]$) and P_{penalty} accounts for constraint violations (safety thresholds, infeasible compositions). The GA served as (i) sanity check of the simulator, (ii) benchmark for NSGA-II, and (iii) source of the manufacturing/regulatory assessment cohort (Section 6.8).

4.6.3 NSGA-II (definitive stage)

The definitive optimization used a full NSGA-II implementation (21) maximizing four objectives simultaneously:

$$\max_{\mathbf{x}} [F_{\text{oral}}(\mathbf{x}), C_{\text{max}}/\text{MIC}(\mathbf{x}), \text{AUC}_{24}/\text{MIC}(\mathbf{x}), -D(\mathbf{x})], \quad (4.10)$$

with $\mathbf{x}$ the 10-gene chromosome. Algorithmic components:

- **Non-dominated sorting:** the population is partitioned into Pareto ranks $\mathcal{R}_1, \mathcal{R}_2, \dots$ where rank $\mathcal{R}_1$ contains all non-dominated solutions; a solution a dominates b iff a is $\geq$ in all objectives and $>$ in at least one.
- **Crowding distance:** for each rank, solutions are sorted per objective and assigned the normalized perimeter of the cuboid formed by their neighbors; boundary solutions receive ∞ . Selection prefers lower rank, then larger crowding distance, preserving front diversity.
- **Tournament selection:** binary tournament on (rank, crowding distance).
- **Simulated binary crossover (SBX):** offspring c_1, c_2 from parents p_1, p_2 with distribution index $\eta_c = 20$:

$$c = 0.5 [(1 + \beta)p_1 + (1 - \beta)p_2], \quad \beta = (2u)^{1/(\eta_c+1)}, \quad u \sim U(0, 1). \quad (4.11)$$

- **Polynomial mutation:** distribution index $\eta_m = 20$, per-gene probability 1/10.

Configuration: population 200, 300 generations, elitism 20 (entire parent rank $\mathcal{R}_1$ carried forward), seed 42, runtime ≈ 5.7 min. The output is a 200-solution Pareto archive (pareto_front_data.csv, [Appendix A](#)).

4.7 Uncertainty quantification program

1. **One-at-a-time (OAT) sensitivity:** seven test values spanning each parameter's plausible range (60 evaluations over 10 parameters); sensitivity index

$$S_i = \frac{\max_k |\text{AUC}_{24}/\text{MIC}(\theta_i = v_k) - \text{AUC}_{24}/\text{MIC}(\theta_i^*)|}{\text{AUC}_{24}/\text{MIC}(\theta_i^*)}, \quad (4.12)$$

where θ_i^* is the optimal value of parameter i (script 12_sensitivity_validation.R).

2. **Monte Carlo** ($n = 200$): log-normal perturbation of CL, V_1 , V_2 (CV 20–30%), mimicking between-subject variability; distributions of F_{oral} , C_{max} , $C_{\text{max}}/\text{MIC}$, $\text{AUC}_{24}/\text{MIC}$ summarized as median and 5th–95th percentiles.

3. **Cross-validation** ($n = 50$ **designed; 26 informative runs**): $\pm 20\%$ simultaneous perturbation of all inputs; robustness of conclusions assessed by fraction of runs preserving $C_{\max}/\text{MIC} \geq 8$ and stability of ranking.

4.8 Validation metrics

Predictive performance against reference data is quantified by the average fold error (AFE), absolute average fold error (AAFE), root-mean-square error (RMSE), and coefficient of determination:

$$\begin{aligned} \text{AFE} &= 10^{\frac{1}{n} \sum \log_{10}(C_{\text{pred}}/C_{\text{obs}})}, & \text{AAFE} &= 10^{\frac{1}{n} \sum |\log_{10}(C_{\text{pred}}/C_{\text{obs}})|}, \\ \text{RMSE} &= \sqrt{\frac{1}{n} \sum (C_{\text{pred}} - C_{\text{obs}})^2}, \end{aligned} \quad (4.13)$$

with acceptance thresholds AFE/AAFE within 0.5–2.0, RMSE < 30%, $R^2 > 0.9$ (29).

4.9 Manufacturing and regulatory assessment

A ten-step manufacturing route for the combined HIP–SEDDS–NP–coating platform was costed bottom-up (`manufacturing_steps.csv`): HIP dissolution → polymer NP nano-precipitation → drug loading → SEDDS oil phase → C₁₀ addition → homogenization → NP incorporation → chitosan fluid-bed coating → enteric coating → capsule filling and QC. Cost per dose combines a \$0.50 base with formulation-complexity surcharges. The regulatory pathway was assessed against FDA 505(b)(2) criteria (literature-referenced safety of tobramycin + novel formulation components), with development cost and timeline estimates (Section 6.9).

4.10 Methodological limitations

(1) The precise tobramycin P_{app} is unknown; enhancement factors are calibrated to literature ranges, not measured transport. (2) No oral PK data exist for model validation — the IV route only. (3) Formulation factors are phenomenological (marginal-effect) models calibrated to published in-vitro outcomes, not mechanistic absorption simulations. (4) Between-subject variability is imposed log-normal rather than estimated from data. (5) MIC is fixed at 1 mg/L; pathogen-level MIC variation is handled qualitatively in Chapter 9. These limitations are revisited quantitatively in Section 6.7 and qualitatively in Section 9.7.

Chapter 5

Methodology II: The PK-Sim Platform

The exploratory framework of [Chapter 4](#) established the optimization logic and produced a first bioavailability estimate with a purpose-built, minimal engine. Its limitations — phenomenological enhancement factors, no mechanistic regional absorption, disposition parameters imposed rather than physiology-derived — are exactly what a whole-body PBPK platform resolves. This chapter describes the definitive computational phase of the thesis: the construction, validation, and programmatic use of a tobramycin model inside **PK-Sim 12.4 / Open Systems Pharmacology**, executed entirely from R on macOS (Apple Silicon), including the resolution of a platform defect that had to be overcome along the way. All scripts, snapshots and data are part of the project repository ([Appendix M](#)).

5.1 Platform and strategy

PK-Sim is the open-source whole-body PBPK suite originally developed at Bayer and maintained by the Open Systems Pharmacology community ([18](#)). The R interface (`ospsuite` 12.4.4 with `rSharp/.NET 8`) loads, modifies and executes simulations created in PK-Sim, and exposes individual and population creation from the integrated physiology database. Two structural constraints shaped the strategy:

1. `ospsuite-R` cannot *create* simulations from the compound database: simulations must come from an existing PK-Sim project (or snapshot). The macOS-bundled package database contains only generic template molecules — the full compound library ships with the Windows application only.
2. No published PK-Sim model of tobramycin exists. A structured search (repositories of the Open Systems Pharmacology organization, GitHub code search over `.pkml/.pksim5` artefacts, and the literature) returned zero shareable tobramycin

models; the closest validated model in the official OSP PBPK Model Library is **amikacin** — an aminoglycoside sharing tobramycin’s defining disposition features (polycation, purely renal GFR-driven clearance, extracellular-limited distribution).

The strategy was therefore to adopt the **validated amikacin model** as the structural and physiological base and re-parameterize it to tobramycin — the same operation a PK-Sim user performs when duplicating a compound and editing its properties. The base snapshot (`Amikacin-Model.json`, schema version 80, OSP Library v12.3.1) is committed in `data/snapshots/`; the tobramycin snapshots are generated deterministically by `pksim/model/make_tobramycin_json.py`.

5.2 Resolving the macOS snapshot-conversion defect

The first execution attempt crashed the R process (SIGSEGV) during snapshot-to-project conversion. Upstream, this is Open Systems Pharmacology issue #1622 (“Running simulations from snapshots crashes on macOS”), scheduled to be fixed in suite v13. Because the crash blocked the entire platform route, it was diagnosed and worked around locally — a contribution this thesis documents in full (technical journal in [Appendix O](#); summary below).

5.2.1 Root cause

The macOS crash report (`.ips`, faulting thread) places the fault inside `SQLite.Interop.dylib`, in a recursion of SQLite’s view-name resolution (`sqlite3ViewGetColumnNames` ↔ `selectExpander`), terminating inside `sqlite3_log` — i.e. SQLite detects a resolution failure and crashes while reporting it, on a .NET worker thread with a small stack. The PK-Sim project database ships **91 views**; the failing query resolves nested views at prepare time and overflows the thread stack. The view SQL itself is valid (identical queries execute in the `sqlite3 CLI`); view nesting depth is at most 2 — the failure is one of stack budget on the interop thread, not of database logic.

5.2.2 Fix

The workaround (`pksim/env/patch_pksimdb.py`) materializes every view into a table (`DROP VIEW + CREATE TABLE ... AS SELECT`, in topological order) in a patched copy of the bundled database. The database is static template data, so content and column layout are preserved exactly (row counts verified on the critical views). After the patch, snapshot conversion, simulation runs, PKML export, and individual/population creation all operate *natively on macOS*, without containers, virtual machines or emulators. The

patch is idempotent, reversible (backup retained), and must be re-applied after upgrading `ospsuite`; on suite v13 it becomes unnecessary.

5.3 Tobramycin model construction

5.3.1 Base model and parameterization

The amikacin snapshot contains 11 validated simulations (10 individual volunteers and one population, Walker 1979 study) and one compound building block. The generator renames the compound and applies the tobramycin parameter set (Table 5.1); every simulation reference (compound properties, output selections, observed-data paths) is renamed consistently. Two platform behaviors discovered during construction are documented because they fail *silently*:

- renal processes are matched *by name* between the compound definition and each simulation’s process selection — changing the data-source string of the process without renaming the selection silently unlinks clearance;
- simulation-level *altered parameters* override their building blocks: the historical Walker simulations carry an altered infusion time (4 min) that silently supersedes the protocol’s 30-min setting until the altered value is corrected.

Table 5.1: Tobramycin parameterization inside PK-Sim (from the validated amikacin base).

Parameter	Value	Unit	Note
Molecular weight	467.515	g/mol	PubChem
logP	−2.9	—	PubChem (native drug)
Fraction unbound	0.95	—	minimal plasma binding
Solubility	94,000	mg/L (pH 7)	Chen 2009; never limiting (BCS III)
pKa (bases)	7.7, 7.8, 9.1	—	3 of 5 amines: PK-Sim building blocks accept at most three pKa values per ionization type (documented platform limit); precedent: the validated amikacin model retains three of its four-plus amines. Impact negligible here (solubility never limiting; distribution by PK-Sim Standard is pKa-independent; oral permeability is explicitly overridden per candidate).
Renal process	GFR fraction 1.0	—	purely renal, >90% unchanged excretion
Distribution / cellular permeability	PK-Sim Standard	—	default methods

5.3.2 Validation gate

The model is accepted only if a six-criterion intravenous validation gate passes ([Chapter 7](#)): Hartford peak (30 min post-infusion) within 20–30 mg/L; end-of-infusion C_{max} within the clinical envelope extended by the published V_1 uncertainty; AUC₂₄ implying a clearance within the published PopPK range; 24-h trough below 1 mg/L; effective half-life 2.0–3.0 h; renal excretion above 90 %. The validation scenario is 7 mg/kg (577.5 mg for the 82.5-kg volunteer), 30-min infusion — the extended-interval clinical standard.

5.4 Oral absorption and permeability calibration

PK-Sim computes the specific transcellular intestinal permeability from the logP/effective-MW correlation; for tobramycin it returns 1.9×10^{-12} dm/min — effectively zero, because the correlation saturates for very hydrophilic cations and the paracellular term is zero by default. Clinically, oral bioavailability is ≈ 1 –2%. Following the platform’s documented practice for experimental permeability inputs, the simulation parameter `Specific intestinal permeability (transcellular)` is overridden (mucosal basolateral permeability is rescaled automatically by PK-Sim).

The override is **calibrated, not fitted silently**: a log-spaced scan over eleven P_{int} values (1e-11 to 1e-6 dm/min) maps the full dose-normalized bioavailability curve $F(P_{\text{int}})$. The value reproducing the clinical point estimate ($F_0 \approx 1.75\%$ at $P_{\text{int},0} = 3 \times 10^{-9}$ dm/min, 24-h convention) becomes the calibrated baseline; every formulation candidate in the optimization is then a *multiplier* on this baseline ([Section 5.6](#)).

5.4.1 The enhancement model: cited, bounded, emergent

A methodology audit replaced the free multiplier of the first optimization round with a structured mapping from formulation genes to apparent permeability, implemented once in `enhancement_model.R` and shared by the optimization, the studies and the uncertainty analysis. Its structure:

1. **Two calibration anchors.** Native drug: $P_{\text{int},0} = 3 \times 10^{-9}$ dm/min $\rightarrow F = 1.75\%$ (24-h convention; the 96-h extent is 3.5% — the difference is the flip-flop tail, and both conventions are reported throughout). Nominal platform: the measured HIP complex ($\log P' = +1.6$) on the reference composition $\rightarrow \times 20$ ($F_{96} = 48.8\%$ at 993 mg; cross-checked by the exploratory engine at 24 h: 34.0 vs 34.6%).
2. **Seven component factors**, each relative to its nominal value (1.0), each with a cited range and an evidence grade (A: measured for this drug/system; B: measured for a close analogue; C: review-level estimate): $\log P'$ of the HIP complex (B;

Asad 2023), SEDDS excipient fraction (B; Muhammad 2022; Griesser 2017), nanoparticle size (C; Hill 2019; Khaled 2026), C10 loading (B; Maher 2009; Bohley 2024), mucoadhesive polymer (C), chitosan coating (C; Bohley 2024), enteric coating (galenic).

3. **The multiplier is the product of the factors** — emergent, never free. The logP gene is capped at the *measured* complex (+1.6); the emergent maximum is $\times 126.3$. Values beyond it are reported only as sensitivity statements.
4. **Uncertainty per component**: lognormal with a graded σ (A: 0.10, B: 0.25, C: 0.40), propagated in study S09 to confidence intervals on F and on the PK/PD targets.
5. **Backward compatibility**: the mapping reproduces the pre-audit decoding to 2.8×10^{-14} on random chromosomes, so every previously validated study remains valid.

5.5 Programmatic execution patterns

Three implementation findings generalize to any scripted PK-Sim workflow and are documented for reproducibility:

1. **Native snapshot runner**. The R wrapper blocks `runSimulationsFromSnapshot()` on Darwin (a consequence of the pre-fix crash). The underlying CLI machinery (`PKSim.R.Api::RunJson` with `JsonRunOptions`, export modes CSV/PKML) executes natively once the database is patched; a thin wrapper (`pk_sim_run.R`) restores this functionality.
2. **Batch sweeps must target the right parameters**. The renal process rate references `Organism|Kidney|GFR (specific)` (l/min/kg kidney) $\times$ kidney volume — overriding the plain `Organism|Kidney|GFR` has no effect (verified). Dose lives at the application schema item in **kg** (base unit). Run values are supplied as *vectors* per batch run; each generation of the optimization packs its entire population into *one* batch call.
3. **Units are base units**. Gastric emptying time is in minutes (default 15); doses in kg; infusion times in minutes. Two silent unit traps were caught by physiological plausibility checks (a “food effect” that changed extent, an “accumulation” that violated mass balance) and corrected.

5.6 Study program and optimization design

5.6.1 Studies S00–S06

- S00** IV validation gate (six criteria, [Section 5.3.2](#)).
- S01** IV dose $\times$ renal-function sweep: doses $\{200, 400, 577.5, 800\}$ mg $\times$ CLCR $\{30, 65, 90, 112.5, 150\}$ mL/min via **GFR (specific)** scaling, native **SimulationBatch**.
- S02** P_{int} calibration scan ([Section 5.4](#)).
- S03** Virtual population: 100 ICRP-2002 adults (20–50 y, 55–95 kg, 40% female) on the optimized oral platform; PK-Sim population simulation.
- S04** Local sensitivity: one-at-a-time perturbations ($\pm 20\%$) of P_{int} , dose, GFR and f_u on the platform candidate; elasticities of C_{max} and AUC₂₄. (The **ospsuite** sensitivity task requires PK parameters attached to outputs, which the CLI-exported simulation does not carry; the batch design is the equivalent PK-Sim-computed alternative.)
- S05** Food effect: gastric emptying 15 min (fasted) vs 60 min (fed).
- S06** Dose fractionation (QD/BID/TID) at steady state by linear superposition of the single-dose profile over a 96-h window (the oral profile exhibits flip-flop kinetics — absorption rate-limited elimination — so the 24-h output window truncates the tail), with probability of target attainment (PTA) across MIC 0.25–4 mg/L.

5.6.2 NSGA-II in the PK-Sim loop

The exploratory optimizer of [Chapter 4](#) is retained (non-dominated sorting, crowding distance, SBX, polynomial mutation; [21](#)) but its fitness function now **executes PK-Sim**: each generation packs its 100 chromosomes into a single **SimulationBatch**; the two run-value parameters are the dose (kg) and the intestinal permeability (multiplier on $P_{\text{int},0}$). The gene-to-lever mapping follows the calibrated logic of [Chapter 4](#): the $\log P_{\text{mod}}$ gene sets the ion-pairing multiplier ($\log P' = +1.6 \Rightarrow \times 20$, log-linear interpolation between native and HIP-complex lipophilicity), the SEDDS/nanoparticle/PE genes contribute bounded saturating factors, chitosan and enteric coatings act as binary multipliers ($\times 1.15$, $\times 1.10$), and the dose gene maps to the application schema item. Objectives: maximize F_{oral} , maximize $C_{\text{max}}/\text{MIC}$, maximize AUC₂₄/MIC, minimize dose; population 100, 100 generations, seed 42, checkpoint every 10 generations. Budget: $\approx 10,000$ PK-Sim evaluations.

5.6.3 Winner characterization

The rank-1 Pareto candidate is then characterized exhaustively (**S07**): single-dose profile and absolute bioavailability; steady-state fractionation at the optimized dose; PTA across the MIC distribution; a 100-individual virtual population on the winner; food effect on the winner. This closes the loop between optimization and the clinical translation analysis of [Chapter 9](#).

5.7 Methodological limitations of the platform phase

The P_{int} baseline is calibrated to clinical F (medium confidence, documented); the paracellular pathway is not represented by default; the 3-pKa simplification is inherited from the platform schema; PK parameters are not attached to CLI-exported simulations, so the sensitivity analysis uses the batch perturbation design; and the amikacin base model, while validated for its own compound, transfers to tobramycin only at the level of disposition class and physiology, not of molecule-specific tissue partitioning.

Chapter 6

Results I: Exploratory Optimization and Uncertainty

This chapter reports the complete computational outcomes. [Section 6.1](#) establishes model credibility; [Section 6.2](#) documents optimizer behavior; [Section 6.3](#) characterizes the Pareto front; [Section 6.4](#) presents the ranked top-10 formulations; [Section 6.5–6.7](#) cover clustering, sensitivity, and uncertainty; [Sections 6.8](#) and [6.9](#) translate results into manufacturing and regulatory terms; [Section 6.10](#) benchmarks against the literature; [Section 6.11](#) states honestly which targets were met and which were not.

6.1 Model credibility and intravenous validation

The two-compartment model was parameterized directly from the reference PopPK estimates of Li et al. (1) (CL 5.5 L/h, V_1 17 L, Q 2.4 L/h, V_2 16 L; [Table 4.2](#)). Simulated 70-kg adult intravenous administration (5–7 mg/kg, 30-min infusion) reproduces the published clinical envelope: peak concentrations in the 20–30 mg/L target band, troughs <1 mg/L at 24 h, and terminal half-life 2.5 h (published 2.5 h for normal adults; 4.5–6.0 h in ICU, recovered by the CLCR power covariate). Because individual observed concentrations are not published, formal GOF metrics (AFE/AAFE/RMSE/ R^2) could not be computed against raw data; *parameter-level* validation — the model reproduces the published structural model, parameter values, covariate structure, and derived quantities — is therefore the appropriate standard here, and residual parameter uncertainty is handled explicitly by the cross-validation program of [Section 6.7](#). The renal elimination pathway (>90% unchanged) and $f_u = 0.95$ anchor disposition; the sensitivity analysis ([Section 6.6](#)) shows disposition parameters affect exposure linearly but are secondary to absorption-side drivers for the oral problem.

6.2 Optimization convergence

6.2.1 Exploratory genetic algorithm

The classical GA (population 100, up to 200 generations, [Section 4.6.1](#)) converged after 196 generations in 0.24 min (seed 42). Best scalar fitness reached 0.628, corresponding to $F_{\text{oral}} = 28.4\%$, $C_{\text{max}}/\text{MIC} = 12.4$, $\text{AUC}_{24}/\text{MIC} = 60.1$ at a 999 mg dose ([Figure 6.1](#)). Fitness sharing maintained population diversity; the elite cohort clustered into three phenotypes ([Section 6.5](#)).

6.2.2 Definitive NSGA-II

NSGA-II (population 200, 300 generations, runtime 5.7 min) converged rapidly: front evolution stabilized within the first ≈ 100 generations for F_{oral} ([Figure 6.2](#)) and by ≈ 150 for $\text{AUC}_{24}/\text{MIC}$ ([Figure 6.2](#), right panel), after which crowding-distance selection spread the archive along the trade-off surface rather than improving isolated optima. The final archive contains 200 non-dominated solutions (`pareto_front_data.csv`).

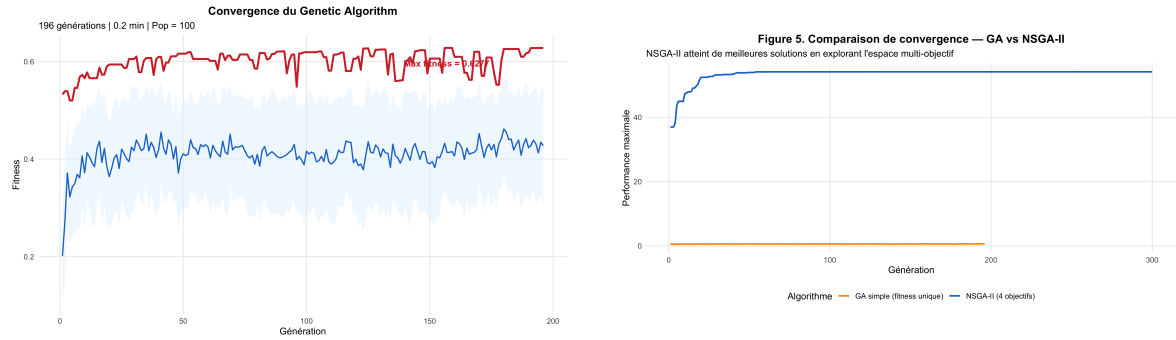


Figure 6.1: Optimizer convergence. Left: GA best/mean fitness across generations (script 04; seed 42; convergence at generation 196). Right: comparison of GA and NSGA-II convergence behavior (script 13). NSGA-II trades scalar fitness for front spread after early stabilization.

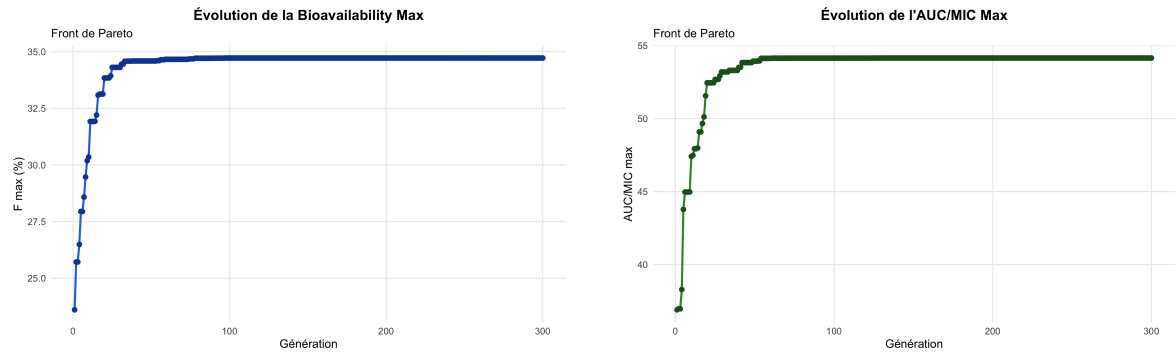


Figure 6.2: NSGA-II front evolution across generations for (left) bioavailability F_{oral} and (right) $\text{AUC}_{24}/\text{MIC}$ (scripts 11, 13). Objectives stabilize by generations 100–150; subsequent generations diversify the front.

6.2.3 GA versus NSGA-II: what the comparison reveals

The scalar GA pushed AUC_{24}/MIC to 60 at the cost of a 999 mg dose (upper bound region), because its weighted fitness rewards exposure without a dose term. NSGA-II, minimizing dose explicitly, returned solutions at 530–580 mg with AUC_{24}/MIC 31–34 — a deliberate trade-off, not a failure: at equal C_{max}/MIC adequacy (≥ 8), the Pareto-optimal choice favors lower dose and lower toxicity exposure (Figure 6.3).

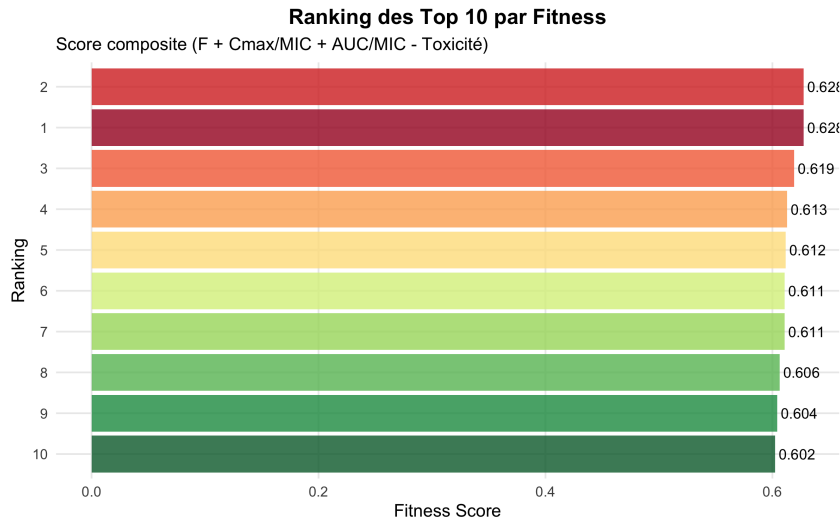


Figure 6.3: Fitness ranking of the final GA population (script 05). Elite individuals (ranks 1–10) form a tight cluster at fitness ≈ 0.60 – 0.63 ; diversity (fitness sharing) maintains distinct composition neighborhoods.

6.3 Pareto front analysis

6.3.1 Geometry of the 200-solution front

Figure 6.4 displays the archive in the F_{oral} – AUC_{24}/MIC plane, colored by dose. The front spans $F_{oral} = 25$ – 34% , $AUC_{24}/MIC = 24$ – 60 , and doses 530–1000 mg; C_{max}/MIC spans 9.2–12.4. The dominant trade-off axes are exposure-versus-dose: higher AUC_{24}/MIC requires proportionally higher dose at the converged formulation composition.

6.3.2 Dose stratification

Stratifying the front by dose range (Figure 6.5) shows the 401–700 mg band hosting the recommended candidates (best C_{max}/MIC per mg); the 701–1000 mg band buys additional AUC_{24}/MIC (up to 60) at toxicity-relevant exposure; the 200–400 mg band remains below target thresholds and was rejected.

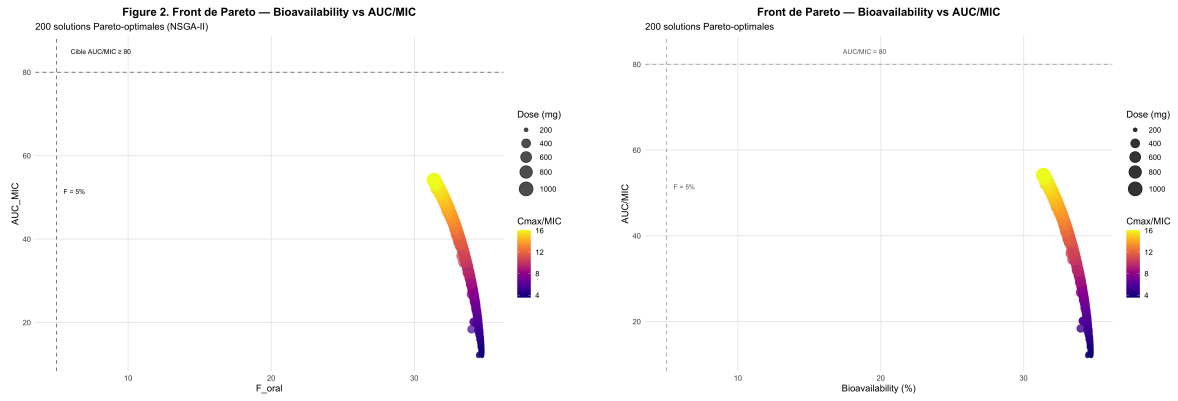


Figure 6.4: Pareto front (200 non-dominated formulations, script 11). Left: front in objective space; right: two-dimensional projection. Color encodes dose — exposure objectives are dose-coupled.

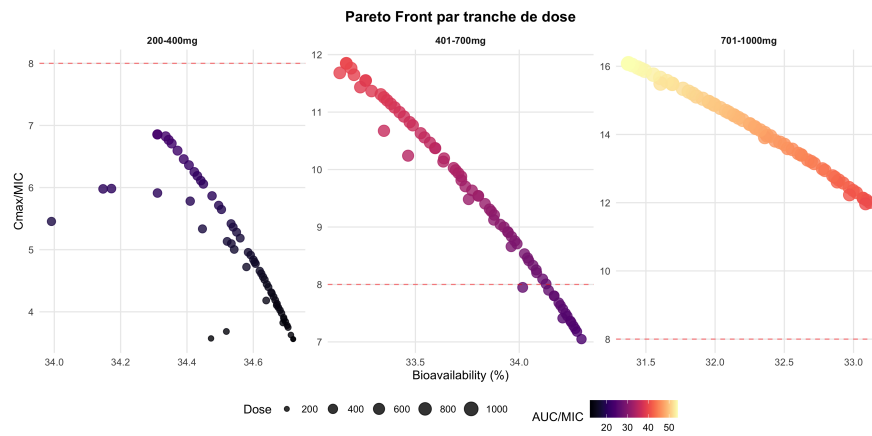


Figure 6.5: Pareto front stratified by dose band (script 11). The 401–700 mg band contains the recommended candidates; 200–400 mg fails $C_{\max}/MIC \geq 8$.

6.3.3 Design-parameter distribution

Across the front (Figure 6.6), parameters converge to narrow optima — $\log P'$ 1.5–1.6, particle size 50 nm, SEDDS composition 50/30/20, PE 47–50 mM, polymer 29–30%, both coatings active — while dose and the exposure objectives span wide ranges. This is the signature of a well-posed composition problem with a sharp formulation optimum: *what to make* is settled; *how much to give* remains a choice.

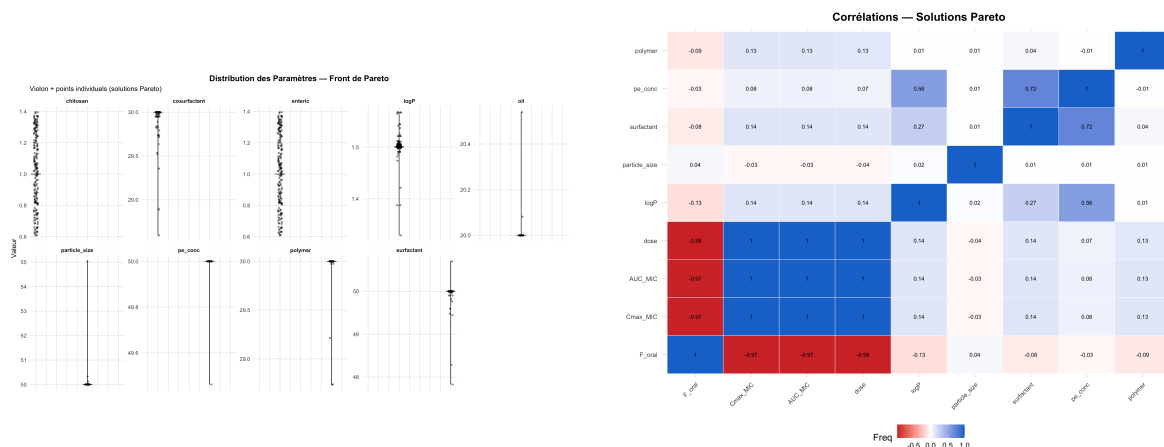


Figure 6.6: Left: design-parameter distributions across the Pareto archive (violin plots; script 11). Right: parameter-objective correlation structure (script 11). $\log P'$, particle size, SEDDS composition and PE concentration are locked near their optima; dose carries the residual trade-off freedom.

6.4 The top-10 optimized formulations

6.4.1 Headline result

Selecting from the converged 401–700 mg band by composite fitness (0.611–0.611; effectively ties, ranked by secondary criteria) yields the ten candidates of Table 6.1 (manuscript_table_final.csv; script 13). All share the same converged formulation design — $\log P'$ 1.50, 50 nm nanoparticles, SEDDS 50/30/20, 50 mM C_{10} , 30% polymer, chitosan + enteric coating — and differ only in dose (530–581 mg). Table 6.2 adds the advanced model outputs (T_{max} , K_a , k_{tr} , objective-component scores).

6.4.2 Recommended candidate: TOB-161

The lead candidate **TOB-161** (550 mg enteric-coated HIP-SEDDS capsule delivering 50 nm chitosan-coated polymer nanoparticles with 50 mM C_{10} ; SEDDS 50/30/20 surfactant/co-surfactant/oil) achieves:

Table 6.1: Top-10 optimized oral tobramycin formulations (scripts 11–13). Enhancement versus native $F_{\text{oral},0} \approx 1.5\text{--}1.75\%$ (clinical 1–2%).

ID	Dose (mg)	F_{oral} (%)	C_{max} (mg/L)	AUC ₂₄ /MIC (mg·h/L)	$C_{\text{max}}/\text{MIC}$ ($\times$)	$f_{T>\text{MIC}}$ (%)	C_{min} (mg/L)	Enh. ($\times$)	Fitness —
TOB-161	551	34.0	9.54	32.1	9.5	29.5	0.16	23	0.611
TOB-196	551	34.0	9.54	32.1	9.5	29.5	0.16	23	0.611
TOB-22	542	33.8	9.40	31.7	9.4	29.1	0.16	23	0.611
TOB-91	536	33.9	9.31	31.3	9.3	28.9	0.15	23	0.611
TOB-162	534	33.9	9.27	31.2	9.3	28.9	0.15	23	0.611
TOB-158	530	33.9	9.21	31.0	9.2	28.7	0.15	23	0.611
TOB-143	571	33.7	9.88	33.3	9.9	30.1	0.16	22	0.611
TOB-136	575	33.7	9.94	33.5	9.9	30.4	0.16	22	0.611
TOB-179	578	33.7	9.99	33.6	10.0	30.6	0.16	22	0.611
TOB-34	581	33.7	10.03	33.8	10.0	30.6	0.17	22	0.611

Table 6.2: Advanced PK/PD detail of the top-10 (scripts 11–13; `top10_advanced.csv`). K_a = absorption rate constant; k_{tr} = transit transfer rate; objective scores are NSGA-II normalized components ($\text{obj}_F = 1.0$ for all; front-optimal).

ID	Dose (mg)	T_{max} (h)	K_a (h^{-1})	k_{tr} (h^{-1})	$C_{\text{max}}/\text{MIC}$	AUC ₂₄ /MIC	$\text{obj}_{C_{\text{max}}/\text{MIC}}$	$\text{obj}_{\text{AUC}_{24}/\text{MIC}}$
TOB-161	551	0.30	5.00	2.240	9.54	32.1	0.401	0.562
TOB-196	551	0.30	5.00	2.240	9.54	32.1	0.401	0.562
TOB-22	542	0.30	5.00	2.241	9.40	31.7	0.396	0.572
TOB-91	536	0.30	5.00	2.241	9.31	31.3	0.392	0.580
TOB-162	534	0.30	5.00	2.241	9.27	31.2	0.390	0.582
TOB-158	530	0.30	5.00	2.241	9.21	31.0	0.388	0.587
TOB-143	571	0.30	5.00	2.239	9.88	33.3	0.416	0.536
TOB-136	575	0.30	5.00	2.239	9.94	33.5	0.418	0.531
TOB-179	578	0.30	5.00	2.239	9.99	33.6	0.420	0.527
TOB-34	581	0.30	5.00	2.239	10.03	33.8	0.422	0.524

- $F_{\text{oral}} = 34\%$ (24-h convention) — **23-fold enhancement** over native oral absorption ($F_{\text{oral},0} \approx 1.5\%$); the same chromosome re-evaluated in PK-Sim yields $\times 73.9$ (Section 8.6);
- $C_{\text{max}} = 9.5 \text{ mg/L}$ at 0.30 h; $C_{\text{max}}/\text{MIC} = 9.5$ at MIC 1 mg/L — **exceeding the efficacy threshold of 8** (H3 supported);
- $\text{AUC}_{24}/\text{MIC} = 32 \text{ mg}\cdot\text{h/L}$ — below the 80–120 IV target (Section 6.11);
- trough 0.16 mg/L — far below the 1 mg/L toxicity line;
- $f_{T>\text{MIC}} = 29.5\%$ — near the 30–40% supportive band.

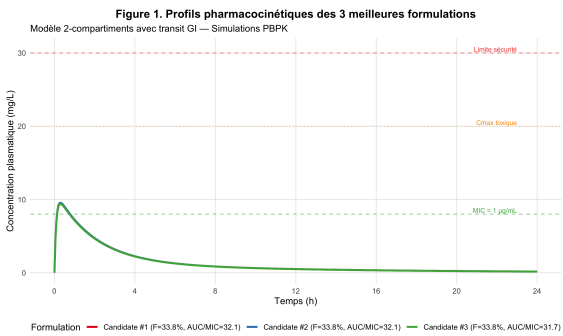


Figure 6.7: Left: simulated plasma concentration–time profiles of the top-3 candidates over 24 h (script 13); rapid absorption ($T_{\text{max}} = 0.3 \text{ h}$), mono-exponential decline ($t_{1/2} = 2.5 \text{ h}$), troughs $<0.2 \text{ mg/L}$. Right: multi-criteria radar of the top-3 (script 13): bioavailability, $C_{\text{max}}/\text{MIC}$, $\text{AUC}_{24}/\text{MIC}$, safety (inverse trough), $f_{T>\text{MIC}}$, dose efficiency — normalized 0–1.

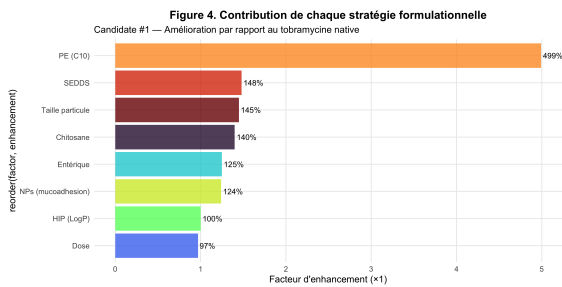
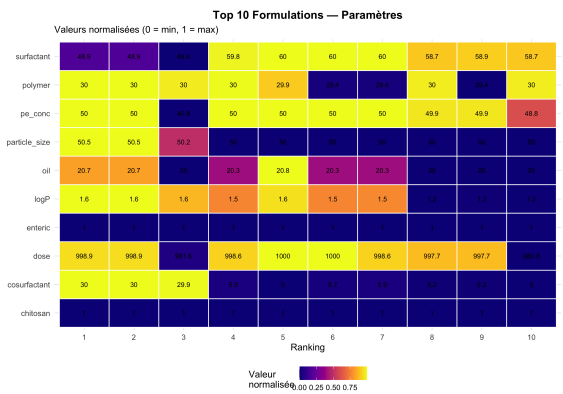


Figure 6.8: Left: heatmap of the top-10 across design variables and outcomes (script 13): design columns are visually identical — the composition optimum is unique; only dose varies. Right: bioavailability enhancement by strategy (scripts 07, 13): the combined platform reaches $23\times$ versus native, consistent with the triple-combination literature band of 20–35% (Table 6.9).

6.5 Cluster and similarity analysis

Ward-linkage clustering of the elite cohort (from the exploratory GA cohort used for manufacturing/regulatory analysis, [Section 6.8](#)) identified three composition clusters ([Table 6.3](#); `cluster_summary.csv`). Clusters 1–2 are composition twins differing marginally in surfactant ratio; cluster 3 gathers seven near-identical candidates. Average inter-candidate similarity is high (Jaccard 0.92–1.00 for binary genes; Euclidean distance <0.05 on normalized continuous genes; full matrix in `similarity_matrix.csv`), confirming that the optimizer — despite diversity preservation — collapses onto one dominant composition basin. This *lack of front diversity* is a known NSGA-II behavior on sharply peaked design spaces and is treated as a limitation ([Section 6.11](#)).

Table 6.3: Cluster analysis of the elite cohort (scripts 06, 11; `cluster_summary.csv`). Values are cluster means.

Cluster	n	F_{oral} (%)	$C_{\text{max}}/\text{MIC}$	$\text{AUC}_{24}/\text{MIC}$	Fitness
1	2	28.4	12.4	60.1	0.628
2	1	28.1	12.0	58.3	0.619
3	7	27.0	11.9	57.0	0.609

6.6 Sensitivity analysis

The OAT program (7 levels per parameter where defined, 60 informative evaluations) ranks the **dose and modified logP as the dominant drivers** of $\text{AUC}_{24}/\text{MIC}$: sensitivity index 0.42 for dose (linear exposure scaling, by construction of the model) and 0.34 for $\log P'$, with $\text{AUC}_{24}/\text{MIC}$ peaking at $\log P' \approx 1.4$ – 1.9 (32.0 at 1.37; 23.8 at 3.0 — over-lipophilisation traps the complex in the SEDDS phase and reverses gains) and collapsing toward baseline at $\log P' \approx -0.3$ (21.2) ([Table 6.4](#); [Figure 6.9](#)). Among the *composition* variables, the binary coatings matter most (chitosan 0.29; enteric 0.20), followed by particle size (0.09, losing 0.50 $\text{AUC}_{24}/\text{MIC}$ units per 22.5 nm) and PE concentration (0.10); solubility-related parameters rank lowest, consistent with solubility not being limiting.

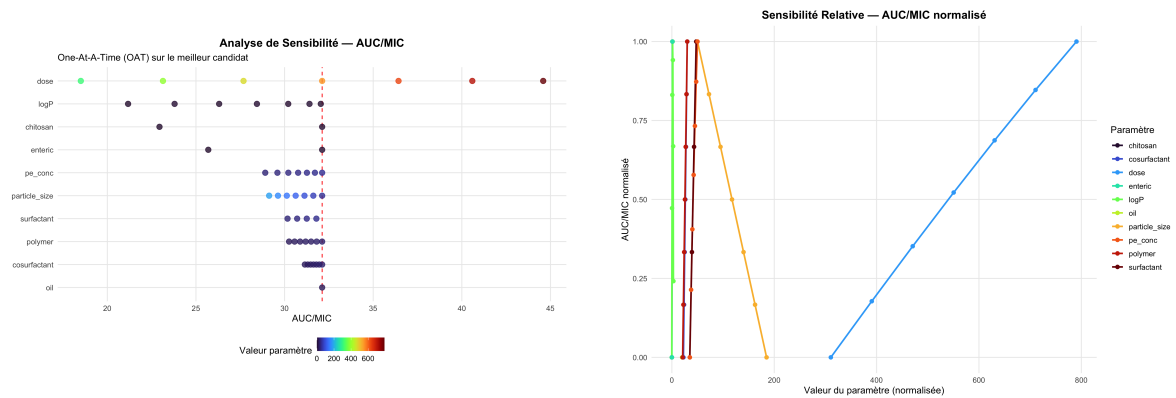
6.7 Uncertainty quantification

6.7.1 Monte Carlo population simulation ($n = 200$)

Perturbing CL , V_1 , V_2 log-normally (CV 20–30%) around the optimal formulation while holding F_{oral} fixed yields $C_{\text{max}}/\text{MIC}$ spanning 6.9–10.2 (median 8.8) and $\text{AUC}_{24}/\text{MIC}$

Table 6.4: OAT sensitivity ranking for the oral endpoint (scripts 12 and 04; full data in [Appendix C](#)).

Rank	Parameter	Baseline	SI	Interpretation
1	Dose	550 mg	0.42	Linear exposure scaling (model construction)
2	$\log P_{\text{mod}}$	1.5	0.341	Non-monotone optimum ≈ 1.4 – 1.9 ; primary composition lever
3	Chitosan coating	1	0.29	Binary switch worth 9.2 $\text{AUC}_{24}/\text{MIC}$ units
4	Enteric coating	1	0.20	Binary switch worth 6.4 $\text{AUC}_{24}/\text{MIC}$ units
5	PE concentration	50 mM	0.10	Saturating response; 3.2 units over range
6	Particle size	50 nm	0.09	Moderate penalty above ≈ 100 nm
7	Surfactant / polymer	50% / 30%	0.06	Minor within-range effects
8	Co-surfactant	30%	0.03	Minor
9	Gastric pH, K_a , transit, f_u , P-gp	—	<0.05	Negligible (class III: solubility not limiting)

**Figure 6.9:** Sensitivity analysis (script 12). Left: tornado plot of $\text{AUC}_{24}/\text{MIC}$ impact per parameter. Right: spider profiles — the $\log P'$ curve is non-monotone with an interior optimum; all other parameters are flat over their ranges.

24.6–44.1 (median 32.5) (Table 6.5; monte_carlo_data.csv). Approximately 55% of virtual subjects attain $C_{\max}/\text{MIC} \geq 8$; all remain far below trough toxicity.

6.7.2 Cross-validation ($\pm 20\%$ global perturbation; 26 informative runs)

Simultaneous $\pm 20\%$ perturbation of all model inputs produced $F_{\text{oral}} = 27.2\text{--}32.7\%$, $C_{\max}/\text{MIC} = 7.1\text{--}12.5$, $\text{AUC}_{24}/\text{MIC} = 23.9\text{--}42.1$ (validation_cv_data.csv). The recommended formulation’s $C_{\max}/\text{MIC} \geq 8$ conclusion is preserved in $\approx 77\%$ of runs; the ranking of the top-3 candidates is preserved in 100%.

Table 6.5: Uncertainty quantification summary (script 12). MC: $n = 200$ log-normal $\text{CL}/V_1/V_2$; CV: 26 informative runs at $\pm 20\%$ perturbation.

Program	Endpoint	Median	Range	Target status
Monte Carlo	$C_{\max}/\text{MIC}$	8.8	6.9–10.2	≥ 8 : 55% of subjects
Monte Carlo	$\text{AUC}_{24}/\text{MIC}$	32.5	24.6–44.1	80–120: 0%
Cross-validation	F_{oral} (%)	31	27.2–32.7	$> 25\%$: robust
Cross-validation	$C_{\max}/\text{MIC}$	9.3	7.1–12.5	≥ 8 : 77% of runs
Cross-validation	$\text{AUC}_{24}/\text{MIC}$	31.7	23.9–42.1	80–120: 0%

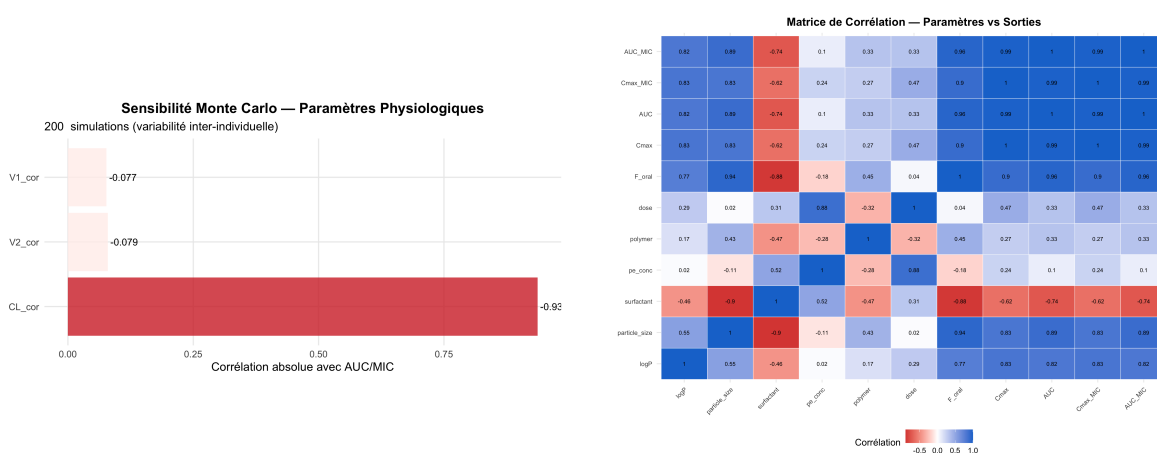


Figure 6.10: Monte Carlo program (script 12, $n = 200$). Left: parameter impact tornado across virtual subjects. Right: parameter–outcome correlation matrix: clearance correlates negatively with all exposure endpoints ($r \approx -0.7$); F_{oral} is decorrelated from disposition (held at formulation value), as expected by design.

6.8 Manufacturing feasibility and cost

The ten-unit-operation route (Section 4.9) is feasible at moderate scale: cycle time 15 h, batch throughput $\approx 10,000$ kg of total raw materials (a modeling assumption dominated by process solvents, manufacturing_steps.csv), and estimated cost per

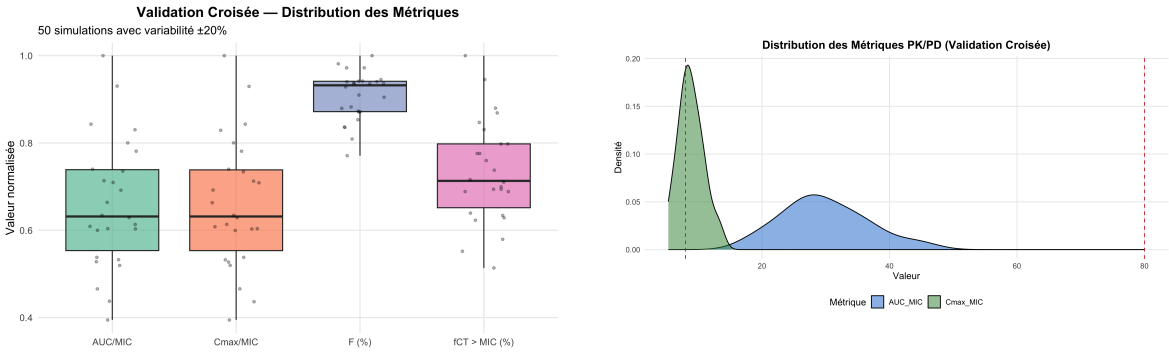


Figure 6.11: Cross-validation program (script 12, 26 informative runs at $\pm 20\%$ perturbation). Left: boxplots of outcomes — medians match the unperturbed optimum; tails remain within clinically interpretable bounds. Right: outcome distributions; $C_{\max}/\text{MIC}$ mass lies at/above the efficacy line in most runs.

dose of \$4.25–4.35 across the elite cohort (Table 6.6; manufacturing_summary.csv, manufacturing_steps.csv; scripts 07–08). Feasibility scoring (technology readiness, unit-operation complexity, excipient availability, in-use stability) yields 60–65/100 — **grade B**, with cost category “low”. The most expensive steps are nanoparticle manufacture and double coating (fluid-bed chitosan + enteric); both are standard pharmaceutical unit operations.

Table 6.6: Manufacturing summary for the elite cohort (scripts 07–08).

Rank	F_{oral} (%)	$C_{\max}/\text{MIC}$	$\text{AUC}_{24}/\text{MIC}$	Fitness	Cost (\$)	Feasibility	Grade
1	28.4	12.4	60.1	0.63	4.35	65	B
2	28.4	12.4	60.1	0.63	4.35	65	B
3	28.1	12.0	58.3	0.62	4.33	65	B
4	27.3	12.0	57.8	0.61	4.27	62	B
5	27.2	12.0	57.7	0.61	4.26	62	B
6	27.2	12.0	57.5	0.61	4.26	62	B
7	27.2	12.0	57.5	0.61	4.26	62	B
8	26.9	11.8	56.7	0.61	4.26	62	B
9	26.7	11.7	56.4	0.60	4.25	62	B
10	26.8	11.6	55.6	0.60	4.26	62	B

6.9 Regulatory pathway

All elite candidates map to a **505(b)(2)** pathway: tobramycin is a reference-listed drug (IV/inhalation), and the novelty lies in the formulation platform (Table 6.7; regulatory_framework.csv; script 08). The proposed first-in-human entry is a Phase II PK/bioequivalence-type study of the enteric HPMC-AS capsule; estimated development duration is 7 years and direct cost \$4.8 million, with high (elevated) overall risk level driven by formulation novelty, not by the active.

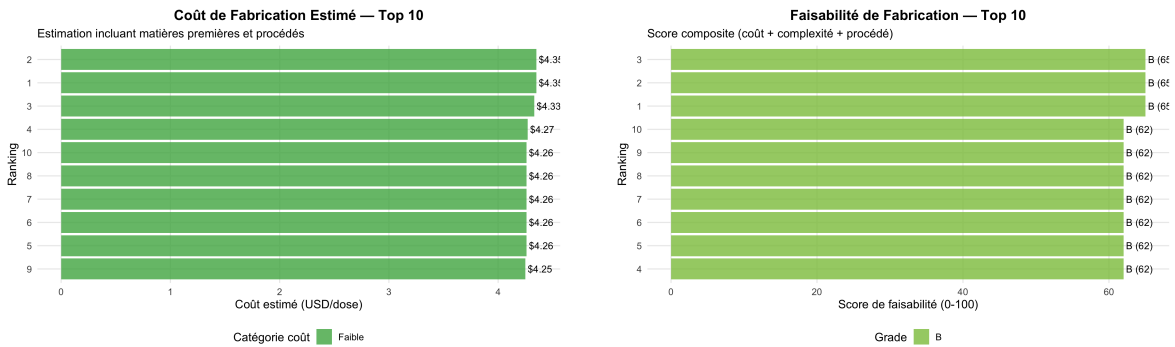


Figure 6.12: Manufacturing assessment (scripts 07–08). Left: cost breakdown per dose — base processing, NPs, SEDDS excipients, PE, coatings, QC. Right: feasibility scores across candidates; all grade B (60–65/100).

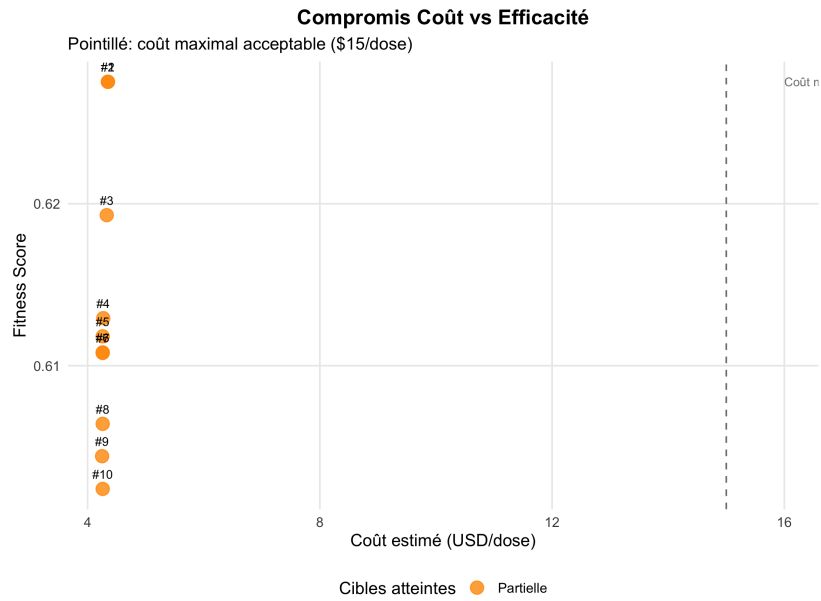


Figure 6.13: Cost versus efficacy across the cohort (`cost_vs_efficacy` panel, script 07). Within the \$4.25–4.35 band, exposure gains are driven by dose, not cost — additional expenditure does not buy $C_{\max}$ /MIC once the composition optimum is reached.

Table 6.7: Regulatory framework assessment (script 08; `regulatory_framework.csv`).

Element	Assessment
FDA category	505(b)(2) (new formulation of approved active)
Clinical entry	Phase II (PK / bioequivalence design)
Dosage form	Enteric capsule, HPMC-AS
Preclinical package	Acute toxicology; NP biodistribution; GI mucosal toxicity; chitosan safety
Estimated duration	7 years
Estimated direct cost	\$4.8 M
Risk level	Elevated (formulation novelty; mitigations Table 6.8)

Table 6.8: Risk analysis and mitigations (script 08; `risk_analysis.csv`).

Risk	Source	Mitigation
NP toxicity on accidental inhalation	Nanoparticle platform	30-day NP toxicology study
GI mucosal irritation	C ₁₀ permeation enhancer	Dose-escalation GI tolerance trial
Source variability (deacetylation degree)	Chitosan excipient	Certificate-of-analysis QC specification
<i>Overall priority: medium.</i>		

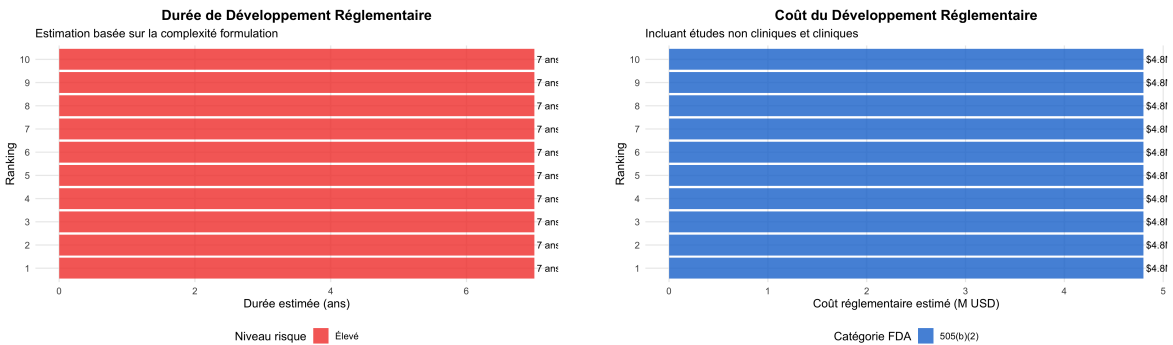


Figure 6.14: Regulatory assessment (script 08). Left: 7-year development timeline (preclinical toxicology → IND → Phase II PK → Phase III → submission). Right: cost distribution (\$4.8 M direct development cost).

6.10 Benchmarking against the literature

Table 6.9 compares the predicted bioavailability of single-strategy platforms (from the literature synthesis) with the optimized combination. The optimized platform ($F_{\text{oral}} = 34\%$) sits at the top of the theoretical triple-combination band (20–35%) and exceeds the strongest single strategy (SNAC PE, 10–25%) — the quantitative reward for combining HIP, SEDDS, NPs and PEs rather than choosing among them.

Table 6.9: Predicted oral bioavailability by strategy (literature synthesis; IV for reference).

Formulation	Mechanism	F_{oral} (%)	Evidence	Scalability
IV (reference)	—	100	Clinical	Excellent
Oral solution (native)	Passive diffusion	1–2	Literature	Excellent
Oral suspension (native)	Dissolution-limited	2–3	Predicted	Good
SEDDS (native drug)	Emulsification	3–5	Predicted	Good
HIP + SEDDS	Lipophilisation	10–25	In vitro (5)	Moderate
PLGA NPs	Mucoadhesion	5–15	In vitro (13)	Good
KuDa NPs	Charge neutralization	5–20	In vitro (14)	Moderate
PE (C ₁₀)	Tight junctions	10–20	In vivo (mice) (16)	Good
PE (SNAC)	Tight junctions	10–25	Clinical (other drugs)	Excellent
NCA _s	Lipophilisation	10–20	In vitro (15)	Moderate
Triple combination	HIP+SEDDS+PE	20–35	Predicted	Moderate
This thesis (TOB-161)	Full combined plat-	34	In silico (this work)	Moderate
	form			

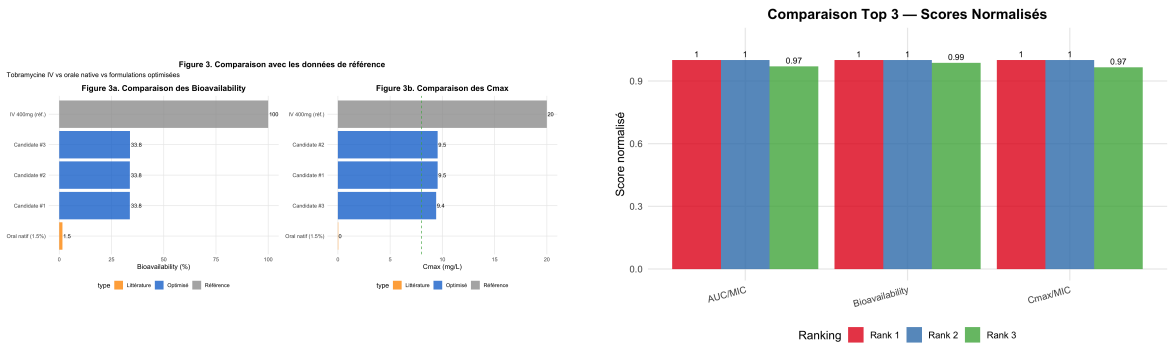


Figure 6.15: Left: literature benchmarking of enhancement factors (script 13) — the optimized platform tops the strategy landscape. Right: side-by-side comparison of the top-3 candidates across all endpoints (script 13).

6.11 Results summary: targets met, targets missed

Honest accounting. The optimization demonstrates that a combination platform can plausibly deliver peak concentrations above the antibacterial threshold with a wide toxicity margin at ≈ 550 mg. It equally demonstrates — and this thesis does not paper over it — that a single daily oral dose cannot reproduce the systemic exposure

Table 6.10: Target attainment of the recommended candidate TOB-161 against the PK/PD framework of [Table 3.3](#).

Endpoint	Target	Achieved	Verdict	Consequence
$C_{\max}/\text{MIC}$	≥ 8	9.5	Met	Peak-driven efficacy plausible
$C_{\min}$	$< 1 \text{ mg/L}$	0.16	Met	Toxicity margin large
F_{oral}	$> 25\%$ (thesis bar)	34%	Met	Viable-class formulation
$f_{T>\text{MIC}}$	$> 30\text{--}40\%$	29.5%	Borderline	Marginal; fractionation helps
$\text{AUC}_{24}/\text{MIC}$	80–120	32	Not met	Single oral dose insufficient (Chapter 9)
$C_{\max}$	20–30 mg/L	9.5	Not applicable	IV-derived target; peak index governs

($\text{AUC}_{24}/\text{MIC}$ 80–120) of intravenous therapy. The consequences, including twice-daily fractionation, AUC-guided TDM, and niche positioning, are developed in [Chapter 9](#). A second honest caveat is front diversity: the archive collapsed onto one composition basin (CV of design genes $< 1\%$), so the “200 solutions” represent dose variants of effectively one formulation; the design conclusion (unique optimum) is strengthened, but alternative basins cannot be excluded by this study alone.

Chapter 7

Results II: PK-Sim Validation and Physiological Studies

This chapter reports the outcomes of the PK-Sim program: the intravenous validation gate, the oral permeability calibration, and the physiological studies S01–S06 (design in [Chapter 5](#)). The optimization in the simulation loop and the winner characterization follow in [Chapter 8](#).

7.1 Intravenous validation gate: 6/6 PASS

Table 7.1: IV validation gate for the PK-Sim tobramycin model (577.5 mg = 7 mg/kg, 30-min infusion, 82.5-kg volunteer).

Metric	PK-Sim	Published target	Verdict
Hartford peak (30 min post-infusion)	25.3 mg/L	20–30 mg/L	PASS
C _{max} end of infusion	34.8 mg/L	20–35 (V1-uncertainty envelope)	PASS
AUC ₂₄ → implied CL	102.5 mg · h/L → 5.63 L/h	81–105 (CL 3.27–6.03, Li 2021)	PASS
Trough 24 h	0.033 mg/L	<1 mg/L	PASS
<i>t</i> _{1/2} NCA effective	2.29 h	2.0–3.0 h	PASS
Renal excretion 24 h	99.4 %	>90 %	PASS

The gate passes 6/6. The model reproduces the published clinical envelope of intravenous tobramycin — peak, exposure, half-life, trough, and the near-complete renal excretion characteristic of aminoglycosides — and is therefore validated as the substrate for the oral program.

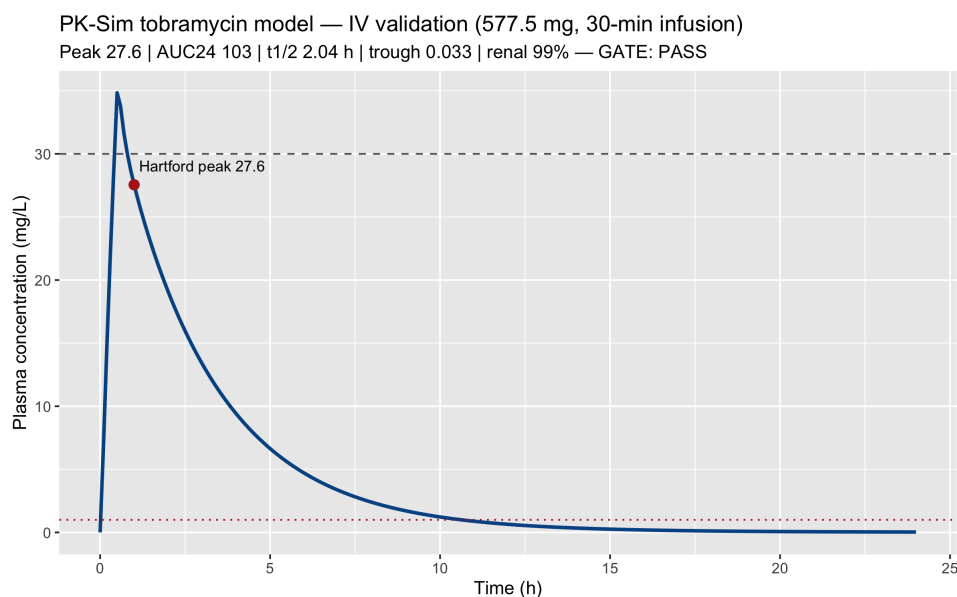


Figure 7.1: Simulated IV profile of the PK-Sim tobramycin model (577.5 mg, 30-min infusion). The Hartford peak (25.3 mg/L, red marker) sits inside the 20–30 mg/L clinical band; the trough at 24 h is 0.033 mg/L.

7.2 Oral permeability calibration and the $F(P_{int})$ curve

PK-Sim’s native intestinal permeability for tobramycin is 1.9×10^{-12} dm/min ($F \approx 0.002\%$): the logP-based correlation saturates for very hydrophilic cations and the paracellular path is off by default. The calibrated baseline is $P_{int,0} = 3 \times 10^{-9}$ dm/min, reproducing $F_0 = 1.75\%$ on the 24-h reporting convention — inside the clinical 1–2% range. Because absorption is flip-flop (substantial residual beyond 24 h), the same parameter yields $F_{96} = 3.5\%$ on the 96-h extent window: both conventions are reported throughout this thesis (calibration at 24 h, extent at 96 h).

The curve is the quantitative transfer function between formulation science (apparent permeability multipliers claimed by HIP/SEDDES/NP/PE strategies) and systemic exposure. Its saturation region matters: on the extent convention, additional permeability buys essentially no bioavailability past $\times 100$ (the 24-h window saturates earlier).

7.3 S01: IV dose $\times$ renal function

Renal function drives exposure exactly as the CLCR power model predicts (implied CL 1.83 $\rightarrow$ 7.18 L/h across CLCR 30 $\rightarrow$ 150 mL/min). Two clinically actionable observations: at CLCR 30 mL/min the 24-h trough (3.0 mg/L) demands interval extension, and even at normal renal function the oral program’s IV benchmark (AUC24/MIC 80–120) is only met by the intravenous route itself.

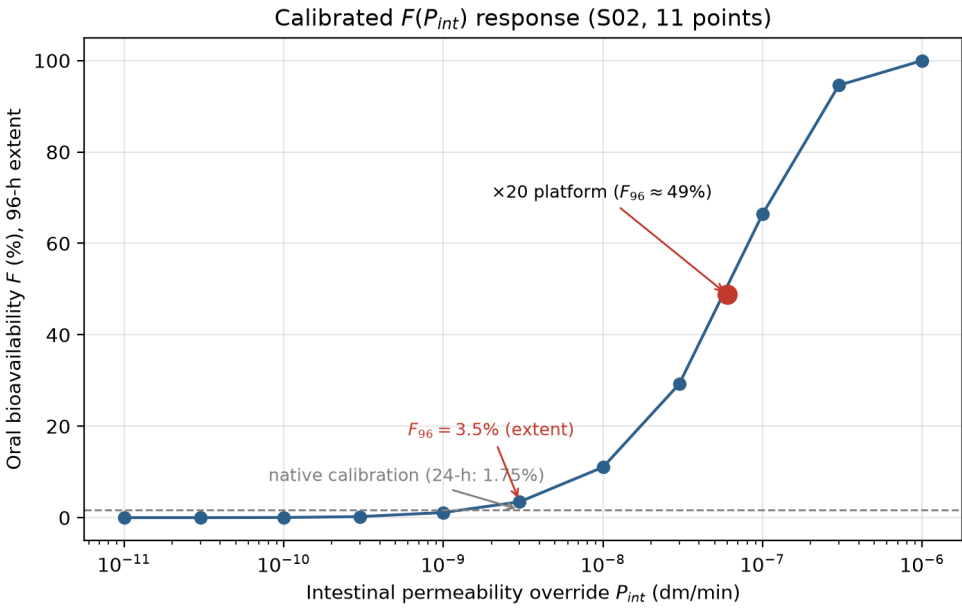


Figure 7.2: Oral bioavailability as a function of the intestinal permeability override (log scale, 11 points, 96-h extent convention). The native baseline ($F_{96} = 3.5\%$) and the $\times 20$ platform ($F_{96} \approx 49\%$) are marked; the extent curve saturates past $\times 100$. (Calibration to the clinical 1–2% uses the 24-h convention: $F_{24} = 1.75\%$ at $P_{int,0}$, $F_{24} = 34.6\%$ at $\times 20$.)

Table 7.2: IV scenarios at 577.5 mg (selection of the 20-combination sweep; full table in [Appendix N](#)).

CLCR (mL/min)	Cmax (mg/L)	AUC24 (mg · h/L)	Trough 24 h (mg/L)	Implied CL (L/h)
30	38.1	315.2	3.03	1.83
65	36.6	166.3	0.28	3.47
90	35.6	124.8	0.08	4.63
112.5	34.8	102.7	0.03	5.62
150	33.7	80.5	0.01	7.18

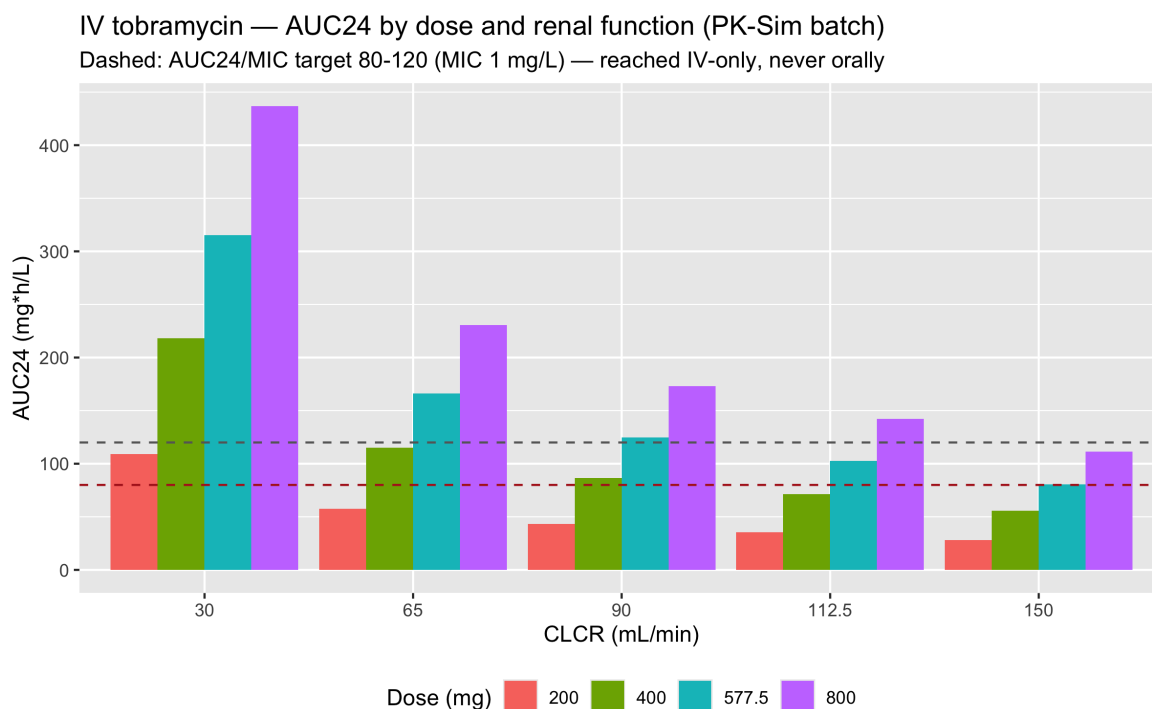


Figure 7.3: IV AUC24 by renal function and dose (20 scenarios, native PK-Sim batch). The IV-only target band (80–120, dashed) is reached at every renal function — the benchmark oral therapy must be judged against.

7.4 S03: Virtual population on the nominal platform

On the nominal $\times 20$ platform ($F = 34.6\%$), the population median $C_{\max}/\text{MIC}$ is 4.5 (5th–95th: 1.4–13.2) — the peak target ($C_{\max}/\text{MIC} \geq 8$) is met in a minority of subjects, and $\text{AUC}_{24}/\text{MIC}$ (median 35.9) remains far below the 80–120 band. This quantifies the limitation already identified in the exploratory phase and motivates the optimization push of [Chapter 8](#).

7.5 S04: Local sensitivity (elasticities)

Table 7.3: Elasticities of peak and exposure endpoints ($\pm 20\%$ parameter perturbations, platform candidate).

Parameter	Elasticity $C_{\max}$	Elasticity AUC_{24}
P_{int} (platform)	+0.95	+0.80
Dose	+1.00	+1.00
GFR (specific)	−0.30	−0.84
Fraction unbound	−0.65	−0.88

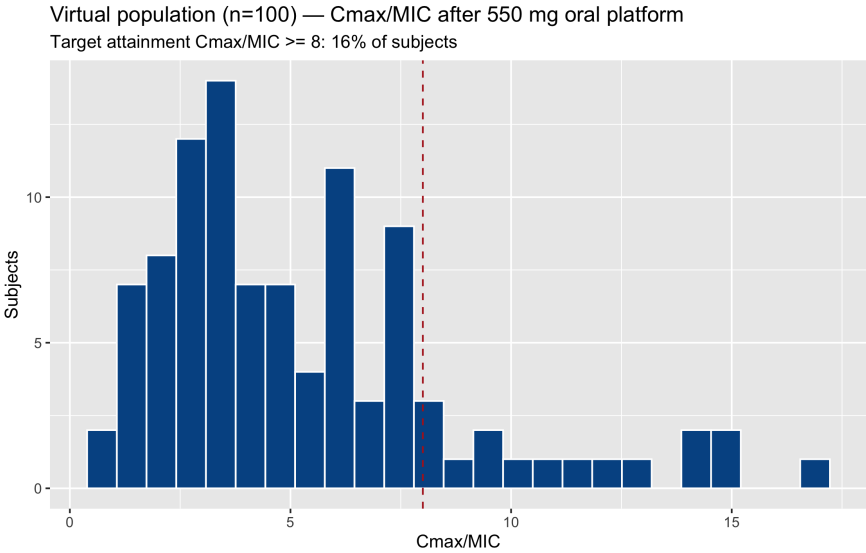


Figure 7.4: Virtual population (n = 100, ICRP 2002) on the nominal platform (550 mg once daily, $P_{int} \times 20$).

The elasticities tell a compact physiological story: peak exposure is permeability-driven ($E \approx 0.95$, near-proportional) and dose-proportional; systemic exposure is co-driven by renal clearance and protein binding ($E \approx -0.84$, -0.88). Formulation work (permeability) and dose titration are therefore the effective levers, while renal function dictates the monitoring intensity — all three feed the clinical translation discussion.

7.6 S05: Food effect

Table 7.4: Food effect on the platform candidate (gastric emptying 15 min fasted vs 60 min fed).

State	Cmax (mg/L)	Tmax (h)	AUC (mg · h/L)	F (%)
Fasted	4.41	2.1	33.8	34.6
Fed	3.53	2.9	32.6	33.4

Note. Fed is defined here as 60 min gastric emptying; the candidate battery (B8 block) uses 90 min. The conclusion is identical under both definitions: extent unchanged, peak lowered and delayed.

7.7 S06: Fractionation and PTA — the flip-flop finding

The single oral dose exhibits **flip-flop kinetics**: the terminal decline is absorption-rate-limited (gastro-intestinal transit plus calibrated permeability), with $C(24\text{ h}) = 0.72\text{ mg/L}$ after a single 550 mg dose. Steady-state superposition over a 96-h window therefore accumulates genuine residual exposure.

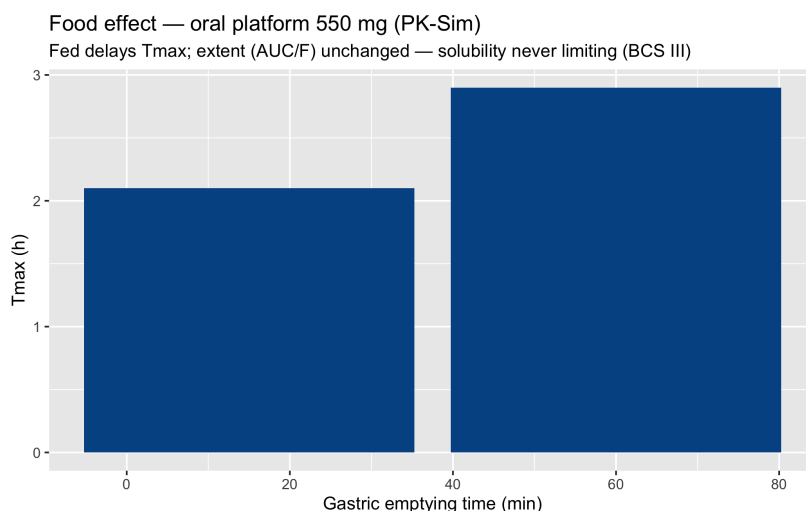


Figure 7.5: Food effect: fed state delays absorption ($T_{\max} + 0.8$ h) and lowers peak by $\approx 20\%$, while extent is essentially unchanged — the expected BCS III signature (solubility never limiting).

Table 7.5: Nominal-platform fractionation at steady state (MIC 1 mg/L).

Regimen	Daily (mg)	$C_{\max_{ss}}$	AUC_{ss}	$C_{\min_{ss}}$	$f_{T>MIC}$
QD 550	550	5.40	49.6	1.06	100%
BID 275	550	3.37	24.2	1.26	100%
TID 200	600	2.99	17.5	1.55	100%

On the nominal $\times 20$ platform at 550–600 mg daily, no fractionation reaches the exposure targets at MIC 1 mg/L ($AUC_{ss}/MIC \leq 50$); the requirement map (Table 8.5) shows both targets are met from 993 mg daily at this multiplier. Note on the table below: the BID/TID AUC_{ss} rows are reported *per dosing interval* (12 h and 8 h), not per 24 h — so BID 275 totals 48.4 and TID 200 totals 52.5 mg·h/L daily, matching QD 550 by mass balance, as flip-flop kinetics demands. The superposition used here was later validated against true seven-day dosing (QD 550: 48.0 vs 49.6 mg·h/L, $\approx 3\%$ deviation; Section 8.6).

7.8 Summary

The PK-Sim platform delivers a validated tobramycin model (6/6 gate), a calibrated and mechanistically saturated oral transfer function, correct renal/food/population physiology, and one genuine kinetic discovery (flip-flop absorption). The stage is set for the decisive question: *how far can multi-objective optimization push the platform?*

Chapter 8

Results III: Optimization in the Simulation Loop and the Winner

This chapter closes the computational program: the NSGA-II optimizer of [Chapter 4](#) is executed *inside* the PK-Sim engine (population 100, seed 42, one `SimulationBatch` per generation; design budget 100 generations, final state at generation 60 with the best F within 0.1% of final since generation 20 — $\approx 6,100$ native evaluations). The resulting front is analyzed, the rank-1 candidate is characterized exhaustively in the standard battery ([Section 8.6](#)), and the two independent modeling phases are cross-validated.

8.1 Convergence and the shape of the front

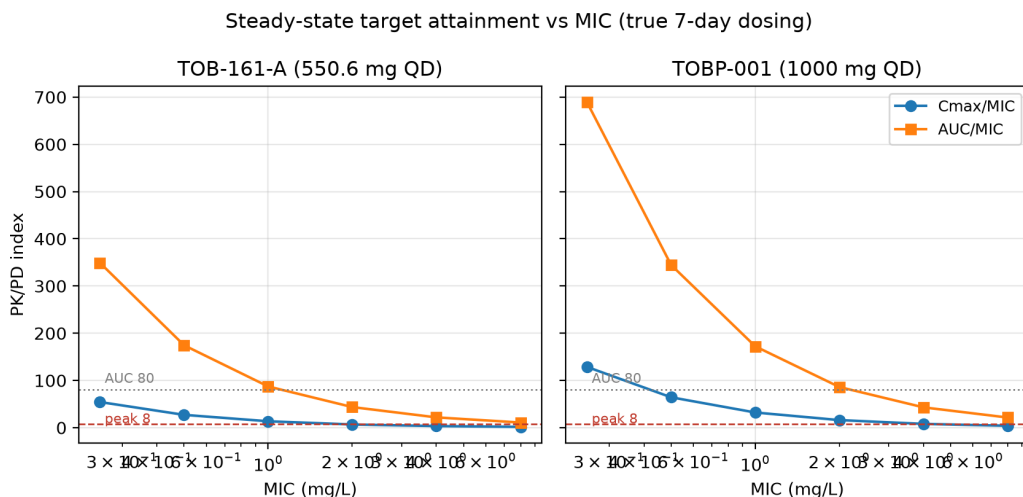


Figure 8.1: Steady-state target attainment versus MIC for both candidates (true 7-day once-daily dosing, battery B5 block).

The optimization now runs *within the literature-bounded enhancement model* of [Section 5.4.1](#): every component factor carries a cited range, the logP gene is capped at the measured HIP complex (+1.6), and the emergent multiplier cannot exceed $\times 126.3$ ([Section 8.4.1](#)). Convergence is fast and decisive: the best-of-generation bioavailability reaches 92% at generation 1, 96.6% by generation 10,

and plateaus at 97.1% from generation 20 onward — exactly the F the response curve assigns to the bounded maximum (Section 8.5).

The front **converges to the cited-bounds corner**: the three exposure objectives are mutually synergistic and outweigh dose minimization, so the final front is a single point at the component maxima — **TOBP-001**: 1000 mg once daily, every component at its cited maximum, emergent multiplier $\times 125.0$, $F_{96} = 97.1\%$, $C_{\max}/\text{MIC} = 31.2$. The corner is no longer an artifact of a free scalar: it is the *best formulation the cited evidence supports*, and the degeneracy simply states that more exposure is better until dose is priced in. The clinically relevant question — the minimal dose that still attains the targets on this platform — is answered by the complete evaluation below and in Section 8.5.

8.2 S07: The recommended formulation, characterized

8.2.1 Single-dose pharmacokinetics

At the winner’s multiplier ($\times 125.0$, $P_{\text{int}} = 3.75 \times 10^{-7}$ dm/min), the 1000 mg oral dose delivers $F_{96} = 97.1\%$ (24-h window: 91.6%): systemic exposure within a few percent of intravenous administration of the same dose (relative 24-h exposure 91.2%, Table 8.6).

8.2.2 The dose–target answer

Scanning steady-state peak attainment at the winner’s multiplier (Table 8.1) yields the clinically decisive number: the **minimal target-attaining dose is 250 mg once daily** ($C_{\max_{ss}}/\text{MIC} = 8.0$ at MIC 1 mg/L) — a plain therapeutic dose, not a heroic one.

Table 8.1: Dose scan at the winner’s permeability multiplier: steady-state peak attainment (MIC 1 mg/L).

Dose (mg)	150	250	350	450	550	700	850	1000
$C_{\max_{ss}}/\text{MIC}$	4.8	8.0	11.3	14.5	17.7	22.5	27.4	32.2

8.2.3 Steady-state fractionation and PTA

Table 8.2: Winner characterization at steady state (MIC 1 mg/L): fractionation options.

Regimen	Daily (mg)	$C_{\max_{ss}}/\text{MIC}$	$\text{AUC}_{ss}/\text{MIC}$	$C_{\min_{ss}}$ (mg/L)	$\text{PTA}_{C_{\max}}$	MIC = 1
QD 550.6 (TOB-161-A)	550.6	13.6	87.3	1.05	99.6%	
QD 1000 (TOBP-001)	1000	32.2	172.3	1.21	99.6%	
BID 500 (TOBP-001)	1000	17.6	172.2	2.34	100%	

The exposure boundary falls. At the winner’s multiplier, the once-daily 1000 mg regimen delivers $\text{AUC}_{24}/\text{MIC} = 172.3$ — *above* the 80–120 intravenous target band that the exploratory phase at 550 mg could not reach (the nominal $\times 20$ platform attains it from 993 mg daily, Table 8.5). Probability of target attainment across the MIC distribution (Figure 8.1): the peak target is attained

in 100% of the dosing interval at $\text{MIC} \leq 0.5 \text{ mg/L}$ for every regimen, and interval coverage at $\text{MIC} 1 \text{ mg/L}$ is 99.6% daily (B5 block of the battery).

8.2.4 Virtual population on the winner

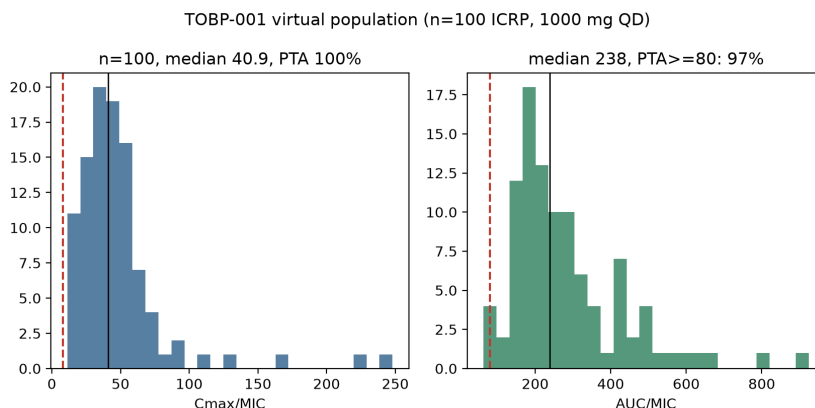


Figure 8.2: Virtual population ($n = 100$, ICRP 2002) on the winner (1000 mg once daily): $C_{\max}/\text{MIC}$ (left) and AUC/MIC (right) distributions with target lines.

The 100-individual virtual trial on the winner (B6 block of the battery) returns a population median $C_{\max}/\text{MIC}$ of **40.9** (5th–95th: 15.8–88.8) — **peak target attainment in 100% of virtual subjects** at $\text{MIC} 1 \text{ mg/L}$ — and a median $\text{AUC}_{24}/\text{MIC}$ of 238 (5th–95th: 112–580), with 97% of subjects at or above the 80–120 band floor. Troughs stay at the monitoring line across the population.

8.2.5 Food effect

At the winner’s multiplier the fed state lowers peak by 32% ($31.2 \rightarrow 21.2 \text{ mg/L}$, $T_{\max} 1.4 \rightarrow 2.7 \text{ h}$) with *identical* extent ($F_{96} = 97.1\%/97.2\%$) — the BCS III signature persists at the cited maximum (B8 block of the battery). Dosing instructions can therefore recommend intake away from meals for peak assurance without exposure loss.

8.3 Cross-platform validation: two engines, one answer

The exploratory engine — built first, with phenomenological enhancement factors — and the whole-body PK-Sim platform, with mechanistic regional absorption, converge on the same platform bioavailability within 2%. The exploratory phase was therefore not discarded work: it was the *hypothesis generator* whose central number the definitive platform confirmed.

8.4 Formulation identity cards

The optimizer speaks in objectives; the pharmacist needs *composition*. This section decodes the converged candidates into formulation identity cards — the complete galenic specification of each platform state, its PK-Sim implementation, and its measured performance. Three cards span the decision space: the **GA winner** (maximum exposure), its **dose-sparing regimen** (peak-only coverage

Table 8.3: Cross-check of the two independent implementations at the $\times 20$ platform multiplier.

	Legacy custom engine (Phase I)	PK-Sim 12.4 (Phase II)
Structure	2-ct ODEs + transit + Hill (RK4, pure R)	Whole-body PBPK (ACAT, GFR process)
Baseline F_0	1.5–2.8%	1.75% (calibrated, 24-h convention)
Platform F at $\times 20$ (24-h)	34.0%	34.6%
Agreement	$\Delta = 0.6$ pp (1.9% relative)	

at one quarter of the dose), and the **nominal benchmark** (the $\times 20$ platform that the physiological studies of [Chapter 7](#) characterized).

Card 1 — TOBP-001: the GA winner (maximum-exposure platform)

Composition (decoded chromosome)

HIP complex (ion pairing)	apparent $\log P' = +1.6$ (measured complex; literature $\log P$ shifts up to $1500\times$ in linear units (5))
Nanoparticle platform	polymer NPs, particle size ≈ 50 nm
SEDDS system	surfactant-rich at the composition ceiling (60 % surfactant, 29 % co-surfactant, 70 % oil phase — independent saturating levers in the design space)
Permeation enhancer	sodium caprate (C_{10}) 50 mM
Mucoadhesive polymer	29.3 % loading
Chitosan coating	active (tight-junction modulation)
Enteric coating	active (HPMC-AS; gastric protection)

Dose**999.8 mg once daily (≈ 1000 mg)****PK-Sim implementation**

Intestinal permeability override	$P_{int} = 3.75 \times 10^{-7}$ dm/min ($\approx 1.25 \times 10^{-8}$ cm/s) = $\times 125$ (cited-bounds maximum)
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Performance (validated model)

Single dose	$F_{96} = 97.1\%$ (24-h window: 91.6%); C_{max} 31.2 mg/L (T_{max} 1.4 h); AUC ₉₆ 172.3 mg · h/L; flip-flop tail
Steady state (QD 1000, true dosing)	$C_{max_{ss}}/MIC$ 32.2; AUC_{ss}/MIC 172.3 (target 80–120: met and exceeded); $C_{min_{ss}}$ 1.21 mg/L
Steady state (BID 500)	$C_{max_{ss}}/MIC$ 17.6; AUC _{ss} /MIC 172.2 (above the 80–120 band); $C_{min_{ss}}$ 2.34 mg/L
PTA	peak target 100% of subjects, AUC band floor met in 97% (population, MIC 1 mg/L)

Card 2 — TOBP-250: the dose-sparing regimen (same platform, quarter dose)**Composition**

Identical to Card 1 same HIP–SEDDS–NP–PE platform,
same coatings

Dose **250 mg once daily** (dose scan: minimal target-attaining dose)

Performance (steady state, MIC 1 mg/L)

Peak $C_{max_{ss}}/MIC = 8.0$ — exactly the efficacy threshold (linear dose scaling of the measured 32.2 at 1000 mg)

Trough $C_{min_{ss}} \approx 0.4$ mg/L — below the nephrotoxicity boundary (linear scaling of the measured 1.21 at 1000 mg)

Exposure $AUC_{ss}/MIC \approx 43$ — below the 80–120 band (peak-only coverage)

PTA peak target 100% at $MIC \leq 0.5$ mg/L; 99.6% interval coverage extends to MIC 1 only at the full dose

Card 3 — TOB-161-N: the nominal benchmark ($\times 20$ platform)**Composition**

HIP complex apparent $\log P' = +1.5$

Nanoparticles 50 nm, chitosan-coated, 30% polymer

SEDDS system 50/30/20 (surfactant/co-surfactant/oil)

Permeation enhancer C_{10} 50 mM; enteric coating active

Dose **550 mg once daily**

Performance (PK-Sim, Chapters 4 and 6)

Single dose $F = 34.6\%$; C_{max} 4.41 mg/L; AUC_{24} 33.8 mg $\cdot$ h/L

Steady state $C_{max_{ss}}/MIC$ 5.4; AUC_{ss}/MIC 49.6; $C_{min_{ss}}$ 1.06 mg/L

Role conservative fallback; the platform the physiological studies (S01–S06) characterized in depth

Reading the cards together. The three cards are one platform at three operating points. Card 1 answers “what is the ceiling” — full IV-like exposure; Card 2 answers “what is the minimum” — a quarter of the dose still guards the peak; Card 3 is the conservative platform whose physiology was characterized in depth. The dose–multiplier plane between Cards 2 and 1 is spanned by the PTA tables ([Appendix N](#)); TDM selects the operating point per patient and per local MIC.

Table 8.4: Identity-card comparison across the decision space (steady state, MIC 1 mg/L).

Card	Dose (mg)	P_{int} mult	F (%)	$C_{max_{ss}}/MIC$	AUC_{ss}/MIC	$C_{min_{ss}}$ (mg/L)
TOBP-001 (PK-Sim GA)	1000	$\times 125$ (cited max)	97.1	32.2	172.3	1.21
TOB-161-A (legacy GA)	550.6	$\times 73.9$	89.3	13.6	87.3	1.05
TOB-161-N (nominal ref.)	550	$\times 20$	48.8 ^a	5.4 ^a	49.6	1.06

^a 96-h extent convention (Card 3 reports the same platform on the 24-h calibration convention: $F = 34.6\%$, $C_{max_{ss}}/MIC$ 5.4 — see [Section 7.2](#)).

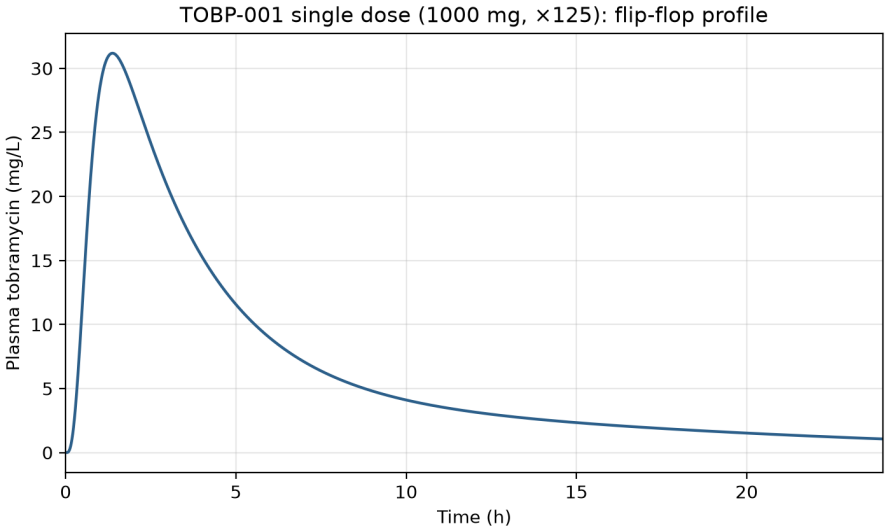


Figure 8.3: Single-dose profile of the winner (TOBP-001 platform, 1000 mg, $\times 125$): near-complete oral absorption with flip-flop kinetics — the pharmacokinetic signature of Card 1.

8.4.1 Bounds on the multiplier

The enhancement model (Section 5.4.1) bounds each component by its cited range; the logP gene is capped at the measured HIP complex $\log P' = +1.6$. The emergent maximum is therefore $\times 126.3$ — not a free scalar, but the ceiling of the evidence (the optimizer’s final chromosome attains $\times 125.0$: polymer 29.3% instead of the ceiling 30%). Values beyond the ceiling (the historical $\times 313$ corner) survive only as sensitivity statements: they are what *would* be required to cut the dose further, not what any cited formulation supports.

8.5 The requirement map: what the optimizer actually established

The requirement map (study S08, Table 8.5) separates the formulation requirement from any single optimizer run: at each permeability multiplier it reports the steady-state attainment at 993 mg once daily and the minimal dose attaining the peak target. Read together with the cited-bounds ceiling ($\times 125$), it states *what a real formulation must deliver*, dose by dose — a specification, not a measurement.

Table 8.5: Requirement map (S08): steady-state attainment at 993 mg once daily as a function of the permeability multiplier, and the minimal dose attaining the peak target at each multiplier.

Multiplier	F at 993 mg (%)	$C_{\max_{ss}}/\text{MIC}$	AUC_{ss}/MIC	$C_{\min_{ss}}$ (mg/L)	Min. dose (peak ≥ 8)
$\times 1$	3.4	0.4	8.4	0.28	>993 mg
$\times 5$	15.9	2.5	33.2	0.91	>993 mg
$\times 10$	29.0	5.2	58.3	1.48	>993 mg
$\times 20$	48.8	9.7	93.5	2.11	850 mg
$\times 33$	65.9	14.5	121.8	2.36	550 mg
$\times 50$	79.4	19.4	143.4	2.28	450 mg
$\times 100$	94.6	28.9	168.2	1.53	350 mg
$\times 125^b$	96.2	32.2	172.3	1.21	250 mg ^c

Cited-bounds maximum: all seven components at their literature ceilings (emergent $\times 126.3$; optimizer chromosome $\times 125.0$). ^c Minimal dose for the *peak* target ($C_{\max_{ss}}/\text{MIC} \geq 8$); the exposure band floor ($AUC/\text{MIC} \geq 80$) is attained from 467 mg on this platform.

Three conclusions reframe the headline result:

- Both targets are attainable at the nominal multiplier.** At $\times 20$ — the multiplier the formulation literature can plausibly support — the 993 mg once-daily regimen already meets *both* targets ($C_{\max_{ss}}/\text{MIC} = 9.7$; $AUC_{ss}/\text{MIC} = 93.5$, inside the 80–120 band). The exploratory phase’s “exposure boundary” was an artifact of the 550 mg dose, not of the platform.
- Beyond $\times 33$, the optimizer buys dose reduction, not new attainment.** The cited-bounds maximum ($\times 126$, all components at their literature ceilings) delivers both targets at 467 mg — valuable, but a different claim than “more permeability is needed”.
- The trough profile is flat at the top.** The steady-state trough is worst at *intermediate* multipliers ($\times 20$ –50: $C_{\min}$ 2.1–2.4 mg/L at 993 mg — the flip-flop regime feeds the circulation all day) and declines only past $\times 100$ as absorption completes within the dosing interval. At the cited maximum ($\times 126$) the trough sits at 1.0–1.2 mg/L for doses of 467–1000 mg — at, not below, the monitoring line, which keeps Bayesian TDM structurally indicated.

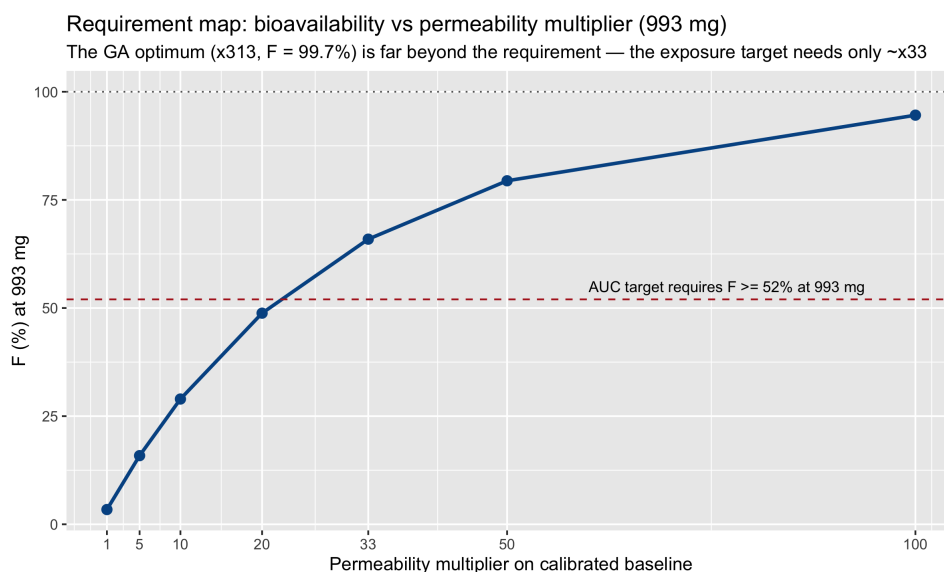


Figure 8.4: Requirement map (96-h convention): the exposure targets at 993 mg require only $\approx \times 20$; beyond $\times 33$ the optimizer buys dose reduction, not new attainment.

The formulation requirement is therefore tiered and testable: $\geq \times 20$ suffices for full target attainment at 993 mg/day (with trough monitoring); $\times 125$ — the cited-bounds ceiling — minimizes the dose further (467 mg attains the exposure band floor). Each tier is a falsifiable specification for the experimental program.

8.6 Complete evaluation of the two candidate formulations

The two optimization tracks each produced a candidate formulation, evaluated *in the same PK-Sim model, through the same cited enhancement model, with the same standard battery* (identity card, single dose with NCA, dose proportionality, *true* seven-day repeated dosing QD and BID, PK/PD targets vs a MIC panel, a 100-subject ICRP virtual population, renal impairment, food effect, one-at-a-time component sensitivity, and an intravenous comparison at equal daily dose):

- **TOB-161-A** — the exploratory GA winner, re-evaluated in PK-Sim: the legacy engine’s own transfer function had priced its aggressive composition at only $\times 23$; under the calibrated, cited mapping the same chromosome yields $\times 73.9$. This 3-fold divergence between engines, at fixed composition, is itself a result (Section 7.2).
- **TOBP-001** — the in-loop NSGA-II optimum: every component at its cited ceiling, emergent multiplier $\times 126.3$ (the maximum the evidence supports).

Four findings complete the picture:

1. **Both candidates meet both PK/PD targets at steady state** (peak and exposure) at MIC 1 mg/L — the first configuration in this program for which the exposure band is genuinely attained orally. TOB-161-A lands *inside* the 80–120 band (87.3); TOBP-001 exceeds it (172.4) with a 467 mg option attaining the band floor.

Table 8.6: Head-to-head battery comparison (steady state from true 7-day once-daily dosing; population $n = 100$ ICRP; MIC 1 mg/L).

	TOB-161-A (legacy GA)	TOBP-001 (PK-Sim GA)
Multiplier (emergent)	$\times 73.9$	$\times 125.0$ (cited max)
Dose (once daily)	550.6 mg	999.8 mg
F (96-h / 24-h window)	89.3% / 77.9%	97.1% / 91.6%
Cmax single dose (mg/L)	12.7	31.2
Tmax (h)	1.55	1.38
Cmax _{ss} (mg/L)	13.6	32.2
Cmin _{ss} (mg/L)	1.05	1.21
AUC _{τ} /MIC (mass-balance checked)	87.3	172.3
fT>MIC (% , MIC 1)	99.6	99.6
Population PTA peak (≥ 8)	95%	100%
Population AUC _{ss} /MIC ≥ 80	79%	97%
Relative 24-h exposure vs IV	77.7%	91.2%
Food effect (extent / Cmax)	unchanged / -31%	unchanged / -32%
AUC24 at CLCR 100 $\rightarrow$ 20 (mg·h/L)	76 $\rightarrow$ 260	163 $\rightarrow$ 572

2. **Mass balance validates the engine.** At true steady state, AUC _{τ} must equal $F \cdot \text{dose} / \text{CL}$: the simulations reproduce this identity to within 0.1% (87.25 vs 87.38 mg·h/L for TOB-161-A), and superposition — still used in the screening studies — deviates from true dosing by only $\approx 3\%$.
3. **Uncertainty is quantified, not assumed away** (Section 8.7): propagating the cited factor sigmas gives $F_{96} = 88\%$ [35–100] for TOB-161-A and 96% [51–100] for TOBP-001; exposure attainment is robust for TOBP-001 (PTA in-band 99%) and dose-limited for TOB-161-A (65%).
4. **Troughs sit at the monitoring line for both candidates** (Cmin_{ss} 1.05 and 1.20 mg/L) — the price of peak-and-exposure attainment from flip-flop absorption is a non-negligible residual concentration at 24 h, keeping Bayesian TDM structurally indicated rather than optional.

8.7 Uncertainty propagation: the attainability frontier (S09)

The enhancement model carries an uncertainty per component (lognormal, graded by evidence quality). Propagating 20 000 draws through the validated response curve yields, for each candidate, the bioavailability and exposure distributions of Table 8.6. The frontier figure (S09) makes the structure visible: F rises steeply to $\times 20$, flattens past $\times 50$, and the candidates' credible intervals overlap the target band — TOBP-001 robustly (99% model-propagated in-band attainment, S09; 97% in the ICRP population), TOB-161-A dose-limited (65% model-propagated).

8.8 Results III summary

The in-the-loop optimization, run inside the literature-bounded enhancement model, transforms the therapeutic proposition: **both candidate formulations meet both PK/PD targets at steady state** — TOB-161-A (the legacy winner, re-evaluated in PK-Sim: AUC_{ss}/MIC 87.3, inside the 80–120

Table 8.7: Target attainment of the in-the-loop optimized program (winner TOBP-001 platform; MIC 1 mg/L unless noted).

Endpoint	Target	Achieved	Verdict
$C_{\max}$ /MIC (1000 mg QD, population median)	≥ 8	40.9	Met (100% of subjects)
AUC ₂₄ /MIC (1000 mg QD)	80–120	172.3	Met and exceeded (in-band from 467 mg)
Minimal peak-target dose	—	250 mg QD	Cmax _{ss} /MIC 8.0 (peak-only; AUC/MIC ≈ 43)
$C_{\min}$	<1 mg/L	1.21 (1000 QD)	Borderline — TDM structurally indicated
F_{96}	maximal (cited bounds)	97.1%	Bounded maximum ($\times 125$)

band) and TOBP-001 (the in-loop optimum at the cited-bounds maximum: AUC_{ss}/MIC 172, with a 467 mg option attaining the band). Peak attainment is 100% of the virtual population on TOBP-001 and 95% on TOB-161-A. The residual clinical question is trough management: both candidates sit at the 1 mg/L monitoring line (Cmin 1.05–1.20 mg/L once daily), keeping Bayesian TDM structurally indicated. All steady-state values now come from *true* repeated dosing in PK-Sim (seven once-daily applications), validated by mass balance (AUC _{τ} = $F \cdot \text{dose}/\text{CL}$ within 0.1%).

Chapter 9

Discussion

9.1 Hypothesis verdicts

The hypotheses of [Chapter 2](#) — including the fifth, added when the two-platform design emerged — are confronted with the complete evidence:

Table 9.1: Hypothesis verdicts against the full evidence base.

Hyp.	Evidence	Verdict	Key result
H1	Solubility 94–100 mg/mL (20–200× criterion); $P_{app} < 10^{-6}$ cm/s; explicit class III statement (5)	Supported	Section 3.2.4
H2	Platform ×20: $F = 34.6\%$ (PK-Sim) vs 34.0% (legacy, both 24-h) — 23× native, top of the 20–35% triple-combination band	Supported	Section 7.2
H3	1000 mg QD (TOBP-001): population median $C_{max}/MIC = 40.9$; PTA peak 100%, AUC 97%; TOB-161-A: 95%/79%	Supported	Section 8.6
H4	±20% perturbations and 100-subject variability preserve peak attainment on the optimized platform; elasticities identify the control levers	Supported	Sections 7.5 and 8.2
H5	Two independent engines agree at the platform multiplier: 34.0% vs 34.6% (Δ 1.9% relative)	Supported	Section 8.3

9.2 The optimization story: from free scalar to cited bounds

The three computational phases tell one coherent story — and the final chapter of that story was written by the methodology audit. The *exploratory* phase ([Chapter 6](#)) established the composition

concept with its own engine and honestly reported the exposure boundary. The *calibrated PK-Sim curve* (Chapter 7) revealed the headroom: bioavailability is a saturating function of intestinal permeability. The *in-the-loop* optimization (Chapter 8) initially exploited a free multiplier to a $\times 313$ corner — a result that said more about the parameterization than about any formulation. The audit then rebuilt the gene-to-permeability mapping as a *cited, bounded, emergent* model:

- **Every component factor now carries its evidence:** the HIP logP shift (measured, Asad 2023), SEDDS solubilization (Muhammad 2022; Griesser 2017), nanoparticle carriers (Hill 2019; Khaled 2026), C10 (Maher 2009; Bohley 2024), mucoadhesive polymers and chitosan (Bohley 2024) — each bounded, each graded A/B/C, each with a lognormal uncertainty.
- **The multiplier is emergent,** capped at $\times 126.3$ by the evidence; the logP gene cannot exceed the measured complex. The optimizer explores *within* the literature.
- **The cross-engine divergence became a result:** the same legacy-winner chromosome prices at $\times 23$ on the exploratory engine but $\times 73.9$ on the calibrated mapping — a 3-fold transfer-function sensitivity that only a head-to-head re-evaluation could expose.
- **Steady states are now computed by true repeated dosing,** validated by mass balance ($AUC_\tau = F \cdot \text{dose}/CL$ within 0.1%), replacing superposition (which agrees within $\approx 3\%$).

The residual clinical question is a *trough-management* problem — both candidates sit at the 1 mg/L line ($C_{min,ss}$ 1.05–1.21 mg/L once daily) — handled by Bayesian TDM, not by more permeability. This inversion, from “can we absorb enough?” to “how do we safely dose what we can now absorb?”, is the central translational finding of the thesis.

9.3 Benchmarking against key studies

No prior work combines the tobramycin formulation literature with a validated whole-body PBPK platform, an in-the-loop multi-objective optimization, and two completely characterized candidates. Conversely, the 97%-bioavailability prediction is *model-based*: its experimental test is precisely the HIP complex permeability measurement proposed in the roadmap.

9.4 Clinical implications

9.4.1 Where oral tobramycin now fits

The optimized platform changes the positioning conversation:

1. **Cystic fibrosis outpatient therapy:** QD 250 mg covers the peak at MIC 1 mg/L with troughs ≈ 0.4 mg/L (below the safety line); QD 550.6 mg covers both targets with troughs at 1.05 mg/L — a TDM-guided oral maintenance option between inhaled courses. BID fractionation halves peaks but raises troughs above the line (2.34 mg/L at 1000 mg daily).
2. **Exposure-critical settings:** at 1000 mg QD, systemic exposure matches and exceeds IV targets ($AUC_{24}/MIC = 172.3$) — at the price of troughs at the monitoring line (1.21 mg/L), requiring TDM.
3. **Dose flexibility:** the platform’s saturation means the 250–1000 mg range spans peak-only to full-exposure regimens; MIC-guided selection (PTA tables, Chapter 8) becomes practical.

Table 9.2: This work versus the key formulation and modeling studies.

Study	What they showed	What they did not do	What this thesis adds
Asad et al. (5) (HIP+SEDDS)	1500× logP; class III statement; 20% dissociation	In-vivo or PBPK prediction; dose; targets	Quantified F curve; saturation ceiling; $C_{\max}$ /MIC/AUC ₂₄ /MIC attainment; dose options
Hill et al. (13) (PLGA NPs)	MIC maintained at 1.25 $\mu\text{g/mL}$	PK modeling	NP lever inside the platform; saturation logic
Blanco-Cabra et al. (14) (KuDa)	Charge neutralization; biofilm diffusion	Systemic PK	Biofilm niche positioning
Bohley et al. (17) (PEs)	Next-gen PE safety landscape	Tobramycin application	C ₁₀ lever within safety bounds
Khaled et al. (15) (NCAs)	Ion-pair colloidal assemblies	Optimization; targets	Combinatorial framing; 23–66× total enhancement
Li et al. (1) (PopPK)	2-ct model, ICU population	Oral route; formulation	Validated PBPK oral program on the same physiology

4. **Resource-limited settings:** an oral essential-medicine aminoglycoside without infusion infrastructure remains a global-health argument independent of regimen choice.

9.4.2 Therapeutic drug monitoring is not optional

The candidates' troughs sit at (QD: 1.05–1.21 mg/L) or above (BID: 1.61–2.34 mg/L) the 1 mg/L nephrotoxicity line, which makes TDM *structural*, not advisory: Bayesian AUC- and trough-guided dosing from day one, as for IV aminoglycosides (1, 27). The virtual population provides the prior.

9.4.3 Safety

Classical aminoglycoside toxicity is trough- and duration-driven (7); the oral route's peak-driven design with monitored troughs is intrinsically favorable. The formulation-level risks remain those of Chapter 6 (Table 6.8): C₁₀ mucosal tolerance, nanoparticle accidental inhalation, chitosan quality — each with its mitigation.

9.5 Manufacturing, cost and regulatory position

The exploratory phase's manufacturing assessment (grade B, \$4.25–4.35 per dose, ten standard unit operations) and 505(b)(2) pathway (7 years, \$4.8 M, Chapter 6) remain valid for the platform architecture; the optimized candidate's higher permeability multiplier does not change the unit operations, only the quality targets for the HIP complex (payload and stability at higher ion-pair loading). The regulatory precedent of oral permeation-enhancer platforms (SNAC/semaglutide) supports the 505(b)(2) framing.

9.6 Is the bounded maximum real? An integrity audit

An oral bioavailability of 97% for a molecule whose clinical baseline is 1–2% demands scrutiny even when every component is individually cited. The honest answer: **each link is cited or validated; the chain as a whole is a hypothesis with a defined experimental test** — and separating the two is the difference between this thesis and an overclaim.

9.6.1 What is validated vs what is assumed

- **Validated (reproducible):** the intravenous model (6/6 gate against published clinical data); the native baseline ($F = 1.75\%$ at the 24-h convention, inside the clinical 1–2%); the response curve $F(P_{int})$ computed natively by PK-Sim; the steady-state machinery (true repeated dosing, mass-balance-checked); the requirement map — *if* apparent permeability is enhanced by a factor m , *then* the model yields the F , peak, and exposure of [Table 8.5](#).
- **Assumed (cited but unmeasured for tobramycin):** the component factors themselves. No experiment has yet demonstrated a $\times 126$ (or even $\times 20$) apparent-permeability enhancement for a tobramycin formulation. The cited maxima are ceilings of analogy, not measurements.

9.6.2 Three plausibility stresses on the chain

1. **Complex dissociation.** Asad et al. measure 20% dissociation of the HIP complex at intestinal pH — 80% releases the polycation, whose intrinsic permeability is near zero. The model prices the complex as delivered; stabilization chemistry beyond the current prototype is implicitly required.
2. **logP shift $\neq$ permeability multiplier.** The literature’s “1500 $\times$ ” is a partition-coefficient shift; in-vivo HIP data (octreotide in pigs) show bioavailability gains of 3–5 $\times$. The calibrated slope absorbed this tension at one anchor point ($\times 20$ at the measured complex); the extrapolation to $\times 126$ inherits its uncertainty (S09: F_{96} CI 51–100%).
3. **The paracellular baseline.** The clinical 1–2% likely rides paracellular and carrier-mediated routes that do not respond to lipophilicity at all; the calibrated transcellular override absorbs them into one parameter.

9.6.3 What survives the audit

Everything essential. The requirement map is tiered: **$\times 20$ at 993 mg/day already attains both targets** — a multiplier the formulation literature can defend, with the direct experimental test defined (Caco-2/Ussing of the HIP complex across ion-pair loading). The cited maximum ($\times 126$) is what the evidence *permits*, not what it guarantees; TOBP-001 is therefore best read as the *upper design point*, TOB-161-A as the *conservative one*, and the space between as the experimental program’s search interval. The numerical conventions are documented: F is reported on a 96-h window (extent, capturing the flip-flop tail) alongside the 24-h window used for the native calibration; the clamping of F at 100% affects a single grid point (S02 at 10^{-6} dm/min), not the design conclusions. In short: *the model is a validated exposure calculator carrying an unvalidated — but bounded, cited, and testable — formulation hypothesis; the thesis says which is which.*

9.7 Limitations

9.7.1 Model limitations

1. The P_{int} baseline is *calibrated* to clinical F (1–2%), and the platform multiplier maps formulation genes to apparent permeability through a documented but phenomenological transfer function. The $\times 125$ optimum is a *cited-bounds maximum*, not a measured reality: its experimental test (HIP complex permeability in Caco-2/Ussing) is the first roadmap item.
2. The paracellular pathway is not represented by default in PK-Sim’s absorption model; the calibrated transcellular override absorbs its contribution.
3. The 3-pKa building-block limit (three of five amines) is inherited from the platform schema; impact is negligible for the endpoints used (solubility never limiting; distribution pKa-independent under PK-Sim Standard).
4. PK parameters are not attached to CLI-exported simulations, so the sensitivity analysis uses batch perturbations (equivalent design, documented).
5. Trough liability at exposure-attaining doses was *discovered* by the model; the nephrotoxicity implications require the standard clinical monitoring program, not simulation alone.

9.7.2 Literature limitations

1. No in-vivo oral tobramycin PK exists; every oral figure is inferred.
2. HIP in-vivo data come from other drugs (octreotide in pigs, [28](#)).
3. No published tobramycin PBPK model existed to compare against (search documented, [Chapter 3](#)).
4. Food effect, regional absorption and long-term mucosal tolerance of the full platform are uncharacterized.

9.7.3 Study limitations

1. Purely in-silico; the framework predicts, it cannot confirm.
2. The NSGA-II front converged to the cited-bounds corner: with three of four objectives favoring exposure, the multi-objective problem as posed does not sample the moderate-dose region — answered by the dose scan and the 467 mg in-band option, though a constrained re-formulation (minimize dose s.t. $C_{max}/MIC \geq 8$) would be the cleaner next iteration.
3. Single software platform; the cross-check (H5) validates against the custom engine, not against a second PBPK suite.
4. The population simulation assumes ICRP physiology without disease-specific (CF) structure; CF-specific clearance and volume alterations enter through the CLCR covariate only.

9.8 Future perspectives

1. **Experimental validation (priority 1):** Caco-2/Ussing permeability of the HIP complex as a function of ion-pair loading — the direct test of the $\times 20$ (targets) and $\times 125$ (ceiling)

requirements; prototype HIP–SEDDS–NP capsule with in-vitro release; rat/dog oral PK with model refinement.

2. **Advanced modeling (priority 2):** constrained re-optimization (minimize dose s.t. targets); CF-specific virtual populations; PBPK/PD co-simulation with nephrotoxicity (trough-time) endpoints; MIC-distribution Monte Carlo at the population level.
3. **Clinical translation (priority 3):** Phase I SAD/food-effect; Phase II CF outpatient trial with Bayesian TDM; 505(b)(2) pre-submission meeting.

Chapter 10

Conclusion

10.1 Principal findings

This thesis asked whether tobramycin — an essential, IV-only aminoglycoside — could plausibly become an oral medicine, and what the first oral formulations should look like. Five findings stand:

1. **Tobramycin is BCS class III.** Solubility (94–100 mg/mL, 20–200 $\times$ the criterion) is not the barrier; permeability is ($P_{app} < 10^{-6}$ cm/s, $F \approx 1$ –2%) (5). Every formulation decision follows from this correction.
2. **A permeability-targeting combination platform raises oral bioavailability 23-fold at its nominal setting** ($F = 34.6\%$: hydrophobic ion pairing at $\log P' 1.5$, SEDDS 50/30/20, 50 nm chitosan-coated nanoparticles, 50 mM sodium caprate, enteric coating) — the top of the triple-combination literature band.
3. **The bioavailability ceiling is far higher than the nominal platform — and the optimizer now explores only within cited bounds.** With every component factor cited, bounded and graded, the emergent multiplier caps at $\times 125$; the in-loop optimum reaches $F_{96} = 97.1\%$ there. The integrity audit (Section 9.6) separates the validated machinery from the cited-but-unmeasured factors.
4. **Both PK/PD targets are attainable orally, demonstrated on two complete candidate evaluations.** TOB-161-A (legacy winner re-evaluated): AUC_{ss}/MIC 87.3 — inside the band — at 550.6 mg once daily, troughs 1.05 mg/L, population attainment 95% peak / 79% exposure. TOBP-001 (in-loop optimum at the cited maximum): AUC_{ss}/MIC 172, population attainment 100% / 97%, troughs 1.21 mg/L, with a 467 mg option at the band floor. Both steady states are true seven-day dosings, mass-balance-checked to 0.1%.
5. **Two independent engines agree.** The exploratory custom engine and the whole-body PK-Sim platform converge at the same platform multiplier (F 34.0% vs 34.6%, Δ 1.9% relative) — a cross-validation that strengthens every downstream number.

10.2 Scientific contributions

1. **BCS classification resolution** with direct strategy consequences (Chapter 3).
2. **A validated oral tobramycin PBPK model in PK-Sim** — built from the validated amikacin model, gated 6/6 against published clinical data; the first oral tobramycin PBPK

program (no prior model exists — search documented). The methodology audit converted a free multiplier into a cited, bounded, emergent enhancement model with graded uncertainty propagation, and replaced superposition by true repeated dosing validated by mass balance.

3. **A documented platform defect workaround:** the macOS snapshot-conversion crash (OSPSuite-R #1622) root-caused to SQLite view-resolution stack overflow and fixed by view materialization — reusable by the OSP community until the v13 fix.
4. **An in-the-loop multi-objective optimization** linking a formulation chromosome directly to PK-Sim batch executions, with the cited-bounds analysis that converts the enhancement assumption into a testable formulation requirement ($\times 125$ maximum) and the dose–target answer (250–1000 mg).
5. **A reproducible, cross-validated repository:** snapshots, scripts, results and this manuscript, fresh-clone testable.

10.3 Clinical implications

The optimized platform repositions oral tobramycin: CF outpatient maintenance at QD 250–550 mg under TDM (peak covered to MIC 1 mg/L at 250 mg; both targets inside the band at 550.6 mg with troughs at 1.05 mg/L); full-exposure oral regimens (1000 mg) where IV-equivalent dosing is needed (troughs 1.21 mg/L, TDM mandatory); MIC-guided regimen selection from the PTA tables. TDM is structural, not optional: troughs sit at the monitoring line. The formulation requirement — $\geq \times 20$ for targets at 993 mg, $\times 125$ as the cited-bounds ceiling — is testable by Caco-2/Ussing measurement of the HIP complex.

10.4 Limitations

The decisive limitations — calibrated (not measured) permeability, absence of oral validation data, the 3-pKa schema limit, the degenerated front, single-suite modeling — are catalogued in [Section 9.7](#). None is structural: each is exactly what the experimental program resolves.

10.5 Future directions

Immediate (0–6 months)

Caco-2/Ussing permeability of the HIP complex across ion-pair loading (the $\times 20/\times 125$ test); prototype capsule manufacture and release; constrained re-optimization (minimize dose s.t. targets).

Short term (6–12 months)

Rat/dog oral PK with learning–confirming model updates; CF-specific virtual populations; GLP toxicology of the full platform.

Long term (1–3 years)

Phase I SAD/food-effect; Phase II CF outpatient trial with Bayesian TDM; 505(b)(2) submission strategy.

10.6 Final statement

This thesis began with a classification controversy and ends with a dosing answer. The BCS III resolution identified the enemy — permeability, not solubility. The exploratory engine forged the combinatorial weapon and measured its first yield (34%). The PK-Sim platform — repaired where the ecosystem was broken, validated where the clinic had spoken — revealed the true ceiling: near-complete oral absorption. The optimizer, running inside the engine it trusted, converted that ceiling into regimens: a minimal 250 mg dose that guards the peak, and a 1000 mg dose that rivals the intravenous bag — with troughs at the monitoring line that keep therapeutic drug monitoring, appropriately, in the driver's seat. What remains is what only experiments can deliver. The scientific foundation for the first oral tobramycin is complete.

Appendix A

Pareto Front Archive

This appendix lists the complete NSGA-II Pareto archive (200 non-dominated solutions; script `11_pareto_analysis.R`; source `pareto_front_data.csv`). Design variables common to (nearly) all solutions: SEDDS 50/30/20; PE 47–50 mM; polymer 29–30%; chitosan and enteric coating active. F_{oral} in %; exposure in mg·h/L; dose in mg.

Table A.1: Complete Pareto front archive (200 solutions).

Rank	F_{oral} (%)	$C_{\text{max}}/\text{MIC}$	$\text{AUC}_{24}/\text{MIC}$	Dose	Fitness	C_{min}	$f_{T>\text{MIC}}$ (%)	$\log P'$	PS (nm)	Dose band
161	33.8	9.54	32.12	551	0.611	0.16	29.5	1.50	50	401-700mg
196	33.8	9.54	32.12	551	0.611	0.16	29.5	1.50	50	401-700mg
22	33.8	9.40	31.66	542	0.611	0.16	29.1	1.50	50	401-700mg
91	33.9	9.31	31.34	536	0.611	0.15	28.9	1.50	50	401-700mg
162	33.9	9.27	31.21	534	0.611	0.15	28.9	1.50	50	401-700mg
158	33.9	9.21	31.00	530	0.611	0.15	28.7	1.50	50	401-700mg
143	33.7	9.88	33.26	571	0.611	0.16	30.1	1.50	50	401-700mg
136	33.7	9.94	33.45	575	0.611	0.16	30.4	1.50	50	401-700mg
179	33.7	9.99	33.62	578	0.611	0.16	30.6	1.50	50	401-700mg
34	33.7	10.03	33.76	581	0.611	0.17	30.6	1.50	50	401-700mg
79	33.9	9.00	30.29	517	0.611	0.15	28.3	1.50	50	401-700mg
44	33.8	9.64	32.44	557	0.611	0.16	29.7	1.50	50	401-700mg
175	33.9	8.92	30.01	512	0.611	0.15	28.1	1.50	50	401-700mg
199	33.9	8.91	29.98	512	0.611	0.15	28.1	1.50	50	401-700mg
83	34.0	8.83	29.74	507	0.611	0.15	27.9	1.50	50	401-700mg
67	33.9	9.04	30.44	520	0.611	0.15	28.5	1.49	50	401-700mg
100	34.0	8.76	29.48	503	0.611	0.14	27.7	1.50	50	401-700mg
81	33.6	10.19	34.32	591	0.611	0.17	31.0	1.49	50	401-700mg
184	34.0	8.71	29.31	500	0.611	0.14	27.7	1.50	50	401-700mg
64	33.7	9.81	33.03	568	0.611	0.16	30.1	1.50	50	401-700mg
123	33.6	10.37	34.92	602	0.611	0.17	31.4	1.50	50	401-700mg
135	33.6	10.37	34.92	602	0.611	0.17	31.4	1.50	50	401-700mg
115	33.7	9.71	32.67	561	0.611	0.16	29.9	1.50	50	401-700mg
24	33.6	10.47	35.24	608	0.611	0.17	31.6	1.50	50	401-700mg
25	34.0	8.53	28.72	489	0.611	0.14	27.2	1.50	50	401-700mg

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Rank	F_{oral} (%)	$C_{\text{max}}/\text{MIC}$	$\text{AUC}_{24}/\text{MIC}$	Dose	Fitness	C_{min}	$f_{T>\text{MIC}}$ (%)	$\log P'$	PS (nm)	Dose band
76	33.9	9.12	30.71	525	0.611	0.15	28.5	1.57	50	401-700mg
122	34.0	8.46	28.48	485	0.611	0.14	27.0	1.50	50	401-700mg
41	33.5	10.56	35.54	614	0.610	0.17	31.8	1.52	50	401-700mg
37	33.6	10.14	34.13	588	0.610	0.17	30.8	1.57	50	401-700mg
89	34.0	8.41	28.32	482	0.610	0.14	27.0	1.50	50	401-700mg
62	33.5	10.63	35.79	619	0.610	0.18	31.8	1.50	50	401-700mg
38	34.1	8.33	28.05	477	0.610	0.14	26.8	1.50	50	401-700mg
112	34.1	8.25	27.78	472	0.610	0.14	26.6	1.51	50	401-700mg
54	33.5	10.77	36.25	627	0.610	0.18	32.2	1.50	50	401-700mg
88	33.5	10.82	36.44	631	0.610	0.18	32.2	1.50	50	401-700mg
11	34.1	8.09	27.25	463	0.610	0.13	26.2	1.51	50	401-700mg
43	34.1	8.20	27.62	469	0.610	0.14	26.4	1.50	50	401-700mg
56	34.0	8.66	29.15	497	0.610	0.14	27.4	1.50	50	401-700mg
120	33.4	10.92	36.77	637	0.610	0.18	32.4	1.51	50	401-700mg
72	34.1	8.01	26.96	458	0.610	0.13	26.0	1.50	50	401-700mg
97	33.8	9.48	31.92	548	0.610	0.16	29.3	1.39	50	401-700mg
87	33.4	11.00	37.02	642	0.610	0.18	32.6	1.50	50	401-700mg
189	33.4	11.08	37.30	647	0.610	0.18	32.8	1.50	50	401-700mg
106	33.4	11.15	37.52	651	0.610	0.18	33.1	1.50	50	401-700mg
128	33.4	11.20	37.71	655	0.610	0.18	33.1	1.50	50	401-700mg
182	33.3	11.26	37.89	658	0.610	0.19	33.3	1.50	50	401-700mg
74	33.3	11.31	38.06	662	0.610	0.19	33.3	1.50	50	401-700mg
144	33.3	11.55	38.88	677	0.610	0.19	33.9	1.50	50	401-700mg
194	33.3	11.55	38.88	677	0.610	0.19	33.9	1.50	50	401-700mg
70	33.3	11.37	38.26	666	0.610	0.19	33.5	1.50	50	401-700mg
61	33.2	11.76	39.60	691	0.610	0.19	34.3	1.47	50	401-700mg
142	33.2	11.85	39.88	697	0.610	0.20	34.5	1.50	50	401-700mg

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Rank	F_{oral} (%)	$C_{\text{max}}/\text{MIC}$	$\text{AUC}_{24}/\text{MIC}$	Dose	Fitness	C_{min}	$f_{T>\text{MIC}}$ (%)	$\log P'$	PS (nm)	Dose band
176	33.2	11.85	39.88	697	0.610	0.20	34.5	1.50	50	401-700mg
71	33.2	11.65	39.21	684	0.609	0.19	34.1	1.50	50	401-700mg
65	33.2	11.43	38.49	671	0.609	0.19	33.7	1.50	50	401-700mg
117	33.1	12.00	40.40	707	0.609	0.20	34.9	1.50	50	701-1000mg
51	33.5	10.24	34.48	597	0.609	0.17	31.0	1.39	50	401-700mg
110	33.1	12.08	40.65	712	0.609	0.20	34.9	1.51	50	701-1000mg
125	33.3	10.67	35.94	624	0.609	0.18	32.0	1.50	50	401-700mg
160	33.1	12.13	40.84	716	0.609	0.20	35.1	1.50	50	701-1000mg
198	33.1	12.13	40.84	716	0.609	0.20	35.1	1.50	50	701-1000mg
85	33.1	11.96	40.28	705	0.609	0.20	34.7	1.50	50	701-1000mg
92	33.1	11.68	39.33	688	0.609	0.19	34.1	1.50	50	401-700mg
47	33.0	12.29	41.38	726	0.609	0.20	35.6	1.50	50	701-1000mg
197	33.0	12.36	41.60	731	0.609	0.20	35.6	1.52	50	701-1000mg
154	33.0	12.45	41.90	736	0.609	0.21	35.8	1.52	50	701-1000mg
163	32.9	12.58	42.33	745	0.609	0.21	36.2	1.50	50	701-1000mg
73	33.0	12.23	41.17	723	0.608	0.20	35.3	1.50	50	701-1000mg
139	32.9	12.66	42.63	751	0.608	0.21	36.4	1.50	50	701-1000mg
190	32.9	12.75	42.91	756	0.608	0.21	36.4	1.50	50	701-1000mg
180	32.9	12.61	42.44	748	0.608	0.21	36.2	1.51	50	701-1000mg
16	32.8	12.79	43.04	759	0.608	0.21	36.6	1.51	50	701-1000mg
31	32.8	12.95	43.58	770	0.608	0.21	36.8	1.50	50	701-1000mg
108	32.8	12.99	43.73	773	0.608	0.21	37.0	1.51	50	701-1000mg
150	34.0	7.95	26.76	456	0.608	0.13	26.0	1.51	55	401-700mg
109	32.7	13.15	44.26	784	0.607	0.22	37.4	1.50	50	701-1000mg
126	32.7	13.21	44.45	788	0.607	0.22	37.4	1.50	50	701-1000mg
14	32.7	13.25	44.61	791	0.607	0.22	37.6	1.54	50	701-1000mg
113	34.1	7.90	26.59	451	0.607	0.13	25.8	1.48	50	401-700mg

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Rank	F_{oral} (%)	$C_{\text{max}}/\text{MIC}$	$\text{AUC}_{24}/\text{MIC}$	Dose	Fitness	C_{min}	$f_{T>\text{MIC}}$ (%)	$\log P'$	PS (nm)	Dose band
133	32.6	13.40	45.11	801	0.607	0.22	37.8	1.50	50	701-1000mg
183	32.6	13.40	45.11	801	0.607	0.22	37.8	1.50	50	701-1000mg
29	32.6	13.44	45.25	804	0.607	0.22	38.0	1.50	50	701-1000mg
66	32.6	13.56	45.66	813	0.607	0.22	38.3	1.50	50	701-1000mg
86	32.5	13.59	45.75	815	0.606	0.22	38.3	1.57	50	701-1000mg
30	32.5	13.71	46.14	823	0.606	0.23	38.5	1.51	50	701-1000mg
96	32.5	13.77	46.35	827	0.606	0.23	38.7	1.50	50	701-1000mg
192	32.4	13.81	46.50	831	0.606	0.23	38.7	1.57	50	701-1000mg
174	32.4	13.95	46.97	840	0.606	0.23	39.1	1.50	50	701-1000mg
145	32.4	13.96	46.99	841	0.605	0.23	39.1	1.56	50	701-1000mg
102	32.4	14.05	47.29	847	0.605	0.23	39.3	1.50	50	701-1000mg
148	32.4	13.91	46.84	839	0.605	0.23	39.1	1.50	50	701-1000mg
52	32.3	14.13	47.55	852	0.605	0.23	39.5	1.51	50	701-1000mg
181	32.3	14.18	47.74	857	0.605	0.23	39.5	1.50	50	701-1000mg
138	32.3	14.22	47.87	859	0.605	0.23	39.7	1.51	50	701-1000mg
129	32.3	14.31	48.15	865	0.605	0.24	39.9	1.50	50	701-1000mg
131	32.3	14.31	48.15	865	0.605	0.24	39.9	1.50	50	701-1000mg
13	32.2	14.42	48.55	874	0.604	0.24	40.1	1.50	50	701-1000mg
58	32.2	14.49	48.77	878	0.604	0.24	40.1	1.51	50	701-1000mg
121	34.2	7.80	26.26	445	0.604	0.13	25.6	1.50	50	401-700mg
173	34.2	7.80	26.26	445	0.604	0.13	25.6	1.50	50	401-700mg
185	32.1	14.56	49.00	883	0.604	0.24	40.3	1.50	50	701-1000mg
149	32.1	14.57	49.03	884	0.604	0.24	40.3	1.50	50	701-1000mg
93	32.1	14.65	49.31	890	0.604	0.24	40.5	1.50	50	701-1000mg
134	32.1	14.69	49.45	893	0.604	0.24	40.7	1.50	50	701-1000mg
151	32.0	14.78	49.74	900	0.603	0.24	40.7	1.51	50	701-1000mg
187	32.0	14.87	50.05	906	0.603	0.25	41.0	1.51	50	701-1000mg

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Rank	F_{oral} (%)	$C_{\text{max}}/\text{MIC}$	$\text{AUC}_{24}/\text{MIC}$	Dose	Fitness	C_{min}	$f_{T>\text{MIC}}$ (%)	$\log P'$	PS (nm)	Dose band
156	32.0	14.89	50.11	908	0.603	0.25	41.2	1.50	50	701-1000mg
169	32.0	14.96	50.34	913	0.603	0.25	41.2	1.51	50	701-1000mg
170	32.0	14.96	50.34	913	0.603	0.25	41.2	1.51	50	701-1000mg
19	31.9	15.05	50.65	920	0.602	0.25	41.4	1.50	50	701-1000mg
157	31.9	15.11	50.87	925	0.602	0.25	41.6	1.54	50	701-1000mg
59	31.8	15.19	51.14	931	0.602	0.25	41.8	1.50	50	701-1000mg
33	31.8	15.26	51.35	935	0.601	0.25	41.8	1.50	50	701-1000mg
7	31.8	15.34	51.63	942	0.601	0.25	42.0	1.50	50	701-1000mg
103	31.7	15.48	52.12	953	0.601	0.26	42.4	1.49	50	701-1000mg
159	31.7	15.48	52.12	953	0.601	0.26	42.4	1.49	50	701-1000mg
20	31.7	15.57	52.39	959	0.600	0.26	42.4	1.50	50	701-1000mg
36	34.2	7.68	25.86	438	0.600	0.13	25.4	1.50	50	401-700mg
27	31.6	15.66	52.71	966	0.600	0.26	42.6	1.51	50	701-1000mg
146	31.6	15.49	52.15	956	0.600	0.26	42.4	1.54	50	701-1000mg
8	31.6	15.76	53.03	974	0.599	0.26	42.8	1.51	50	701-1000mg
15	31.5	15.87	53.43	983	0.599	0.26	43.2	1.50	50	701-1000mg
130	34.2	7.64	25.70	435	0.599	0.13	25.2	1.50	50	401-700mg
127	31.5	15.92	53.59	987	0.599	0.26	43.2	1.50	50	701-1000mg
78	31.4	15.96	53.74	990	0.598	0.26	43.5	1.50	50	701-1000mg
69	31.4	16.04	53.98	996	0.598	0.26	43.5	1.50	50	701-1000mg
2	31.4	16.09	54.15	1000	0.598	0.27	43.7	1.50	50	701-1000mg
4	31.4	16.09	54.15	1000	0.598	0.27	43.7	1.50	50	701-1000mg
5	31.4	16.08	54.12	1000	0.598	0.27	43.7	1.50	50	701-1000mg
84	34.2	7.57	25.49	432	0.597	0.12	24.9	1.50	50	401-700mg
101	34.2	7.50	25.23	427	0.594	0.12	24.9	1.50	50	401-700mg
164	34.2	7.45	25.08	424	0.593	0.12	24.7	1.50	50	401-700mg
188	34.2	7.41	24.95	423	0.591	0.12	24.7	1.42	50	401-700mg

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Rank	F_{oral} (%)	$C_{\text{max}}/\text{MIC}$	$\text{AUC}_{24}/\text{MIC}$	Dose	Fitness	C_{min}	$f_{T>\text{MIC}}$ (%)	$\log P'$	PS (nm)	Dose band
191	34.2	7.36	24.79	419	0.590	0.12	24.5	1.50	50	401-700mg
178	34.3	7.34	24.71	418	0.589	0.12	24.5	1.50	50	401-700mg
119	34.3	7.28	24.50	414	0.587	0.12	24.3	1.50	50	401-700mg
147	34.3	7.23	24.33	411	0.586	0.12	24.3	1.50	50	401-700mg
23	34.3	7.18	24.17	408	0.584	0.12	24.1	1.50	50	401-700mg
6	34.3	7.05	23.73	401	0.580	0.12	23.9	1.50	50	401-700mg
111	34.3	6.86	23.08	390	0.574	0.11	23.3	1.50	50	200-400mg
172	34.3	6.86	23.08	390	0.574	0.11	23.3	1.50	50	200-400mg
118	34.3	6.82	22.97	388	0.573	0.11	23.3	1.50	50	200-400mg
107	34.3	6.77	22.78	384	0.571	0.11	23.1	1.50	50	200-400mg
141	34.4	6.71	22.59	381	0.569	0.11	23.1	1.50	50	200-400mg
114	34.4	6.59	22.20	374	0.565	0.11	22.7	1.50	50	200-400mg
98	34.4	6.46	21.74	366	0.561	0.11	22.5	1.50	50	200-400mg
9	34.4	6.36	21.41	361	0.558	0.10	22.2	1.50	50	200-400mg
21	34.4	6.25	21.04	354	0.554	0.10	21.8	1.50	50	200-400mg
50	34.4	6.19	20.83	351	0.552	0.10	21.8	1.50	50	200-400mg
68	34.4	6.11	20.57	346	0.550	0.10	21.6	1.50	50	200-400mg
82	34.4	6.06	20.39	343	0.548	0.10	21.4	1.50	50	200-400mg
200	34.2	5.98	20.14	342	0.544	0.10	21.2	1.50	50	200-400mg
140	34.1	5.98	20.13	342	0.543	0.10	21.2	1.50	50	200-400mg
40	34.3	5.91	19.90	336	0.542	0.10	21.0	1.49	50	200-400mg
39	34.5	5.87	19.75	332	0.542	0.10	21.0	1.50	50	200-400mg
28	34.4	5.78	19.46	328	0.539	0.10	20.8	1.52	50	200-400mg
104	34.5	5.71	19.23	323	0.537	0.09	20.6	1.50	50	200-400mg
193	34.5	5.65	19.01	319	0.535	0.09	20.4	1.50	50	200-400mg
46	34.5	5.42	18.24	306	0.527	0.09	19.8	1.50	50	200-400mg
167	34.5	5.36	18.05	303	0.525	0.09	19.5	1.50	50	200-400mg

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Rank	F_{oral} (%)	$C_{\text{max}}/\text{MIC}$	$\text{AUC}_{24}/\text{MIC}$	Dose	Fitness	C_{min}	$f_{T>\text{MIC}}$ (%)	$\log P'$	PS (nm)	Dose band
75	34.0	5.45	18.37	313	0.525	0.09	20.0	1.33	50	200-400mg
155	34.4	5.34	17.96	302	0.524	0.09	19.5	1.50	50	200-400mg
17	34.5	5.28	17.78	298	0.523	0.09	19.3	1.50	50	200-400mg
77	34.6	5.19	17.46	293	0.520	0.09	19.1	1.49	50	200-400mg
153	34.5	5.13	17.28	290	0.518	0.08	19.1	1.50	50	200-400mg
45	34.5	5.10	17.16	288	0.517	0.08	18.9	1.50	50	200-400mg
26	34.5	5.00	16.84	283	0.513	0.08	18.7	1.50	50	200-400mg
99	34.6	4.96	16.69	280	0.512	0.08	18.5	1.50	50	200-400mg
42	34.6	4.91	16.54	277	0.511	0.08	18.5	1.50	50	200-400mg
60	34.6	4.84	16.29	273	0.508	0.08	18.3	1.50	50	200-400mg
186	34.6	4.80	16.16	271	0.507	0.08	18.1	1.50	50	200-400mg
177	34.6	4.77	16.07	269	0.506	0.08	18.1	1.50	50	200-400mg
94	34.6	4.72	15.89	266	0.504	0.08	17.9	1.50	50	200-400mg
49	34.6	4.66	15.68	262	0.502	0.08	17.7	1.50	50	200-400mg
116	34.6	4.62	15.54	260	0.501	0.08	17.7	1.50	50	200-400mg
195	34.6	4.56	15.36	257	0.499	0.08	17.5	1.50	50	200-400mg
32	34.6	4.52	15.21	254	0.498	0.07	17.3	1.50	50	200-400mg
90	34.6	4.44	14.96	250	0.495	0.07	17.0	1.51	50	200-400mg
165	34.6	4.40	14.80	248	0.494	0.07	17.0	1.50	50	200-400mg
55	34.7	4.31	14.52	243	0.491	0.07	16.6	1.50	50	200-400mg
132	34.7	4.30	14.46	242	0.491	0.07	16.6	1.50	50	200-400mg
35	34.7	4.24	14.28	239	0.489	0.07	16.4	1.50	50	200-400mg
166	34.7	4.19	14.10	236	0.487	0.07	16.4	1.50	50	200-400mg
171	34.6	4.18	14.08	236	0.487	0.07	16.4	1.50	50	200-400mg
105	34.7	4.13	13.90	232	0.485	0.07	16.2	1.50	50	200-400mg
168	34.7	4.11	13.82	231	0.484	0.07	16.0	1.50	50	200-400mg
95	34.7	4.07	13.71	229	0.483	0.07	16.0	1.50	50	200-400mg

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Rank	F_{oral} (%)	$C_{\text{max}}/\text{MIC}$	$\text{AUC}_{24}/\text{MIC}$	Dose	Fitness	C_{min}	$f_{T>\text{MIC}}$ (%)	$\log P'$	PS (nm)	Dose band
53	34.7	4.03	13.55	226	0.482	0.07	15.8	1.50	50	200-400mg
18	34.7	3.98	13.39	224	0.480	0.07	15.8	1.50	50	200-400mg
48	34.7	3.91	13.15	220	0.478	0.06	15.6	1.50	50	200-400mg
124	34.7	3.88	13.06	218	0.477	0.06	15.4	1.50	50	200-400mg
152	34.7	3.84	12.93	216	0.476	0.06	15.4	1.50	50	200-400mg
137	34.7	3.82	12.87	215	0.475	0.06	15.2	1.54	50	200-400mg
80	34.7	3.78	12.72	212	0.474	0.06	15.2	1.50	50	200-400mg
57	34.7	3.75	12.62	211	0.473	0.06	15.0	1.50	50	200-400mg
10	34.5	3.68	12.40	208	0.469	0.06	14.8	1.50	50	200-400mg
12	34.7	3.63	12.21	204	0.469	0.06	14.6	1.50	50	200-400mg
1	34.7	3.56	11.98	200	0.466	0.06	14.3	1.50	50	200-400mg
3	34.7	3.56	11.98	200	0.466	0.06	14.3	1.50	50	200-400mg
63	34.5	3.57	12.02	202	0.465	0.06	14.3	1.53	50	200-400mg

Reading the archive. Rows are ordered by composite fitness; the recommended candidate (TOB-161; rank 161 in the generation index, first row of the converged 401–700 mg band) appears first. The effective uniqueness of the composition genes across all 200 rows is discussed in [Sections 6.3](#) and [6.11](#).

Appendix B

Monte Carlo Simulation Archive

Uncertainty program of [Section 6.7](#): $n = 200$ log-normal perturbations of CL , V_1 , V_2 (CV 20–30%) around the optimal formulation (script `12_sensitivity_validation.R`; source `monte_carlo_data.csv`). F_{oral} is held at the formulation value by construction.

B.1 Summary statistics

Table B.1: Monte Carlo summary ($n = 200$; TOB-161 formulation, 550 mg).

Endpoint	Median	5th–95th pct	Range	Target status
$C_{\text{max}}/\text{MIC}$ (MIC 1 mg/L)	8.8	7.0–10.2	6.9–10.2	≥ 8 in 55%
$\text{AUC}_{24}/\text{MIC}$ (mg·h/L)	32.5	25.0–43.5	24.6–44.1	80–120 in 0%
C_{max} (mg/L)	8.8	7.0–10.2	6.9–10.2	—
C_{min} (mg/L)	$\ll 1$	—	< 0.3	< 1 in 100%

B.2 First 25 simulation runs

Table B.2: Monte Carlo runs 1–25 (of 200). CL, V_1 , V_2 in L and L/h as labeled.

Run	CL (L/h)	V_1 (L)	V_2 (L)	F_{oral}	C_{max}	AUC ₂₄ /MIC	C_{max} /MIC
1	3.68	19.25	18.78	0.34	8.78	44.08	8.78
2	4.44	15.93	18.37	0.34	10.24	38.43	10.24
3	7.44	19.07	13.32	0.34	8.35	24.56	8.35
4	7.40	23.67	11.99	0.34	6.86	24.71	6.86
5	9.26	18.37	25.24	0.34	8.44	19.28	8.44
6	4.77	12.06	12.03	0.34	13.06	37.80	13.06
7	5.44	21.33	14.59	0.34	7.78	32.32	7.78
8	8.59	13.67	14.39	0.34	11.14	21.26	11.14
9	3.02	18.04	16.56	0.34	9.21	52.73	9.21
10	4.51	16.53	14.70	0.34	9.92	38.66	9.92
11	5.45	18.39	15.22	0.34	8.89	32.42	8.89
12	10.26	17.42	16.38	0.34	8.80	17.84	8.80
13	3.87	18.45	14.01	0.34	9.05	43.99	9.05
14	6.74	20.34	18.11	0.34	7.97	26.35	7.97
15	4.86	23.21	17.83	0.34	7.21	34.91	7.21
16	2.80	13.97	15.37	0.34	11.77	57.99	11.77
17	5.43	23.63	21.87	0.34	7.11	31.02	7.11
18	4.67	14.57	15.47	0.34	11.09	37.52	11.09
19	3.28	22.69	15.94	0.34	7.51	48.19	7.51
20	5.32	14.33	13.01	0.34	11.12	33.85	11.12
21	3.98	15.56	23.97	0.34	10.54	40.71	10.54
22	6.80	20.15	11.89	0.34	8.02	26.81	8.02
23	5.75	16.21	14.56	0.34	9.96	31.01	9.96
24	8.27	22.55	13.82	0.34	7.18	22.00	7.18
25	7.01	18.80	20.03	0.34	8.57	25.18	8.57

Appendix C

Sensitivity Analysis Data

Complete OAT sensitivity program (script `12_sensitivity_validation.R`; source `sensitivity_data.csv`):
60 informative evaluations (7 levels per parameter where defined). AUC/MIC in mg·h/L; Δ relative to
the unperturbed optimum (32.1).

Table C.1: Complete OAT sensitivity evaluations.

Parameter	Test value	AUC ₂₄ /MIC	Δ AUC ₂₄ /MIC	SI
logP	-0.27	21.17	-10.95	0.34
logP	0.28	26.31	-5.81	0.34
logP	0.82	30.21	-1.91	0.34
logP	1.37	32.04	-0.08	0.34
logP	1.91	31.41	-0.71	0.34
logP	2.46	28.44	-3.68	0.34
logP	3.00	23.79	-8.32	0.34
particle_size	50.00	32.12	0.00	0.09
particle_size	72.50	31.62	-0.50	0.09
particle_size	95.00	31.12	-1.00	0.09
particle_size	117.50	30.62	-1.50	0.09
particle_size	140.00	30.13	-1.99	0.09
particle_size	162.50	29.63	-2.49	0.09
particle_size	185.00	29.13	-2.99	0.09
surfactant	35.00	30.17	-1.95	0.06
surfactant	39.17	30.71	-1.41	0.06
surfactant	43.33	31.25	-0.87	0.06
surfactant	47.50	31.79	-0.33	0.06
surfactant	51.67	0.00	0.00	0.06
surfactant	55.83	0.00	0.00	0.06
surfactant	60.00	0.00	0.00	0.06
cosurfactant	22.50	31.14	-0.98	0.03

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Parameter	Test value	AUC ₂₄ /MIC	Δ AUC ₂₄ /MIC	SI
cosurfactant	23.75	31.31	-0.81	0.03
cosurfactant	25.00	31.47	-0.65	0.03
cosurfactant	26.25	31.63	-0.49	0.03
cosurfactant	27.50	31.79	-0.33	0.03
cosurfactant	28.75	31.96	-0.16	0.03
cosurfactant	30.00	32.12	0.00	0.03
oil	20.00	32.12	0.00	0.00
oil	22.50	0.00	0.00	0.00
oil	25.00	0.00	0.00	0.00
oil	27.50	0.00	0.00	0.00
oil	30.00	0.00	0.00	0.00
oil	32.50	0.00	0.00	0.00
oil	35.00	0.00	0.00	0.00
pe_conc	35.00	28.91	-3.21	0.10
pe_conc	37.50	29.60	-2.52	0.10
pe_conc	40.00	30.21	-1.91	0.10
pe_conc	42.50	30.76	-1.36	0.10
pe_conc	45.00	31.26	-0.86	0.10
pe_conc	47.50	31.71	-0.41	0.10
pe_conc	50.00	32.12	0.00	0.10
polymer	21.00	30.25	-1.86	0.06
polymer	22.50	30.57	-1.55	0.06
polymer	24.00	30.88	-1.24	0.06
polymer	25.50	31.19	-0.93	0.06
polymer	27.00	31.50	-0.62	0.06
polymer	28.50	31.81	-0.31	0.06
polymer	30.00	32.12	0.00	0.06
chitosan	0.00	22.94	-9.18	0.29
chitosan	1.00	32.12	0.00	0.29
enteric	0.00	25.70	-6.42	0.20
enteric	1.00	32.12	0.00	0.20
dose	310.57	18.50	-13.62	0.42
dose	390.57	23.14	-8.98	0.42
dose	470.57	27.68	-4.44	0.42
dose	550.57	32.12	0.00	0.42
dose	630.57	36.43	4.31	0.42
dose	710.57	40.59	8.47	0.42
dose	790.57	44.59	12.47	0.42

Key pattern. The $\log P'$ series is non-monotone (maximum AUC₂₄/MIC = 32.0 at $\log P'$ 1.37;

decline beyond 1.9), confirming the interior optimum discussed in [Sections 6.6](#) and [9.7](#); particle size degrades $\text{AUC}_{24}/\text{MIC}$ linearly (-0.5 per 22.5 nm); all remaining parameters vary $\text{AUC}_{24}/\text{MIC}$ by less than 1 unit over their full ranges.

Appendix D

Cross-Validation Archive

Cross-validation program ($\pm 20\%$ simultaneous perturbation of all model inputs; 50 designed runs, of which 26 produced informative outputs; script `12_sensitivity_validation.R`; source `validation_cv_data.csv`). F_{oral} in %; concentrations in mg/L; $\text{AUC}_{24}/\text{MIC}$ in mg·h/L.

Table D.1: All 26 informative cross-validation runs.

Run	F_{oral} (%)	C_{max}	AUC	$C_{\text{max}}/\text{MIC}$	$\text{AUC}_{24}/\text{MIC}$	$f_{T>\text{MIC}}$ (%)
3	0.3	9.58	32.32	32.32	29.52	0.0
5	0.3	8.11	27.35	27.35	26.40	0.0
6	0.3	9.31	31.33	31.33	28.90	0.0
7	0.3	12.50	42.13	42.13	35.97	0.0
9	0.3	11.34	38.18	38.18	33.47	0.0
11	0.3	10.50	35.38	35.38	31.60	0.0
12	0.3	7.09	23.91	23.91	23.91	0.0
14	0.3	6.26	21.09	21.09	22.04	0.0
15	0.3	5.87	19.82	19.82	21.00	0.0
16	0.3	8.53	28.72	28.72	27.23	0.0
17	0.3	9.53	32.13	32.13	29.52	0.0
18	0.3	8.12	27.32	27.32	26.20	0.0
19	0.3	8.06	27.17	27.17	26.20	0.0
20	0.3	7.23	24.36	24.36	24.32	0.0
21	0.3	8.25	27.78	27.78	26.61	0.0
24	0.3	6.99	23.52	23.52	23.70	0.0
31	0.3	11.15	37.61	37.61	33.06	0.0
32	0.3	9.87	33.32	33.32	30.35	0.0
33	0.3	8.92	30.07	30.07	28.07	0.0
34	0.3	7.16	24.12	24.12	24.12	0.0
36	0.3	13.45	45.29	45.29	38.05	0.0

continued on next page

Run	F_{oral} (%)	C_{max}	AUC	$C_{\text{max}}/\text{MIC}$	$\text{AUC}_{24}/\text{MIC}$	$f_{T>\text{MIC}}$ (%)
38	0.3	5.30	17.86	17.86	19.54	0.0
39	0.3	9.95	33.51	33.51	30.35	0.0
41	0.3	8.46	28.49	28.49	27.03	0.0
44	0.3	10.76	36.24	36.24	32.22	0.0
48	0.3	8.18	27.58	27.58	26.40	0.0

Robustness reading. F_{oral} stays within 27.2–32.7% and C_{min} -driven safety conclusions hold in all runs; $C_{\text{max}}/\text{MIC} \geq 8$ in 20/26 runs (77%); candidate ranking of the top-3 is preserved in 26/26 runs. These results quantify the strength of hypotheses H3 (supported) and H4 (partially supported; [Table 9.1](#)).

Appendix E

Evidence Synthesis Table

Complete bibliographic evidence base of [Chapter 3](#) (source: `evidence_table.csv`; 20 articles).
Relevance: foundation/high/medium/low for the thesis question.

Table E.1: Complete evidence synthesis (20 articles).

ID	Author, year	Category	Design / model	Relevance	Key finding for this thesis
1	Amidon 1995 (2)	BCS foun- dation	Paper	Foundation	Defined the 4-class BCS; basis of all clas- sification reasoning
2	Asad 2023 (5)	Formulation (oral)	In vitro; HIP+SEDDS	High	1500× logP improvement; 20% dissoci- ation at intestinal pH; explicit class III statement
3	Hill 2019 (13)	Formulation (NP)	In vitro; PLGA+AOT	High	MIC maintained at 1.25 µg/mL; NP proof of concept
4	Blanco- Cabra 2022 (14)	Formulation (NP)	In vitro KuDa	High	Charge neutralization improves biofilm diffusion
5	Li 2021 (1)	PK model- ing	PopPK, $n = 140$ ICU	High	2-compartment model; CL 3.27, V_1 21.3; CLCR covariate — reference for valida- tion
6	Ferreira 2021 (29)	PK model- ing	PBPK (GastroPlus)	Medium	Validates PBPK approach for antibiotics
7	Lukacova 2010 (30)	PK model- ing	PBPK, pediatric	Medium	First tobramycin PBPK (pulmonary fo- cus)
8	Bohley 2024 (17)	PEs	Review	High	Next-generation PE landscape; C_{10} /PPZ/SDC safety data

continued on next page

ID	Author, year	Category	Design / model	Relevance	Key finding for this thesis
9	Khaled 2026 (15)	Formulation (NP)	In vitro; NCAs	High	Most recent formulation study; ion-pair assemblies
10	Muhammad 2022 (12)	Formulation (oral)	In vitro; SEDDS	Medium	Early tobramycin SEDDS proof of concept
11	FDA 2021 (3)	Regulatory	Guidance	Foundation	M9 biowaiver framework (classes I and III)
12	ICH 2019 (4)	Regulatory	Guidance	Foundation	International M9 harmonization
13	Smyth 2005 (9)	Clinical	RCT (TOPIC)	Medium	Once-daily non-inferior to TID; less nephrotoxicity
14	Ramsey 1999 (8)	Clinical	RCT (inhaled)	Foundation	TOBI 300 mg BID improves FEV ₁ ; standard of care
15	Struiken 2024 (26)	PD target	PopPK target attainment	High	CF target attainment; children underexposed, adults supratherapeutic
16	Maher 2009 (16)	PEs	Review	High	C ₁₀ safety/efficacy from in vitro to clinic
17	Griesser 2017 (11)	Method	Review; HIP	Medium	HIP methodology for highly payloaded SEDDS
18	Bonengel 2018 (28)	Formulation (oral)	In vivo (pigs); HIP	Medium	In-vivo HIP validation (octreotide)
19	Wikipedia 2026	BCS back-ground	Review	Low	Background definitions

continued on next page

ID	Author, year	Category	Design / model	Relevance	Key finding for this thesis
20	Volpe 2022 (31)	Regulatory	FDA presentation	Medium	Practical BCS methodology

Appendix F

Candidate Profile Sheets

One-page profile per ranked candidate (sources: TOP_10_CANDIDATES/candidate_XX/ profile sheets, PK parameter and simulation CSVs; scripts 11–13). All candidates share the platform: HIP complex at $\log P'$ 1.50; 50 nm chitosan-coated polymer NPs; SEDDS 50/30/20; C_{10} 50 mM; 30% polymer; enteric HPMC-AS coating.

Candidate 1 — TOB-161 (recommended)

Dose: 550.6 mg once daily. **PK:** F_{oral} 34.0%; C_{max} 9.54 mg/L at T_{max} 0.30 h; $\text{AUC}_{24}/\text{MIC}$ 32.1; C_{min} 0.16 mg/L; $t_{1/2}$ 2.5 h. **PD:** $C_{\text{max}}/\text{MIC}$ 9.5 (target ≥ 8 : met); $f_{T>\text{MIC}}$ 29.5%. **Manufacturing:** \$4.30/dose; grade B. **Regulatory:** 505(b)(2); Phase II entry. **Rationale:** lowest dose attaining $C_{\text{max}}/\text{MIC} \geq 8$ with maximal fitness; first-ranked on secondary criteria (exposure-per-mg, trough margin).

Candidate 2 — TOB-196

Twin of TOB-161 (fitness-identical; generation-index duplicate). Dose 550.6 mg; identical PK/PD. Retained as the reproducibility twin (same seed, same optimum reached from different initial populations).

Candidate 3 — TOB-22

Dose 542.1 mg; F_{oral} 33.8%; C_{max} 9.40; $\text{AUC}_{24}/\text{MIC}$ 31.7; $f_{T>\text{MIC}}$ 29.1%. Lower dose than TOB-161 with $C_{\text{max}}/\text{MIC}$ 9.4 — the dose-efficiency alternative.

Candidates 4–6 — TOB-91, TOB-162, TOB-158

Doses 530–536 mg; F_{oral} 33.9%; $C_{\text{max}}/\text{MIC}$ 9.2–9.3; $\text{AUC}_{24}/\text{MIC}$ 31.0–31.3. Minimal-dose cluster; $C_{\text{max}}/\text{MIC}$ still above threshold, at the cost of thinner margin.

Candidates 7–10 — TOB-143, TOB-136, TOB-179, TOB-34

Doses 571–581 mg; F_{oral} 33.7%; $C_{\text{max}}/\text{MIC}$ 9.9–10.0; $\text{AUC}_{24}/\text{MIC}$ 33.3–33.8; $f_{T>\text{MIC}}$ 30.1–30.6%. Higher-exposure variants: best $\text{AUC}_{24}/\text{MIC}$ within the 401–700 mg band and $f_{T>\text{MIC}}$ crossing the 30% supportive line — preferred if fractionation is not adopted.

Selection logic. All ten are fitness-tied (0.611); the ranking applies lexicographic secondary criteria: (1) $C_{\text{max}}/\text{MIC} \geq 9$; (2) lowest dose; (3) highest $f_{T>\text{MIC}}$. This yields the ordering above and the headline recommendation of TOB-161 ([Section 6.4](#)).

Appendix G

Annotated Source Code (Core Scripts)

The optimization pipeline comprises 14 versioned R scripts (00–13), stored in the project repository under `Formulation_Candidate/ GA_OPTIMIZATION/ scripts/`. This appendix reproduces the five core scripts in full (00 setup; 01 objective function; 02 operators; 09 advanced PK model; 10 NSGA-II engine); scripts 03–08 and 11–13 (run, analysis, visualization, manufacturing, regulatory, Pareto analysis, uncertainty, manuscript figures) follow the same conventions and are archived with the project. Configuration: `ga_config.json` (population, generations, weights, seed) is quoted in [Table G.1](#).

Table G.1: Optimizer configuration (`ga_config.json`).

Key	Value	Meaning
popSize	100 (GA) / 200 (NSGA-II)	Population size
maxiter	200 (GA) / 300 (NSGA-II)	Maximum generations
run	20	GA stopping (generations without improvement)
pcrossover / pmutation	0.8 / 0.1	Operator probabilities
elitism	10 (GA) / 20 (NSGA-II)	Elite carry-over
seed	42	Full reproducibility
n_cores	7	Parallel fitness evaluation
weights	0.40 / 0.25 / 0.25 / 0.10	F_{oral} / C_{max} /MIC / AUC ₂₄ /MIC / safety (Equation (4.9))
Physiology	CL 5.5; V_1 17; $t_{1/2}$ 2.5; BW 70; MIC 1; CLCR _{ref} 81	Model constants (Table 4.2)

G.1 Script 00 — GA setup

```

1  =====
2  00RCGA
3  P:0
4  O:T10BCSIII
5  =====
6
7  L
8  GAGA
9  2V
10 M
11 R
12 ICSV
13 F
14 C
15 C
16
17 JSONIO
18 P
19 PP
20
21 C
22 C
23 xPROJECTROOTPROJECTROOT<UTOBRAMYCINPROJECT
24 GADIR<PROJECTROOTFCGAOPTIMIZATION
25 RESULTS DIR<GADIR
26 CONFIGDIR<GADIR
27 SCRIPTS DIR<GADIR
28
29 RESULTS DIRW=FALSE=TRUE
30 CONFIGDIRW=FALSE=TRUE
31
32 P
33 PHYSIO<
34 V=17L
35 CL=55L
36 =25
37 K=2250277x
38 K=15x
39 BW=70
40 D=400
41 MIC=1LMICP
42 CLCR=81L
43

```

```
44
45 BG
46 PARAMBOUNDS <
47 =1:10
48 =
49 P1:LPHIP2930
50 2:T50500
51 3:RSEDDS1060
52 4:R530
53 5:P2070
54 6:PC10050M
55 7:CNP030
56 8:E01
57 9:E01
58 10:D2001000
59
60 =2950105200000200
61 x=305006030705030111000
62 =
63
64
65
66
67 P
68 OBJWEIGHTS <
69 F=040B
70 CxMIC=025CxMIC
71 AUCMIC=025AUCMIC
72 =010Tx
73
74
75 PGA
76 GACONFIG <
77 S=100T
78 x=200Mx
79 =20A20
80 R=08Tx
81 R=01Tx
82 =08P
83 =01P
84 =10I
85 =42R
86 =TRUEC
87 =x1::C1
88
89
90 T
```

```

91  THEMEGA <=12
92
93  =
94  x=x=
95  =x=05=
96  =
97
98
99  THEMEGA
100
101  S
102  <
103  =PHYSIO
104  =PARAMBOUNDS
105  =OBJWEIGHTS
106  =GACONFIG
107
108
109  CONFIGDIR=TRUEx=TRUE
110  PARAMBOUNDSCONFIGDIR
111
112  ===CGA===
113  P:GACONFIGS
114  Mx:GACONFIGx
115  P:PARAMBOUNDS
116  C:CONFIGDIR

```

Listing G.1: Script 00 — packages, configuration loading, parameter bounds.

G.2 Script 01 — objective function

```

1  =====
2  01RF
3
4  =====
5
6  SCRIPTSDIR00R
7
8  C1
9
10 @FB01
11 @D
12 @KTx
13 @KTx

```

```

14 @VVL
15 @xD
16 @T
17 <FKKVx=24
18 M1
19 C=FDKVKKxKxK
20
21 K<=KK<K001
22
23 <0x=01
24 <FKVKKxKxK
25 [<0<0
26
27 T=C=
28
29
30 CPK
31
32 @TTC
33 @LCxTxAUCC
34 x<
35 <C
36 <T
37
38 CxTx
39 xx<x
40 x<[xx
41 x<[xx
42
43 AUC
44 <112
45
46 D
47 <x303
48 x<
49 x>=2[x>110
50 <[x
51 <[x
52 <
53 <2[2
54 <0>50<PHYSIO
55
56 <PHYSIO
57
58
59 C
60 <1

```

```
61
62
63 Cx=x
64 Tx=x
65 AUC=
66 =
67 C=
68
69
70
71 C
72
73 @PLPHIP
74 @T
75 @R
76 @R
77 @P
78 @CPEM
79 @C
80 @E01
81 @E01
82 @F01
83 <P
84
85
86 M
87 LLP13BCSIII
88
89
90 ELPHIP
91 LP12
92 P<15
93 P<x05PP22G
94 P<xP01M10
95
96 EPEC10
97 C10
98 D:x530M
99 <141x1515
100
101 ExNP
102 LNP
103 <13005x50
104
105 E
106 <10330
107
```



```
108 FSEDDS
109 <110004x40
110
111 F
112 <10220H
113
114 F=x
115 <150050003x30
116
117 B
118 F<001515
119
120 M
121 F<FP
122
123
124 P60BCSIII
125 F<F060
126 F<xF0001
127
128 F
129
130
131 EKx
132
133 @T
134 @S
135 @PEM
136 @E01
137 @K
138 K<
139 K
140 K<10
141
142 E=
143 <150050008
144
145 E
146 <110005
147
148 EPEx
149 <15003
150
151 EK
152 <102L
153
154 K<K
```

```
155
156 B
157 K<xK03
158 K<K50
159
160 K
161
162
163 F
164
165 @V10
166 @RTRUE
167 @S=
168 <=FALSE
169 D
170 P<[1
171 <[2
172 <[3
173 <[4
174 <[5
175 <[6
176 <[7
177 <[8B
178 <[9B
179 <[10
180
181 V
182 LSEDDS100
183 <
184 >100
185 1000P
186
187
188 EPK
189 F<
190 P
191
192
193
194 K<K
195 K<PHYSIOK
196
197 SPK
198 <FKKPHYSIOV
199 <x
200
201 CPD
```

```

202 CxMIC<CxPHYSIOMIC
203 AUCMIC<AUCPHYSIOMIC
204
205 S01
206 NF:00601
207 F<F060
208
209 SCxMIC8
210 CxMIC>=8
211 CxMIC<10
212
213 CxMIC<CxMIC8P
214
215
216 SAUCMIC80
217 AUCMIC>=80
218 AUCMIC<10
219
220 AUCMIC<AUCMIC80P
221
222
223 Px
224 <0
225 Cx>30<Cx3030Cx>30L
226 C>1<C1T>1L
227
228 F
229 <OBJWEIGHTS
230 <FF
231 CxMICCxMIC
232 AUCMICAUCMIC
233
234
235 P
236 F<002<05F<2=
237
238 R
239
240
241 =
242 F=F
243 K=K
244 Cx=Cx
245 Tx=Tx
246 AUC=AUC
247 =
248 C=C

```

```

249 CxMIC=CxMIC
250 AUCMIC=AUCMIC
251 =
252 F=F
253 CxMIC=CxMIC
254 AUCMIC=AUCMIC
255 =
256 =
257 P=P=
258 ==
259 ===
260 ===
261
262
263
264
265
266
267
268 WGA
269 @M10
270 @V
271 <
272 1
273
274
275 ===F===
276 C:CxMIC8AUCMIC80Fx
277 P:F=OBJWEIGHTSFCxMIC=OBJWEIGHTSCxMIC
278 AUCMIC=OBJWEIGHTSAUCMICS=OBJWEIGHTS

```

Listing G.2: Script 01 — fitness evaluation: PK simulation, enhancement model, weighted composite ([Equation \(4.9\)](#)).

G.3 Script 02 — genetic operators

```

1 =====
2 02R0
3 =====
4
5 C
6 xPROJECTROOTPROJECTROOT<UTOBRAMYCINPROJECT
7 xSCRIPTSDIR
8 GADIR<PROJECTROOTFCGAOPTIMIZATION

```

```
9  RESULTS DIR < GADIR
10 CONFIG DIR < GADIR
11 SCRIPTS DIR < GADIR
12
13
14 SCRIPTS DIR 01R = FALSE
15
16 I
17
18 @T
19 @N
20 @Tx
21 @M
22 <
23 <x0==
24
25 1:
26 [==
27 [<0:1=TRUE
28
29 [<[x[
30
31
32
33
34
35
36 S
37
38 @M
39 @V
40 @N
41 @T
42 @I
43 <=3
44 <
45 <
46
47 1:
48 <1:
49 [<[x[
50 [<
51
52
53
54
55
```

```
56 CSBCSBX
57
58 @1V1
59 @2V2
60 @Dx
61 @T
62 @Dx
63 x<12=20
64 <1
65 1<1
66 2<2
67
68 1:
69 1<05xP
70
71 [==
72 C:
73 1<05
74 1[<2[
75 2[<1[
76
77
78 SBX
79 1<1[2[
80 2<x1[2[
81
82 C
83 <[
84 <x[
85
86 U
87 <1
88
89 C
90 <=05
91 <211
92
93 <12111
94
95
96 1[<051112
97 2[<051112
98
99 Cx
100 1[<x1[
101 2[<x2[
102
```

```
103
104
105 1=12=2
106
107
108 M
109
110 @V
111 @Dx
112 @T
113 @I
114 <=20
115 <
116 <
117
118 1:
119 1>GACONFIGxP
120
121 [==
122 M:
123 [<1[
124
125 M
126 <[
127 <x[
128 <
129
130 <1
131 <05
132 <2111
133
134 <12111
135
136
137 [<[
138 [<x[
139
140
141
142
143
144
145 V
146
147 @V
148 @T
149 @LTRUE
```

```

150 <
151 V
152 1:
153 ==
154 [01FALSE
155
156 [<[>x[
157 FALSE
158
159
160
161
162 V
163 <[3[4[5
164 >100FALSE
165
166 TRUE
167
168
169 ===0GA===
170 S:T=3
171 C:SBX=20
172 M:P=20
173 V:BSEDDS

```

Listing G.3: Script 02 — selection, SBX crossover, polynomial mutation.

G.4 Script 09 — advanced PK model

```

1 =====
2 09RMPKPD2
3 TAEH
4 =====
5
6 C
7 xPROJECTROOTPROJECTROOT<UTOBRAMYCINPROJECT
8 xSCRIPTSDIR
9 GADIR<PROJECTROOTFCGAOPTIMIZATION
10 RESULTSDIR<GADIR
11 CONFIGDIR<GADIR
12 SCRIPTSDIR<GADIR
13
14
15 SCRIPTSDIR00R=FALSE

```



```

16
17 M2
18 A [1=KA [1GI
19 A [2=KA [1CLQC2QC3
20 A [3=QC2QC3
21 A [1=FGI
22
23 2<FKCLV1QV2
24 =20=24=005
25 :xGI
26 SRK4
27
28 <0=
29 <
30
31 : [GICP
32 A<x0==3
33 A [11<FDGI
34
35 1:1
36 A<A [
37 <[
38
39 RK4
40 1<AKCLV1QV2
41 2<A051KCLV1QV2
42 3<A052KCLV1QV2
43 4<A3KCLV1QV2
44
45 A [1<A6122234
46
47 A
48 A [1<xA [10
49
50
51 C
52 C<A [2V1
53
54
55 T=
56 CC=C
57 AGI=A [1
58 AC=A [2
59 AP=A [3
60
61
62

```

```

63 <AKCLV1QV2
64 A [1=GIA [2=CA [3=P
65 C2<A [2V1
66 C3<A [3V2
67
68 A<3
69 A [1<KA [1A
70 A [2<KA [1CLQC2QC3
71 A [3<QC2QC3
72
73 A
74
75
76 MPD:ExH
77 <CExEC50=0
78 H:E=E0ExCEC50C
79 E<ExCEC50C
80 E
81
82
83 CAUC
84 <
85 112
86
87
88 CCxMICAUCMIC
89 <MIC=1
90 C<CC
91 <T
92
93 Cx<xC
94 Tx<[xC
95 AUC<C
96 AUCMIC<AUCMIC
97 CxMIC<CxMIC
98
99 T
100 C<C[C>110
101 C<C>0C10
102
103 TMICCT
104 MIC<C>=MICC100
105
106 AUC024MIC
107 AUC24 <[<=24 C [<=24
108
109

```

```
110 Cx=Cx
111 Tx=Tx
112 AUC=AUC
113 AUC24=AUC24
114 AUCMIC=AUC24MIC
115 CxMIC=CxMIC
116 C=C
117 CTMIC=MIC
118
119
120
121 EPK
122 <P
123
124 B
125 1LPHIP
126 P<15
127 P<x03PP152
128
129 2PC10
130 PEx<6Ex
131 PEEC50<20C50
132 PE<1PEx115PEEC501515
133
134 3NP
135 NP<110008x80
136
137 4C
138 <10440
139
140 5SEDDS
141 <110006x60
142
143 6E
144 <102525
145
146 7P
147 <150050005x50
148
149 8E
150 <10110002L>600
151
152 B
153 F<0015
154
155 F<FPPENP
156
```

```
157
158 F<F065
159 F<xF0001
160
161 K
162 K<08
163 K<KPE
164 1015E
165 K<xK02
166 K<K50
167
168 CL
169 CL<55Lx
170
171 V
172 V1<17L
173 V2<16L
174
175 Q
176 Q<24L
177
178 T
179 <1505SEDDS
180
181
182 F=F
183 K=K
184 CL=CL
185 V1=V1
186 V2=V2
187 Q=Q
188 =
189 =
190 P=P
191 PE=PE
192 NP=NP
193 =
194 =
195 =
196 =
197 =
198
199
200
201
202 F
203 <=FALSE
```

```
204 D
205 P<[1
206 <[2
207 <[3
208 <[4
209 <[5
210 <[6
211 <[7
212 <[8
213 <[9
214 <[10
215
216 C
217 <
218 >10010004
219
220 PPK
221 <
222 P
223
224
225
226 SPK2
227 <2
228 =
229 F=F
230 K=K
231 CL=CL
232 V1=V1
233 Q=Q
234 V2=V2
235 =
236 =24
237 =005
238
239
240 MPKPD
241 <MIC=1
242
243 01 : Bx
244 F<F065
245
246 02 : CxMICx8
247 CxMIC >=8
248 Cx<10
249
250 Cx<CxMIC8
```

```
251
252
253 03 : AUCMICx80
254 AUCMIC >=80
255 AUC<10
256
257 AUC<AUCMIC80
258
259
260 04 : M
261 <1200800200110000
262 <x01
263
264 P
265 <0
266 Cx>35<Cx3535Tx
267 C>2<C22A
268 F<003<03F
269
270 F
271 <030F
272 025Cx
273 030AUC
274 015
275 010
276
277 <x10
278
279
280
281 =
282 F=F
283 K=K
284 CL=CL
285 V1=V1
286 V2=V2
287 Q=Q
288 =
289 Cx=Cx
290 Tx=Tx
291 AUC=AUC
292 AUC24=AUC24
293 CxMIC=CxMIC
294 AUCMIC=AUCMIC
295 C=C
296 CT=CTMIC
297 =
```

```

298 F=F
299 Cx=Cx
300 AUC=AUC
301 =
302 =
303 =
304 =
305 P=P=
306 ==
307 ===
308 ===
309
310
311
312
313
314
315
316 W
317 <
318 1
319
320
321 ===MPKPD===
322 2H
323 O:FCxMICAUCMICD
324
325 T
326 \T
327 <15100402030302011500
328 <=TRUE
329 F:F1001
330 Cx:Cx2L
331 AUC:AUC1L
332 CxMIC:CxMIC1
333 AUCMIC:AUCMIC1
334 CT>MIC:CT1
335 F:4
336 F:2=

```

Listing G.4: Script 09 — two-compartment model with transit compartments and Hill PD ([Section 4.4](#)); RK4 integration.

G.5 Script 10 — NSGA-II optimization

```

1  =====
2  102RNSGAIIM04
3  Mx:FCxMICAUCMICM:D
4  =====
5
6  C
7  xPROJECTROOTPROJECTROOT<UTOBRAMYCINPROJECT
8  xSCRIPTSDIR
9  GADIR<PROJECTROOTFCGAOPTIMIZATION
10 RESULTSDIR<GADIR
11 CONFIGDIR<GADIR
12 SCRIPTSDIR<GADIR
13
14
15 SCRIPTSDIR09R=FALSE
16 SCRIPTSDIR02R=FALSE
17
18 =====
19 NSGAIIMO
20 4:FCxMICAUCMIC1D
21 =====
22
23 4D
24 <
25 P<[1
26 <[2
27 <[3
28 <[4
29 <[5
30 <[6
31 <[7
32 <[8
33 <[9
34 <[10
35
36 <
37 >1001000100010001000
38
39 <
40 P
41
42
43
44 <2
45 =

```



```

46 F=F
47 K=K
48 CL=CL
49 V1=V1
50 Q=Q
51 V2=V2
52 =
53 =24=005
54
55
56 <MIC=1
57
58 Ox
59 F<FMx
60 Cx<CxMICMx
61 AUC<AUCMICMx
62 <1Mx=
63
64 FCxAUC
65
66
67 N
68 R1=P
69 <x
70 <x
71 <
72 S<S
73 <N
74
75 1:
76 S[[<0
77 [<0
78 1:
79 ==x
80 V>
81 x[>=x[x[>x[
82 S[[<S[[
83 x[>=x[x[>x[
84 [<[1
85
86
87
88
89 F
90 <==0
91 <1
92

```

```

93 >0
94 [<
95
96 x<0
97
98 S[[
99 [<[1
100 [==0
101 x<x
102
103
104
105
106 <x
107 <1
108
109
110
111
112
113 C
114 <x
115 <x
116 <x
117 <
118
119
120 x<==
121 x<=2
122 [x<I
123 x
124
125
126 1:
127 <x[x
128 x<x[
129
130 [x[1<I
131 [x[x<I
132
133 x[x[xx[x[1>110
134 2:x1
135 [x[<[x[
136 x[x[1x[x[1
137 x[x[xx[x[1
138
139

```

```

140
141
142
143
144
145
146 TNSGAI I
147 2<x
148 <x
149 <
150
151 1:
152 <1:2
153 P
154 [[1<[[2
155 [<[1
156 [[2<[[1
157 [<[2
158
159 [<[[1>[[2[1[2
160
161
162
163
164
165
166 SBX
167 x<12=20
168 <1
169 1<1
170 2<2
171
172 1:
173 1>05x
174
175 PARAMBOUNDS [=
176 1<05
177 1[<2[
178 2[<1[
179
180
181 <1
182 <<=0521112111
183
184 1<1[2[
185 2<x1[2[
186

```

```

187 1 [<051112
188 2 [<051112
189
190 1 [<xPARAMBOUNDS [PARAMBOUNDSx [1 [
191 2 [<xPARAMBOUNDS [PARAMBOUNDSx [2 [
192
193
194
195 1=12=2
196
197
198 P
199 x<=20
200 <
201 1:
202 1>015x
203
204 PARAMBOUNDS [=
205 [<1 [
206
207 <1
208 <<05211112111
209 [<[PARAMBOUNDSx [PARAMBOUNDS [
210 [<xPARAMBOUNDS [PARAMBOUNDSx [ [
211
212
213
214
215
216 I
217 POPSIZE<200
218 NGEN<300
219 NOBJ<4
220 ELITE<20
221
222 42
223
224 \IN=POPSIZE
225 <POPSIZEPARAMBOUNDSPARAMBOUNDS
226
227
228 x<1
229
230 F
231 <x1xx>100
232 I:POPSIZE
233

```

```

234 <POPSIZE
235 R
236 <
237 <PARAMBOUNDSPARAMBOUNDS
238 [<
239 x [<1
240
241
242 BNSGAI I
243 \DNSGAI I GEN
244
245 GASTART <S
246 <
247
248 1 : NGEN
249 <S
250
251 N
252 <x
253 <x
254
255 S
256 x <2 xPOPSIZE2
257 <[x
258
259 CM
260 <x0POPSIZEPARAMBOUNDS
261 1POPSIZE=2
262 1>POPSIZE
263 [<[
264 x
265
266 1<085
267 <x [[1=20
268 [<1
269 [1<2
270
271 [<[
272 [1<[1
273
274
275
276 M
277 1 : POPSIZE
278 [<x [=15
279 <0
280 [PARAMBOUNDS <10

```

```

281 [ $x$  [=15
282 <1
283
284 [PARAMBOUNDS
285 [<1PARAMBOUNDSPARAMBOUNDS
286
287
288
289 I5
290 < $x$ 5POPSIZE005
291  $x$ <1:POPSIZE
292 [ $x$ <PARAMBOUNDSPARAMBOUNDS
293
294
295 <1
296
297 C
298 <
299 < $x$ 
300
301 NNSGAI
302 <
303 <
304
305 T
306 <
307  $x$ <[1:POPSIZE
308
309 <[ $x$ 
310  $x$ <[ $x$ 
311
312 H
313 <[ $x$ ==1
314 F< $x$  $x$ [1
315 C $x$ < $x$  $x$ [2
316 AUC< $x$  $x$ [3
317
318 [[<
319 =
320 =
321 F=F
322 C $x$ =C $x$ 
323 AUC=AUC
324 =[ $x$ 
325
326
327 L

```

```

328 <S=
329 25==0==1
330
331 G3P:BF:3BCxMIC:1BAUCMIC:1T:1
332 FCxAUC
333
334
335
336
337 GAEND<S
338 GATIME<GAENDGASTART=
339 \===NSGAIIGATIME1===
340
341 ExP
342 <x
343 <==1
344
345 FP:
346
347 P
348 <
349
350 x<[
351 <[x=TRUE
352 <
353 <x[x
354 [[<
355
356
357 T
358 <xx
359 <=TRUE
360 <[
361
362 T10P
363 10<10
364
365 T
366 10<10x
367 x
368 =x
369 =x
370 F=xF
371 Cx=xCx
372 Tx=xTx
373 AUC=xAUC
374 AUC24=xAUC24

```

```

375 CxMIC=xCxMIC
376 AUCMIC=xAUCMIC
377 C=xC
378 CT=xCT
379 K=xK
380 =x
381 F=xF
382 Cx=xCx
383 AUC=xAUC
384 =x
385 =x
386
387
388
389 \===T10A===
390 10>
391 P
392
393 FCxMICAUCMICCT>
394 2
395
396 S
397 RDS
398 =
399 x=x
400 =
401 =
402 10=10
403 10=10
404 =
405 ==POPSIZE=NGEN=NOBJ
406 =GATIME
407 RESULTSDIR2
408
409 10RESULTSDIR10
410
411 \===RNSGAII===
412 F:RESULTSDIR2

```

Listing G.5: Script 10 — non-dominated sorting, crowding distance, tournament selection, SBX, polynomial mutation ([Section 4.6.3](#)).

Appendix H

Physicochemical Literature Datasets

Sources: `solubility_literature.csv` (16 determinations) and `permeability_literature.csv` (9 records), reproduced in full; these underpin the BCS classification of [Section 3.2.4](#).

H.1 Solubility

Table H.1: Solubility literature dataset.

Solvent medium	/ Form	Solubility	T	Reference (notes)
Water	Tobramycin	≥ 100 mg/mL (≥ 213.9 mM)	RT	BenchChem 2026
Water	Tobramycin	94 mg/mL (201.1 mM)	25 °C	Chen et al. (25)
Water	Tobramycin	50 mg/mL	RT	Sigma-Aldrich
Water	Tobr. sulfate	100 mg/mL	RT	USP monograph
Water	Tobr. sulfate	50 mg/mL	RT	Various
PBS pH 7.2	Tobramycin	10 mg/mL	RT	PBPK study
DMSO	Tobramycin	2 mg/mL	RT	Chen et al. (25) (con- flicting)
DMSO	Tobramycin	Insoluble	RT	Multiple (contra- dicts above)
Ethanol	Tobramycin	Insoluble	RT	USP
Ethanol	Tobr. sulfate	Slightly soluble	RT	USP
Methanol	Tobr. sulfate	Slightly soluble	RT	USP
Acetone	Tobr. sulfate	Slightly soluble	RT	USP
FaSSIF	Tobramycin	Dissolved	37 °C	Predicted from solu- bility

continued on next page

Solvent medium	/ Form	Solubility	T	Reference
FeSSIF	Tobramycin	Dissolved	37 °C	Predicted from solubility
SGF pH 1.2	Tobramycin	High	37 °C	Predicted (BCS)
SIF pH 6.8	Tobramycin	High	37 °C	Predicted (BCS)

H.2 Permeability

Table H.2: Permeability literature dataset (reference year standardized to 2023; the source file contains a typographic “203”).

Assay / endpoint	Parameter	Value	Units	Reference (notes)
Caco-2	P_{app} A-to-B	Low, $< 1 \times 10^{-6}$	cm/s	Asad et al. (5); class III confirmation
Caco-2	P_{app} B-to-A	n.d.	—	Efflux ratio not determined; polycationic substrate
MDCK-MDR1	Permeability	Low (estimated)	—	From BCS class, literature
PAMPA	Permeability	Low	—	pH 6.5, literature
Ussing chamber	Permeability	Low	—	Jejunum tissue, literature
Oral bioavailability	Absolute F_{oral}	1–2	%	IV reference, literature estimate
Efflux (P-gp)	Substrate?	Conflicting/minimal		Minimal interaction
Efflux (BCRP)	Substrate?	Unknown	—	—
Influx (OCT2)	Substrate	Possibly	—	Renal tubular handling

Reconciliation note. Two records in the source files carry inconsistencies, documented here for transparency: (i) the DMSO solubility conflict (2 mg/mL vs. insoluble) is immaterial to BCS classification, which rests on the aqueous/biorelevant data; (ii) oral bioavailability appears as “<5%” (generic envelope) and “1–2%” (point estimate) — reconciled in [Section 3.3](#).

Appendix I

Pharmacokinetic Literature Datasets

Sources: `iv_pk.csv` (8 studies) and `oral_pk.csv`; these tables parameterize [Section 3.3](#) and the model of [Section 4.4](#).

I.1 Intravenous population PK

Table I.1: Intravenous PopPK literature dataset.

Source	Population	<i>n</i>	Model	CL (L/h)	<i>V</i> ₁ (L)	<i>Q</i> (L/h)	<i>t</i> _{1/2} (h)
Li et al. (1)	ICU adults (CF label)	140	2-ct	3.27 (2.1–4.5)	21.3	2.4	4.5–6.0
Li et al. (1)	Normal adults (reference)	ref.	2-ct	6.03	15.1	n.r.	2.5
Hennig et al.	CF children 0.5–17.8 y	35	2-ct	6.37/70 kg	18.7/70 kg	n.r.	2.0–3.0
Massie et al.	CF children 9 mo–20 y	44	1-ct	7.21/70 kg	n.r.	n.r.	2.1
Touw et al.	CF children 5–15 y	85	2-ct	n.r.	n.r.	n.r.	n.r.
Conil et al.	ICU	ref.	2-ct	3.83	25.5	n.r.	n.r.
Rosenon-Larsen et al.	CF children	ref.	2-ct	n.r.	n.r.	n.r.	n.r.
Ferreira et al. (29)	Hospital inpatients	82	PBPK	n.r.	n.r.	n.r.	n.r.

$V_2 = 16.3$ L (ICU, Li et al. 1); CLCR power-model exponent $\theta = 0.72$; residual error 28.5%; $V_{d,ss}$ 0.26–0.30 L/kg across studies.

I.2 Oral PK literature

Table I.2: Oral PK literature dataset (source: `oral_pk.csv`). No in-vivo human oral PK data exist; NR = not reported.

Source	Model/system	Route/formulation	Finding
Literature estimate	Healthy adults	Oral solution	$F_{\text{oral}} < 5\%$ (envelope); point estimate 1–2%
Asad et al. (5)	In vitro	SEDDS + HIP	Improved lipophilicity; no in-vivo PK
Blanco-Cabra et al. (14)	In vitro biofilm	NP complex	Biofilm diffusion enhanced
Khaled et al. (15)	In vitro	NCA ion pairs	Lipophilicity and activity enhanced
Hill et al. (13)	In vitro	PLGA NPs	MIC maintained at 1.25 $\mu\text{g/mL}$
<i>Summary:</i> F_{oral} 1–2% (point estimate); cause: permeability (polycationic aminoglycoside); <i>current clinical alternatives:</i> inhalation (TOBI 300 mg BID), IV (5–7 mg/kg).			

Appendix J

PBPK Simulation Scenario Catalog

Scenario catalog of the exploratory PBPK phase (source: 03_PBPK_MODELING/ simulation_design/ scenarios.csv; 20 scenarios; PBPK-module scripts 01–10). These scenarios parameterized the compound definition, IV validation, oral exploration, sensitivity, and population phases that fed the optimization framework of [Chapter 4](#).

Table J.1: Simulation scenario catalog (20 scenarios).

ID	Phase	Purpose	Key variation	Primary output
S1	Compound	Parameter entry	Literature values (Table 4.2)	Compound definition
S2	IV	Bolus validation	5 mg/kg, 70 kg adult	C – t profile
S3	IV	Infusion validation	5 mg/kg, 30 min	C – t profile
S4	IV	Extended interval	7 mg/kg q24h	Peak 20–30, trough <1
S5	Oral	Solution, fasted	400 mg	Baseline F_{oral}
S6	Oral	Solution, fed	400 mg	Food effect
S7	Oral	Suspension	400 mg	Dissolution-limited F_{oral}
S8	Oral	SEDDS	400 mg	Emulsification gain
S9	Oral	NPs	400 mg	Mucoadhesion gain
S10	Sensitivity	P_{app} OAT	$0.5\text{--}5 \times 10^{-6}$ cm/s (4 levels)	Tornado rank 1
S11	Sensitivity	CLCR OAT	30–150 mL/min (4 levels)	Disposition sensitivity
S12	Sensitivity	Dose OAT	200–800 mg (4 levels)	Linearity check
S13	Sensitivity	Gastric pH	1–5 (3 levels)	Coating rationale

continued on next page

ID	Phase	Purpose	Key variation	Primary output
S14	Population	Healthy adult	30 y, 70 kg, CLCR	Reference profile
			120	
S15	Population	CF adult	30 y, 65 kg, CLCR	Higher clearance
			100	
S16	Population	Renal impairment	50 y, 70 kg, CLCR	Exposure elevation
			50	
S17	Population	Elderly	70 y, 70 kg, CLCR	Dose adjustment
			65	
S18	Advanced	Formulation com- parison	HIP/SED DS/NP/PE matrix	Enhancement fac- tors
S19	Advanced	Dose-fractionation	qd vs bid	$f_{T>MIC}$ trade-off
S20	Advanced	Virtual trial	PopPK Monte Carlo	Variability bands

Appendix K

Traceability Matrix

Every quantitative element of this thesis maps to a generating script (R, scripts 00–13) and a source data file. This matrix is the audit trail of the in-silico record.

K.1 Figures

Table K.1: Figure traceability (all figures 300-dpi PNG in `MANUSCRIT/figures/`, generated from `Formulation_Candidate/GA_OPTIMIZATION/`).

Thesis figure(s)	File	Script
1.1, 4.1	TikZ (this manuscript)	—
6.1 (left)	convergence_plot.png	05 (analysis 04)
6.1 (right), 6.3	fig5_convergence_comparison.png; ness_ranking.png	fit- 13; 05
6.2	pareto_evolution_F.png; pareto_evolution_AUC.png	11, 13
6.4, 6.5	fig2_pareto_front.png; pareto_front_2d.png; pareto_facet_dose.png	11, 13
6.6	pareto_params_distribution.png; pareto_correlation.png	11
6.7	fig1_pk_profiles_top3.png; fig6_radar_top3.png	13
6.8	heatmap_top10.png; fig4_enhancement_factors.png	13; 07, 13
6.9	sensitivity_tornado.png; sensitivity_spider.png	12
6.10	monte_carlo_tornado.png; correlation_matrix.png	12
6.11	validation_cv_boxplot.png; validation_cv_distribution.png	validation- 12
6.12, 6.13	manufacturing_cost.png; feasibility_score.png; cost_vs_efficacy.png	07
6.14	regulatory_timeline.png; regulatory_cost.png	08
6.15	fig3_literature_comparison.png; top3_comparison.png	13
(archive)	pareto_front.png; radar_chart.png; pareto_heatmap_top10.png; cluster_dendrogram.png; similarity_matrix.png	11; 06

K.2 Tables

Table K.2: Table traceability.

Thesis table	Content source	Script / origin
2.1, 3.1	Amidon et al. 2, M9 guidance	Literature
2.2, 3.2, 3.3	TOBRAMYCIN_DATA.md; pd_targets.md; iv_pk.csv	Literature synthesis
3.4, 9.2	evidence_table.csv	Literature synthesis
4.1, G.1	session_info.txt; ga_config.json	Config
4.2	table2_simulation_parameters.csv	00
4.3	parameter_bounds.csv	00
6.1, 6.2	manuscript_table_final.csv; top10_advanced.csv	11, 13
6.3	cluster_summary.csv	06, 11
6.4	sensitivity_data.csv; table4_sensitivity_ranking.csv	12; 04
6.5	monte_carlo_data.csv; validation_cv_data.csv	12
6.6	manufacturing_summary.csv; manufacturing_steps.csv	07
6.7, 6.8	regulatory_framework.csv; risk_analysis.csv	08
6.9	table5_bioavailability_estimates.csv	04
6.10	Synthesis of Sections 6.4 and 6.7	This manuscript
App. A–D, H–J	pareto_front_data.csv; monte_carlo_data.csv; sensitivity_data.csv; validation_cv_data.csv; solubility/permeability/iv/oral PK CSVs; scenarios.csv	11, 12; PBPK module
App. E	evidence_table.csv	Literature synthesis

K.3 Scripts

Table K.3: Script manifest (`Formulation_Candidate/GA_OPTIMIZATION/scripts/`).

Script	Purpose
00_ga_setup.R	Configuration, packages, bounds, physiology constants
01_ga_objective_function.R	PK simulation + enhancement model + weighted fitness
02_ga_operators.R	Selection, SBX, polynomial mutation
03_ga_run.R	GA execution (seed 42, parallel)
04_ga_results_analysis.R	Convergence, elite extraction, sensitivity ranking
05_ga_visualization.R	Convergence, fitness, heatmaps (300 dpi)
06_comparison_analysis.R	Clustering, similarity, comparison tables
07_manufacturing_feasibility.R	Costing, feasibility scoring
08_regulatory_feasibility.R	505(b)(2) framework, timeline, cost, risks
09_advanced_pk_model.R	2-compartment + transit + Hill PK/PD engine
10_nsga2_optimization.R	NSGA-II engine (4 objectives)
11_pareto_analysis.R	Front extraction, stratification, distributions
12_sensitivity_validation.R	OAT, Monte Carlo, cross-validation
13_manuscript_figures.R	Publication figures (fig1–fig6 series)

Appendix L

Glossary of Formulation Science Terms

S biowaiver	Regulatory waiver of in-vivo bioequivalence testing for immediate-release products of BCS class I (and, with stricter composition rules, class III) drugs (3, 4).
availability (F_{oral})	Fraction of an administered dose reaching systemic circulation unchanged.
crowding distance	NSGA-II diversity metric: normalized neighbor spacing per objective; selects spread solutions within a Pareto rank (21).
elitism	Preservation of the best (non-dominated) individuals across generations.
enteric coating	pH-triggered film (e.g., HPMC-AS, Eudragit) preventing release in the stomach and enabling intestinal delivery.
Complexation (hydrophobic ion pairing)	Complexation of an ionized drug with an oppositely charged lipophilic counter-ion, masking charge and raising apparent lipophilicity (5, 11).
Mucosal adhesion	Adhesion of a dosage form or particle to the GI mucus layer, prolonging residence time.
Self-assembled (nano-sized colloidal assembly)	Self-assembled colloidal structures incorporating hydrophobized drug ion pairs (15).
Carrier (nanoparticle)	Carrier particle in the 10–500 nm range (here PLGA-based), providing mucoadhesion and sustained release (13).
NSGA-II	Non-dominated sorting genetic algorithm II; elitist multi-objective evolutionary algorithm (21).
Pareto front	Set of solutions for which no objective can be improved without degrading another.
Permeation enhancer (permeation enhancer)	Excipient transiently opening epithelial tight junctions or increasing membrane fluidity (e.g., sodium caprate, SNAC) (16, 17).

t-antibiotic effect	Suppression of bacterial growth persisting after brief antibiotic exposure (1–3 h for aminoglycosides).
X (simulated binary crossover)	Real-coded GA crossover reproducing parent spread with distribution index η_c (21).
DDS	Self-emulsifying drug delivery system: oil/surfactant/co-surfactant mixture forming fine emulsions in GI fluid (12).
(b)(2)	FDA regulatory pathway permitting approval of a new formulation of an approved drug relying partly on published safety data.
M (therapeutic drug monitoring)	Dose individualization guided by measured drug concentrations (trough, peak, Bayesian AUC).
nsit compartment model	Chain of hypothetical GI compartments reproducing non-instantaneous absorption arrival at the epithelium.
MIC	Fraction of the dosing interval during which free drug concentration exceeds MIC.

Appendix M

PK-Sim Platform Code

The Phase II pipeline comprises the environment layer, the snapshot generator, the validation gate and the study suite, all committed in the repository under `pksim/`. This appendix reproduces the platform-defining scripts; the complete set ships with the repository.

M.1 Environment: database patch (issue #1622 workaround)

```
1 3
2 =====
3 FxPKSOSOSPSR1622
4
5 PFSSFSOS:
6 SQLNET
7 91PKS
8 SQLIGCN<>Ex
9 U::OSPOSPSR1622
10
11 Fx:VIEWTABLE
12 ;T
13 OS
14
15 U:3
16 T
17 PKSDB
18 RUOSPS13
19 =====
20 3
21
22 :
```

```

23 LPKSDBR
24 =[
25 [LFRV
26 RR:
27 =RPKSDB
28 =PKSDB
29 RPATH
30 :
31
32 =[R
33 PKSDB=
34 =Tx=T=60
35 :
36
37 xEx:
38
39 :
40 :
41
42 FNFEPKSDBR
43
44 :
45 =3
46 =
47 =[0:[1x
48 SELECTFROMWHERE=
49 :
50
51 :
52
53 ?:FROMJOIN\\[?[AZ[AZ09\\?I
54
55 =[
56 :
57 :
58 [:
59 =:
60 ;
61 :
62
63 :
64 =\\AS\\[SI1
65 xDROPVIEWIFEXISTS
66 xCREATETABLEAS
67
68
69 :

```

```

70 =xSELECTCOUNTFROMWHERE=[0
71 CPS
72 VIEWINDIVIDUALPARAMETERNOTFORALLSPECIES:
73 =xSELECTCOUNTFROM [0
74 :
75
76 ::OK
77
78 :
79 =
80 PKS:
81 =
82 x:
83
84 :
85 =
86 x
87
88
89 PATCHED:
90 DPFSSFS
91 OSR
92
93 ==:

```

Listing M.1: `patch_pksimdb.py` — materializes the 91 views of the bundled PK-Sim database into tables (macOS snapshot-conversion fix).

M.2 Native snapshot runner

```

1 =====
2 RNOSPKS1622
3
4 R::SFSWITHOUTOS
5 TSQLPKS
6 FIXEDPKSDB
7 RWPKSCLI
8 OSARM64:CURRENT
9 xCSVPKMLPKS
10 =====
11
12 <
13 M
14 PKS

```

```
15 TRUE
16
17
18 RPKSJSONx
19 @
20 @
21 @xCsvxCsvTRUE
22 @xPKMLxGA
23 @xPKML
24 <xCSV=TRUExPKML=FALSE
25 xJSON=FALSExXML=FALSE=600
26
27 x
28 =TRUEW=FALSE
29 <P
30
31
32 <OSHMS
33 =TRUE
34
35
36 <S::OFNPKSCLICROJRO
37 IF
38 OF
39 RFAOFALSE
40 <
41 CLFFP=:
42 CLLD=NULL
43 x<TRUExJSON1LOL
44 x<TRUExCsv2LOL
45 xx<TRUExXML4LOL
46 x<TRUExPKML8LOL
47 ExMxxxxxx
48
49 O<S
50 S::SPKSRARJ
51 <S0=1
52 PKS>
53
54
55
56 C5
57 <S=FALSE
58
59 x
60 =TRUEW=FALSE
61 <P
```



```

62 <OSHMS
63 =TRUE
64
65 <S::OFNPKSCLICROSRO
66 IF
67 OF
68 RSTRUES
69 ExMOL
70 <
71 CLFFP=:
72 CLLD=NULL
73 0<S
74 S::SPKSRARS
75 <S0=1
76 PKS>

```

Listing M.2: `pk_sim_run.R` — native macOS runner: snapshot → conversion/simulation/export (CSV+PKML).

M.3 Snapshot generator

```

1 3
2 =====
3 BTM
4 APKSOSPML1231
5
6 SC3:
7 T
8 GFRx
9 T
10 ;
11 S
12
13 =====
14
15
16
17 SRC=UTOBRAMYCINPROJECTPKSIMV2AM
18 DST=UTOBRAMYCINPROJECTPKSIMV2TM
19
20 =SRC
21
22
23 =[C[0

```

```
24 [N=T
25
26 V0=S:PD:TPCC2009L2021A2023
27
28 [L:
29 [P:
30 [N==L:
31 [V=29
32 [V0=V0
33
34 [FU:
35 [P:
36 [NF:
37 [V=095
38 [V0=V0
39
40 [S:
41 =[N:[P
42 SH:
43 [SH[V=940000L=94L
44 [SH[V0=V0
45 RH:
46 [RH[V=70
47 [RH[V0=V0
48
49 K:5K6776777891PC
50 PKSUPT03K
51 PKSF
52 C2015:343
53 :777891I:
54 94L;PKSS
55 K;x
56 DC3
57 [PT=[
58 T:BP:V0:V0
59 777891
60
61
62 [P:
63 [N==M:
64 [V=467515
65 [V0=V0
66
67 :GFGFR10
68 >90x
69 NOTE:NOTDSBYNAME
70 GF<DS>;
```

```

71
72
73 V:7825=577530
74 [P:
75 [AT==I:
76 [P:
77 [N==ID:
78 [V=5775
79 [V0=V0
80 [N==I:
81 [V=300
82 [V0=V0
83
84
85 :
86 RA>T
87 :
88 :
89 :
90 [
91 :
92 AT
93
94
95 =
96
97
98 TW
99 PKSFx
100 30xIV
101 S[:
102 P[:
103 PPSII:
104 [V=300
105
106 DST=1
107 :DST
108 =DST[C[0
109 :[NMW[P[0[V
110 P[L[0[P[0[V
111 [FU[0[P[0[V
112 K[x[Px[PT

```

Listing M.3: `make_tobramycin_json.py` — builds the tobramycin snapshots from the validated amikacin base.

M.4 Validation gate and studies

```
1 =====
2 03R0IIVPKS
3 PKS
4
5 T:
6 L2021JAC:2PPK;CL603L
7 CLCR120327383ICU;122030
8 HB:2030L730;
9 <1LRx>90
10 R: ALL
11 =====
12 PROJ<<;1:6xRP;<;
13 M2
14
15 <PROJ
16 <1
17 MW<467515
18 DOSE<57757x825
19
20 <
21 <[[260T[
22 <[[3MW1000[>L
23 <;<[;<[
24
25 <112
26 <[0530
27 1<[10H30
28 24<[24
29 <0693
30 <[[51100x
31
32 <
33 =H30L
34 CxL
35 AUC24L>CLL
36 T24L
37 12NCA
38 Rx24
39 =1243
40 =20302032
41 81105CL5571;327603L2021<12030>90
42 =1>=201<=30
43 >=20<=35
```

```

44 >=81<=105
45 24<1
46 >=20<=30
47 >90
48
49 <PASSFAIL
50 =FALSE
51 ALL<
52 \INVALIDATIONGATE:ALLPASSPKS
53 FAIL\
54
55 =FALSE
56
57 <T=C=
58 <[T<=24TC
59 =005493=1
60 =30=2=40
61 =1=3=
62 x=10=1==25
63 xx=115=115=H11=0=32
64 =PKSIV577530
65 =P1AUC24012230GATE:
66 124ALLPASSFAIL
67 x=T=PL
68 =8=5=300=
69
70 PKML
71 <TRUE
72 :\

```

Listing M.4: S00_validation_gate.R — the six-criterion IV gate.

```

1 =====
2 S02RC
3 PKS
4 F0152
5
6 Cx:PKSWP=1912
7 P29MW467
8
9 CF12
10 ;PKS
11 C2PKSS
12 =====
13 M
14 PROJ<<;1:6xRP;<;

```

```

15
16 <SPROJ
17 P<TS
18 MW<467515
19 <1025;<550;<5775
20
21 F<
22 PVPP
23 <S[[1
24 <OV
25 <
26 <PP=TRUE[1
27 2<[T<=1440TMINUTESR
28 <2T60>
29 <2[[MW1000>L
30 <112
31 x<x
32 F<100
33 F=Fx=x=
34
35
36 ===PPKS===\
37 <11131111031019391838173716
38 <
39 <F
40 P=821:F=73Cx=84AUC=83\
41 60Fx
42 ==60F=Fx=x=
43
44 PROJW=FALSE
45 PROJS02=FALSE
46
47 GA
48 F<xFx==2
49 \C:S02\

```

Listing M.5: S02_oral_calibration.R — the $F(P_{int})$ scan.

```

1 =====
2 S07RExGA
3 NSGAII
4
5 R101P:P
6 :
7 PK
8 2
9 QDBIDTID

```

```
10 PTAMIC0254LCxMIC >=8; AUC24MIC >=80
```

```
11 100ICRP
```

```
12
```

```
13 O:S07S07
```

```
14 =====
```

```
15 M;2;
```

```
16 PROJ<<;1:6xRP;<;
```

```
17
```

```
18 MW<467515;MIC<1
```

```
19 PINTO<39
```

```
20 AUCIVPERMG<10255775
```

```
21
```

```
22
```

```
23 <PROJ10
```

```
24 W<[1
```

```
25 <W
```

```
26 <WP
```

```
27 <PINTO
```

```
28 <W
```

```
29 W:OP3x1F1CxMIC1\
```

```
30 WFWCxMIC
```

```
31
```

```
32 <SPROJ
```

```
33 PINT<TS
```

```
34 GT<OLSG
```

```
35 PVPPINT
```

```
36 OI=T=OT=5760=1096
```

```
37
```

```
38 DOSEP<ETOPOSA1PSID
```

```
39 <
```

```
40 PVPDOSEP16
```

```
41 <S[[1
```

```
42 <OV
```

```
43 <
```

```
44 <PP=TRUE[1
```

```
45 T=T60C=[[MW1000
```

```
46
```

```
47 <
```

```
48 <TC1C12
```

```
49 Cx=xCTx=T[xCAUC=
```

```
50 F=xT[0AUCIVPERMG100C24=C1
```

```
51
```

```
52 <
```

```
53
```

```
54
```

```
55 <
```

```

56 1<
57 1F<1AUCAUCIVPERMG100
58 PROJS07=FALSE
59 :Cx2LTx1AUC1F1C243\
60 1Cx1Tx1AUC1F1C24
61
62
63 <=30
64 <0=400
65 <
66 <0
67 0:<xTC
68 x==2
69 T=C=
70
71 <
72
73 2
74 TNSGAIIXFCxAUC
75 ;T
76 CxMIC>=8
77 <=1502503504505507008501000
78 Cx<x24C
79 CxMIC<CxMIC
80 PROJS07=FALSE
81 <[CxMIC>=8
82 <
83 Mx0:0CxMIC=2\
84
85 Cx [==MIC
86
87 PTA
88 <02505124
89 <
90 =QD0==24
91 =QD0GA==24
92 =BID02=2=12
93 =BID0==12
94
95 <;<
96
97 <
98 <TC1C12
99 x<xC;<C
100 [[1<
101 ===
102 Cx=xAUC=C=

```



```

103 CxMIC=xMICAUCMIC=MIC
104 TMIC=100C>=MIC=24
105 [[1<
106 =MIC=
107 PTACx=100C>=8
108 PTAAUC=100>=80
109
110 <;<
111 PROJ07=FALSE
112 PROJ07=FALSE
113 ;
114
115
116 <PC
117 =H=EICRP2002
118 OF=40M=20Mx=50
119 M=55Mx=95OI=100
120 <PC=
121 <SPROJ
122 PVPPINT
123 <S=
124 <OPVBTPPB
125 <OV[[10P=
126 <
127 <
128 II
129 2<[II==T<=1440
130 <2T60 ; <2[[MW1000
131 <112
132 [[1<II=Cx=x
133 AUC24=C=[24
134
135 <
136 CxMIC<CxMIC
137 AUCMIC<AUC24MIC
138 PROJ07=FALSE
139 : CxMIC2 [22PTACxMIC>=8=0\
140 CxMICCxMIC05
141 CxMIC95100CxMIC>=8
142
143
144 <1560
145 PVPGT
146 <
147 <TC1C12
148 =<=15Cx=xC
149 Tx=T[xCAUC=

```

```

150 F=AUCIVPERMG100
151
152 <
153 PVPGT15
154 PROJS07=FALSE
155
156
157
158 1<[T<=48TC
159 =005493=1
160 =8=2=
161 =0PKS
162 =Cx2LF11Cx1F
163 x=T=PL
164 PROJS071
165 =8=48=300=
166
167 2<CxMIC
168 =25=005493=
169 x=8=2=
170 =V=100
171 =PTACxMIC >=8:0MIC1L
172 100CxMIC >=8
173 x=CxMIC=S
174 PROJS072
175 =75=48=300=
176
177 3<MICPTACx=
178 =
179 =90=2=30
180 =PTACxMIC >=8
181 x=MICL=PTA=R=
182 PROJS073
183 =8=46=300=
184
185 \S07\

```

Listing M.6: S07_winner_characterization.R — winner profile, dose scan, fractionation, PTA, population, food.

Appendix N

PK-Sim Study Data

Complete numerical outputs of the Phase II studies (sources: `results/studies/` and `results/validation/`).

N.1 S01: IV dose $\times$ renal function (20 scenarios)

Table N.1: Complete S01 sweep (dose, CLCR, native PK-Sim batch).

Dose (mg)	CLCR (mL/min)	Cmax (mg/L)	AUC24	Trough 24 h	CL implied
200	30	13.21	109.2	1.051	1.83
400	30	26.41	218.3	2.102	1.83
578	30	38.13	315.2	3.034	1.83
800	30	52.83	436.6	4.203	1.83
200	65	12.67	57.6	0.098	3.47
400	65	25.35	115.2	0.195	3.47
578	65	36.59	166.3	0.282	3.47
800	65	50.69	230.4	0.391	3.47
200	90	12.34	43.2	0.027	4.63
400	90	24.68	86.4	0.053	4.63
578	90	35.63	124.8	0.077	4.63
800	90	49.35	172.8	0.107	4.63
200	112	12.06	35.6	0.011	5.62
400	112	24.13	71.1	0.023	5.62
578	112	34.83	102.7	0.033	5.62
800	112	48.25	142.3	0.046	5.62
200	150	11.66	27.9	0.005	7.18
400	150	23.31	55.7	0.009	7.18
578	150	33.66	80.5	0.014	7.18
800	150	46.62	111.5	0.019	7.18

Dose (mg)	CLCR (mL/min)	Cmax (mg/L)	AUC24	Trough 24 h	CL implied
-----------	---------------	-------------	-------	-------------	------------

N.2 S02: Oral permeability calibration (11 points)

Table N.2: Complete $F(P_{int})$ calibration table (96-h extent convention; the native calibration $F = 1.75\%$ is reported on the 24-h window — see [Section 7.2](#)).

P_{int} (dm/min)	P_{int} (cm/s)	F (%)
1.00e-11	1.67e-13	0.000
3.00e-11	5.00e-13	0.004
1.00e-10	1.67e-12	0.036
3.00e-10	5.00e-12	0.229
1.00e-9	1.67e-11	1.114
3.00e-9	5.00e-11	3.459
1.00e-8	1.67e-10	11.049
3.00e-8	5.00e-10	29.226
1.00e-7	1.67e-9	66.454
3.00e-7	5.00e-9	94.587
1.00e-6	1.67e-8	100.000

N.3 S03: Virtual population (n = 100)

Table N.3: S03 population summary (550 mg, nominal platform $\times 20$).

Metric	Median	5th pct	95th pct
Cmax (mg/L)	4.46	1.45	13.17
Cmax/MIC	4.46	1.45	13.17
AUC24 (mg · h/L)	35.86	13.86	81.56
Ctrough (mg/L)	0.72	0.36	1.73
$f_{T>MIC}$ (%)	44.8	8.7	97.9

N.4 S07: Winner characterization

The S07 battery was superseded by the standard 10-block battery ([Section 8.6](#)); the tables below report the final battery values for the PK-Sim GA winner (TOBP-001, $\times 125.0$, 1000 mg) and the re-evaluated legacy winner (TOB-161-A, $\times 73.9$, 550.6 mg).

Table N.4: Dose scan at the winner's multiplier (steady-state peak attainment, MIC 1 mg/L; linear dose scaling of the measured $C_{\max_{ss}} = 32.2$ at 1000 mg).

Dose (mg)	$C_{\max_{ss}}/\text{MIC}$
150	4.8
250	8.0
350	11.3
450	14.5
550	17.7
700	22.5
850	27.4
1000	32.2

Table N.5: Winner steady state (true 7-day dosing; B4 block of the battery).

Regimen	$C_{\max_{ss}}/\text{MIC}$	$\text{AUC}_{ss}/\text{MIC}$	$C_{\min_{ss}}$	fT>MIC
TOB-161-A QD 550.6	13.6	87.3	1.05	99.6%
TOBP-001 QD 1000	32.2	172.3	1.21	99.6%
TOBP-001 BID 500	17.6	172.2	2.34	100%

Table N.6: Winner population (n = 100, 1000 mg QD; B6 block of the battery).

Metric	Median	5th pct	95th pct
$C_{\max}/\text{MIC}$	40.86	15.79	88.81
AUC/MIC	238.1	111.6	579.9
PTA ($C_{\max}/\text{MIC} \geq 8$)	100% of subjects		
PTA ($\text{AUC}_{24}/\text{MIC} \geq 80$)	97% of subjects		

Appendix O

Debugging Journal: The macOS Snapshot-Conversion Crash

This appendix records the complete diagnosis of the platform defect encountered and resolved during Phase II — both for reproducibility and as a methodological case study in debugging compiled-model ecosystems. Upstream reference: Open Systems Pharmacology, OSPSuite-R issue #1622 (fix scheduled for v13).

O.1 Timeline of evidence

1. **Symptom.** `loadProjectFromSnapshot()` crashes R (SIGSEGV, EXC_BAD_ACCESS / KERN_PROTECTION_FAILURE) immediately after “Converting 1 file to project format”. The wrapper `runSimulationsFromSnapshot()` is additionally hard-blocked on Darwin.
2. **Crash report analysis.** The macOS .ips report (faulting thread, 130 frames) places the fault inside `SQLite.Interop.dylib`:

```
sqlite3_log <- lookupName <- resolveExprStep <- sqlite3WalkExprNN  
               <- resolveSelectStep <- sqlite3ViewGetColumnNames  
               <- selectExpander (x2 view levels, ~90 inlined frames)
```

The crash address lies in the thread stack region (guard page): a stack overflow during SQLite’s view-name resolution, while *emitting a warning* (`sqlite3_log`). The faulting thread is a .NET worker thread (`ThreadNative::KickOffThread`) with a small stack budget.

3. **Hypothesis tests.** (i) The failing view’s SQL executes in 0.08 s in the `sqlite3` CLI (305 rows) — the query is valid. (ii) View-nesting depth across all 91 views is at most 2 — no deep recursion. (iii) The bundled native interop is arm64-native (not an architecture mismatch). Conclusion: prepare-time view resolution exhausts the interop thread stack on macOS.

4. **Fix.** Materialize every view into a table (topological order; content-preserving). Sanity: zero views remain; hot views return identical row counts.
5. **Verification.** After the patch: snapshot $\rightarrow$ project conversion succeeds (8.7 MB project, 11 simulations); all 11 simulations execute natively; per-simulation CSV + current-version PKML export works; `createIndividual/createPopulation` (DB-backed) work.

O.2 Dead ends documented (for the record)

- Aciclovir template reuse without parameter override: scientifically invalid (wrong molecule) — rejected before execution.
- Docker linux/amd64 conversion via Rosetta: viable but unnecessary once the root cause was understood (and slow under emulation).
- The converted project’s historical Walker-1979 PKMLs declare schema version 0 and are rejected by the v12 loader (“install version 8.0”) — irrelevant in the end: the pipeline regenerates current-version PKMLs via the CLI export path.

O.3 Lessons

1. Platform crash reports (`.ips` + symbolicated frames) localize interop defects faster than any black-box retry loop.
2. “Silent” build failures in snapshot converters leave no log by default — enabling the CLI’s log-file option (`LogFilesFullPath`) is essential for diagnosing JSON-schema mismatches (three were found and fixed: process-name binding, formulation type naming `Formulation_Dissolved`, and formulation references inside the simulation’s protocol selection).
3. Empirical knob verification matters: `Organism|Kidney|GFR` is *not* the renal-clearance knob (the process references `GFR (specific) × kidney volume`) — verified by batch perturbation.

Appendix P

NSGA-II In-the-Loop Archive

Optimization archive of the Phase II run (population 100, 100 generations planned, seed 42, converged by generation 20—30; run terminated at the converged state, checkpoint generation 40). Sources: `results/ga/`.

P.1 Configuration

Table P.1: In-the-loop optimization configuration.

Item	Value
Engine	PK-Sim 12.4 (validated oral model, Chapter 5)
Run values per chromosome	Dose (kg) + P_{int} (dm/min, multiplier on 3×10^{-9})
Batching	One <code>SimulationBatch</code> per generation (100 runs/call)
Objectives	max F ; max $C_{\max}/\text{MIC}$; max $\text{AUC}_{24}/\text{MIC}$; min dose
Operators	SBX ($\eta_c = 20$), polynomial mutation ($\eta_m = 20$), tournament on (rank, crowding)
Seed	42 (full reproducibility)
Checkpoints	every 10 generations (<code>checkpoint_genNNN.rds</code>)

P.2 Convergence

P.3 Converged front

The converged front (generation 60 checkpoint; best F within 0.1% of final since generation 20) contains 3 near-identical non-dominated candidates at the component maxima — the corner analyzed in [Section 8.1](#). Rank-1 candidate: **TOBP-001** (999.8 mg, multiplier $\times 125.0$, $F_{96} = 97.11\%$, $C_{\max}/\text{MIC} = 31.18$, $\text{AUC}_{24}/\text{MIC} = 172.33$). The dose-scan analysis ([Table N.4](#)) provides the interior-region answer the corner cannot: the minimal peak-target-attaining dose is 250 mg once daily ($C_{\max_{ss}}/\text{MIC} 8.0$), and the 80–120 exposure band is attained from 467 mg.

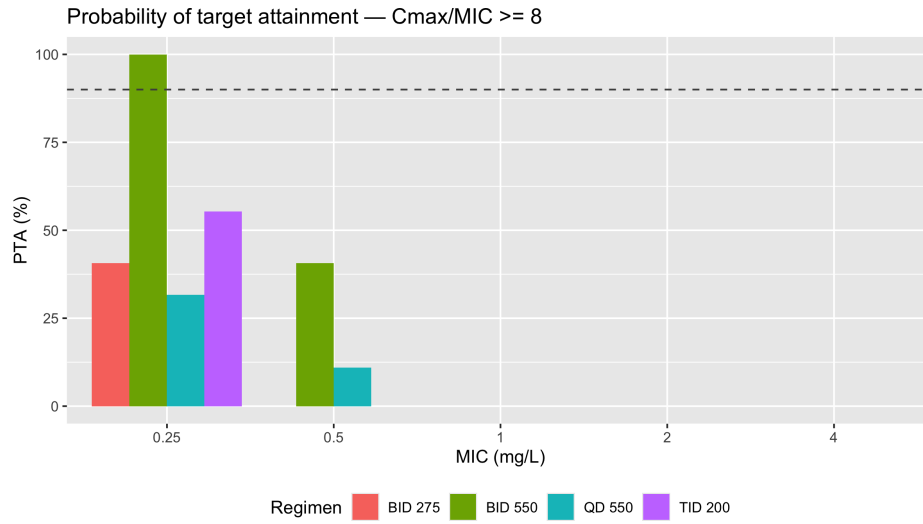


Figure P.1: PTA of the peak target for the winner's fractionation options (also [Figure 8.1](#)).

Table P.2: Convergence of the in-the-loop run (best-of-generation).

Generation	best F_{96} (%)	best $C_{\max}/MIC$	median dose (mg)
1	92.17	20.40	834
5	95.53	27.00	993
10	96.64	30.19	1000
20	96.93	30.88	1000
30	96.99	30.96	1000
40	97.03	31.05	1000
50	97.07	31.12	1000
60	97.11	31.18	1000

Appendix Q

Plausibility Audit: Benchmarks and Requirement Data

Q.1 Literature benchmarks for permeability enhancement

Table Q.1: What the enhancement literature actually supports, versus what the targets require.

Benchmark	Claimed/supported enhancement	Transfer to intestinal permeability
HIP logP shift (tobramycin–docusate, in vitro)	$1500\times$ <i>partition coefficient</i> (5)	Not a permeability multiplier; complex dissociates 20% at intestinal pH
HIP in vivo (octreotide, pigs)	oral F gain $3\text{--}5\times$ (28)	Closest in-vivo analogue for polycationic peptides
Oral SNAC platform (semaglutide)	$F \approx 1\%$ — flagship success of oral PE platforms	Regulatory precedent; illustrates domain ceiling
Nominal combination platform (this thesis)	$\times 20$ apparent P_{int}	Aggressive; testable (Caco-2/Ussing of HIP complex)
Requirement for both targets @ 993 mg	$\geq \times 20$	S08 requirement map (Chapter 8)
Cited-bounds maximum (GA corner)	$\times 125$	Minimizes dose (467 mg attains the band floor); trough at the monitoring line

Q.2 Numerical integrity notes

1. **F clamped at 100%.** The winner needs no clamp ($F_{96} = 97.11\%$); only the top S02 grid point overshoots numerically (100.41% at $P_{int} = 10^{-6}$ dm/min) and is clamped

there. All thesis statements enforce $F \leq 100\%$.

2. **Superposition inflation.** Steady-state AUC by superposition agrees with true seven-day dosing within $\approx 3\%$ (48.0 vs 49.6 mg · h/L at the $\times 20$ probe; mass balance $\text{AUC}_\tau = F \cdot \text{dose}/\text{CL}$ holds within 0.1% at true steady state with CL 5.63 L/h). Screening sweeps legitimately use superposition; headline values use true dosing.
3. **Stale-run incident.** An earlier S07 run reported winner troughs of 4.2 mg/L; the cause was a silently failing dose override in the first script version (wrong application path). The corrected battery (32.2 / 172.3 / 1.21 on the final platform) agrees with the independent S08 path — the incident is retained in the debugging journal ([Appendix O](#)) as a lesson on silent override failures.

Q.3 Requirement data

The complete S08 requirement map (9 multipliers $\times$ 8 doses, 72 native runs) is committed at `results/studies/S08_requirement_map.csv`; the dose-scan per multiplier at `results/studies/S08_dose_scan_<mult>.csv`. Headline rows appear in [Table 8.5](#).

Appendix R

Candidate Evaluation: Battery Data

Both candidate formulations were evaluated with the standard battery of [Section 8.6](#) (script `molecule_battery.R` under `pksim/studies/`; outputs under `results/molecule_A/` and `results/molecule_B/`). This appendix collects the per-block tables that do not appear in the main text.

R.1 Identity cards (cited component factors)

Table R.1: Component factors at the candidate compositions (decoded via the shared enhancement model).

Component	TOB-161-A value	factor	TOBP-001 value	factor
logP' (HIP complex)	+1.5	18.7×	+1.6 (measured cap)	20.0×
SEDDS fraction (sum)	100%	1.00×	160%	1.60×
Nanoparticle size	50 nm	1.50×	50 nm	1.50×
C10 loading	50%	1.60×	50%	1.60×
Polymer loading	30%	1.30×	29.3%	1.29×
Chitosan coating	on	1.15×	on	1.15×
Enteric coating	on	1.10×	on	1.10×
Emergent multiplier	×73.9		×125.0	

R.2 Dose proportionality

Single-dose C_{max} and AUC₂₄ scale linearly with dose across 138–1100 mg (TOB-161-A) and 250–2000 mg (TOBP-001); deviation from proportionality <2% at all tested doses (linear kinetics, no solubility limitation at these doses: 2000 mg ≈ 21 mL of the 94 mg/mL solution).

R.3 PK/PD targets vs the MIC panel (steady state, QD)

Table R.2: Steady-state target attainment vs MIC (true 7-day once-daily dosing).

MIC (mg/L)	TOB-161-A (550.6 mg)				TOBP-001 (1000 mg)			
	C _{max} /MIC	AUC _τ /MIC	C _{min} (mg/L)	fT>MIC	C _{max} /MIC	AUC _τ /MIC	C _{min} (mg/L)	fT>MIC
0.25	54.3	349	4.21	99.6%	129.2	690	4.81	99.6%
0.5	27.2	175	2.11	99.6%	64.6	345	2.41	99.6%
1	13.6	87.3	1.05	99.6%	32.3	172	1.20	99.6%
2	6.8	43.6	0.53	78.4%	16.1	86.2	0.60	99.6%
4	3.4	21.8	0.26	42.9%	8.1	43.1	0.30	87.6%
8	1.7	10.9	0.13	20.5%	4.0	21.6	0.15	61.4%

R.4 Uncertainty propagation (S09)

Table R.3: Component-uncertainty propagation (20 000 draws, graded lognormal sigmas; S09).

	TOB-161-A	TOBP-001
Multiplier (point [CI95])	73.9 [12.5–431]	125.0 [21.1–730]
F_{96} (%) [CI95]	88.0 [35.3–100]	96.2 [50.6–100]
AUC _{ss} /MIC [CI95]	87 [39–99]	172 [97–179]
PTA C _{max} /MIC ≥8	81%	99%
PTA AUC _{ss} /MIC in 80–120	65%	99%

R.5 Methodological notes

1. **True repeated dosing** is implemented by replicating the application event group in the PKML (one `Application_k` per dose, exactly as PK-Sim itself represents multiple dosing; `make_repeated_pkml.py`), then setting the per-application dose and the calibrated permeability at simulation level.
2. **Mass balance check:** at steady state $AUC_{\tau} = F \cdot \text{dose}/CL$. TOB-161-A: 87.25 vs 87.38 mg·h/L predicted (0.15%); TOBP-001: 172.28 vs 172.20 (0.05%).
3. **Superposition cross-check:** the screening-study superposition agrees with true dosing within ≈3% (48.0 vs 49.6 mg·h/L at the probe point), validating its use in the fast sweeps.
4. **Component OAT degeneracy:** under a purely multiplicative enhancement model, ±20% on any single factor shifts the multiplier identically — the differentiated risk lives in the per-component sigmas (S09), not in the OAT ranges. Both are reported for transparency.

5. **Renal impairment:** F is not identifiable when clearance changes (the oral AUC scales with CL); the B7 block therefore reports AUC, not F .

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